

Single-Stage PFC LLC Converter Design

with the STMicroelectronics L6790A

Worked design

90 to 264 Vac, 47 to 63 Hz

25 V / 26.3 A = 657.5 W

Single-Stage PFC LLC Converter Design

Before you start

The L6790A behaviour described here comes from a **draft** datasheet dated 30 April 2026. That draft has items marked TBD and at least one place where it contradicts itself. Check every controller constant against the released datasheet before the design goes to production.

The worked design has not been built. Only the numbers marked as measured are measured; the rest are calculated.

Some of the LLC theory figures are taken from the application notes in the reference list. Each one names its source in the caption.

Revision history

Version	Date	Change
V1.0	August 2026	First issue. Worked design: 25 V / 26.3 A, 657.5 W.
V1.1	September 2026	Background section on LLC and power factor correction added, with circuit figures. Transformer design, the closed form for $f_{sw}(\theta)$, the compensator design method, the gain-margin check and the input capacitor added. Worked design moved to the $N_p:N_s = 15:2$ transformer.
V1.2	September 2026	Renamed single-stage PFC LLC throughout. Verification margins defined. Worked values moved out of the design procedure and into the design example, where each one names the equation it comes from and shows the numbers put into it.

Conventions

- Frequencies are switching frequencies unless written f_l , which is the line frequency.
- $f_n = f_{sw}/f_r$ is the switching frequency normalised to the series resonance. Gain curves are always drawn against it, so that one curve serves every tank. Figures reproduced from other documents may write f_s for f_{sw} ; that is the same quantity.
- M is the voltage gain of the resonant network: the output voltage referred to the primary, divided by the fundamental of the square wave driving it. **$M = 1$ at the series resonance**, whatever the load.
- $Z_0 = \sqrt{L_r/C_r}$ is the characteristic impedance of the tank, and $Q = Z_0/R_{ac}$ is its quality factor — how heavily the tank is loaded. $Q = 0$ is no load; larger Q is more power and a flatter, lower gain curve.
- $V_{o,eff} = V_{out} + N_{rect} V_f$ is the output voltage the primary actually sees, with the rectifier drop added. N_{rect} is the number of rectifier drops in the conduction path: 1 for a centre tap, 2 for a full bridge.
- n is the equivalent-model turns ratio used in the gain equation; n_T is the physical turns ratio of a transformer whose leakage is integrated into L_r .
- $\lambda = L_r/L_m$, and $m = 1 + 1/\lambda$ is the ratio of total primary to resonant inductance used in older literature.
- θ is the line phase angle, zero at the mains zero crossing.
- “Equivalent input” is what the resonant tank sees after topology morphing, which is not the mains voltage.

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1 Introduction

A conventional universal-input LLC power supply has two stages: a boost power-factor corrector that holds a 400 V bus, and an LLC converter that steps that bus down. The capacitor between them does two jobs. It stores the energy that a unity-power-factor input cannot deliver evenly, and it gives the LLC an almost constant input, so the resonant tank only has to cover a narrow range.

A **single-stage PFC LLC** removes both the boost stage and that capacitor. The rectified mains feeds the resonant tank directly. The result is a converter whose input is a 100/120 Hz half sine, whose gain is no longer something the controller can set, and whose output capacitor must absorb everything the bus capacitor used to. It is a different converter, not an LLC with one stage removed, and most of the design habits from the two-stage case do not apply.

This note explains how such a converter works and how to design one with the STMicroelectronics **L6790A** controller. One worked design runs through it, from specification to component values: **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, a low-voltage, high-current dc rail.

Scope. Covered: the power stage, the resonant tank, the output bank, the controller network and the voltage loop.
Not covered: EMI filtering, magnetics construction, layout and safety approval.

2 The two converters this one is made of

What comes later is easier to follow if the two halves are understood separately first. This section covers an ordinary LLC and ordinary power factor correction. A reader who has designed both can skip to Section 3.

2.1 What an LLC converter is

An LLC converter is a square-wave generator that drives a resonant network; the network feeds a transformer and a rectifier. The switches only chop the input into a square wave whose frequency can be varied. There is no duty cycle to set and no inductor current to program.

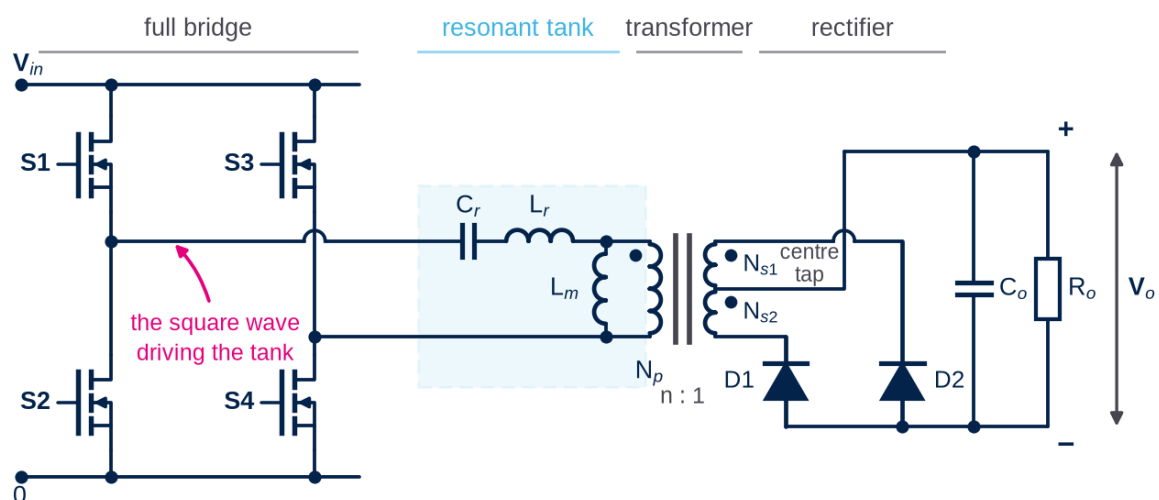


Figure 1: An LLC stage: a square-wave generator, the resonant network, a transformer and a rectifier. The switches only set the frequency — the tank decides how much power flows and the transformer sets the voltage.

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The three tank elements are named in the order they appear in “LLC”: the series inductance L_r , the magnetising inductance L_m of the transformer, and the series capacitance C_r . The middle one is not a separate part: it is the transformer’s own magnetising inductance. So only C_r and L_r are added components, and L_r is often the leakage inductance of the transformer itself.

- **The square wave sets the frequency, and nothing else.** The output voltage is controlled by moving f_{sw} , because the impedance of the tank, and so its voltage division, depends on frequency.
- **The tank current is almost a sine wave.** That is the purpose: the switches turn on and off while the current is small, or while it flows through the body diode, so most of the switching loss disappears.
- **The transformer steps the voltage and provides isolation,** and its leakage inductance can be used as L_r instead of being a problem. That is the usual construction at these power levels.

2.2 Why an LLC, and not something simpler

An LLC is more work than a forward or a flyback converter, and it does not regulate by duty cycle, which is what most designers are used to. At a few hundred watts and above, it is worth the extra work for these reasons:

- **Soft switching over the whole load range, on both sides.** The primary switches turn on at zero voltage, and below resonance the secondary rectifiers turn off at zero current. A phase-shifted full bridge loses ZVS at light load; an LLC does not, because the magnetising current that does the work is present at any load.
- **The frequency can be raised.** In a hard-switched converter, switching loss limits the frequency, and with it the size of the magnetics and the filter. Remove that loss and hundreds of kHz become practical.
- **The transformer leakage is used, not fought.** In a forward converter the leakage inductance is a problem that needs a snubber. Here it can be L_r , so the resonant inductor costs nothing and the snubber is not needed.
- **No output inductor.** The rectifier feeds the output capacitor directly. That removes a wound component, and its loss, from the high-current side, where it hurts most.
- **Low noise.** The currents are sine waves rather than trapezoids, and the switching edges are slow and damped, so the conducted and radiated emissions are much lower than in a hard-switched converter of the same power.

The costs are real too. Regulation is by frequency, so the operating frequency moves with line and load, and the magnetics must work over that whole range. The magnetising current circulates whether or not the load takes power, so light-load efficiency is not a strong point. And the design does not come from a duty-cycle expression: it comes from the gain curve, which is what the rest of this section is about.

2.3 Why resonance buys anything

In a hard-switched converter a switch turns on with the full input voltage across it, so the energy stored in its output capacitance is lost in the channel every cycle. That loss grows with frequency. It is what limits the frequency, and with it the size of the magnetics.

An LLC that runs **in its inductive region** presents an inductive load to the bridge. The tank current then lags the drive voltage. In the dead time between the two switches of a leg, that lagging current discharges the mid-point capacitance by itself, and the next switch turns on at zero volts. **That is zero-voltage switching, and it is the whole reason for the topology.** Section 2.10 shows the mechanism and what it requires of the design.

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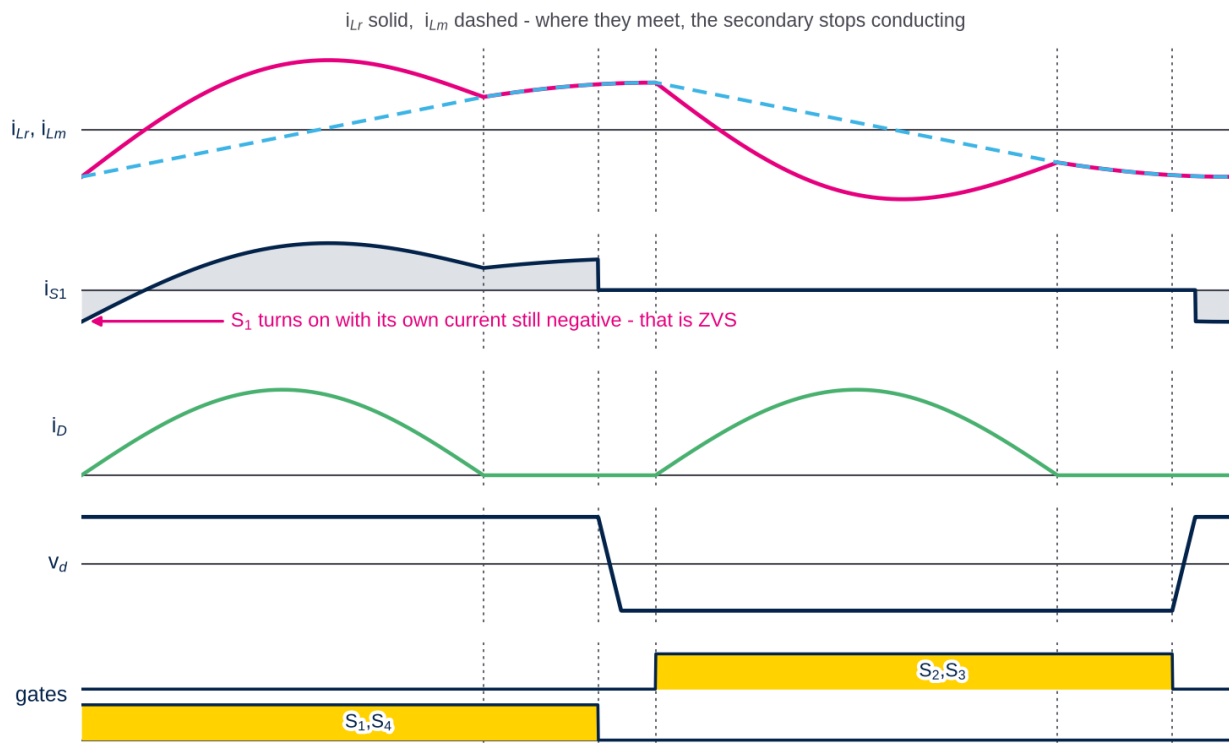


Figure 2: One switching period below resonance. i_{Lr} is the tank current and i_{Lm} the magnetising current; the difference between them is what crosses to the secondary, and it is gone once the two meet. The switch turns on while its own current is still negative, and that is what discharges the mid-point.

The two frequencies that bound this are the subject of Section 2.5, and the operating regions they divide are in Section 2.6.

The condition is *inductive operation*, not a particular frequency. Go low enough in frequency and the tank becomes capacitive: the current leads, the dead time works against the transition instead of for it, and the bridge hard-switches into a low impedance. It is the one failure mode that destroys parts rather than just heating them. **Where that happens is not a fixed frequency:** the boundary moves with load, and Section 2.9 is about exactly that.

2.4 How the circuit becomes $M(f_n, Q)$

Every gain curve in every LLC note comes from three simplifications, applied in order. They are worth doing once, because each one throws something away, and it is useful to know what.

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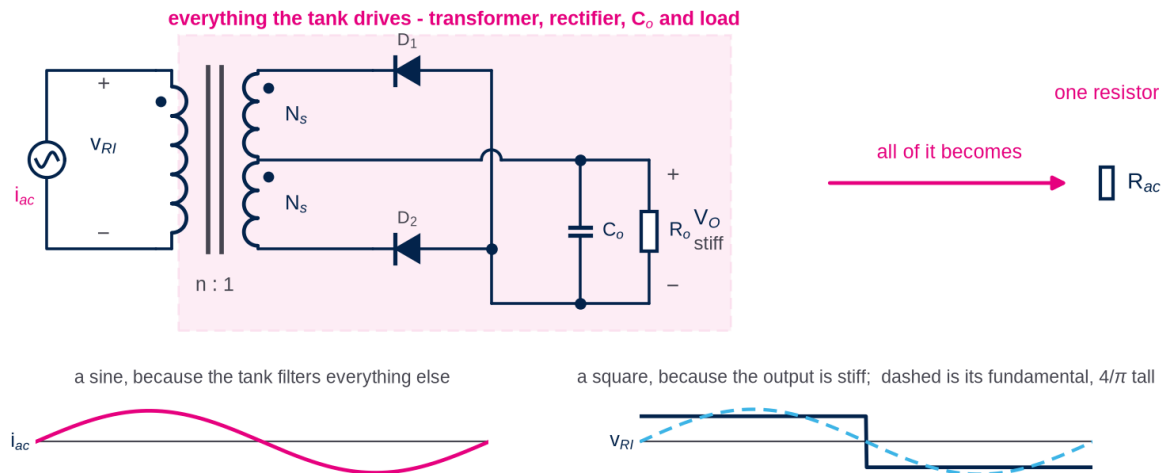


Figure 3: Why the rectifier and the load can be replaced by one resistance. The output is stiff, so the voltage the tank sees is a square wave; the tank filters everything except the fundamental, so the current is a sine wave. A sine of current in phase with a square of voltage delivers the same fundamental power as a resistor would take.

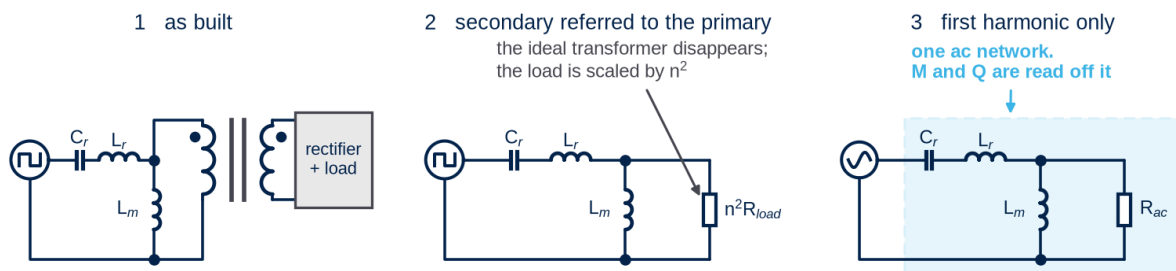


Figure 4: The reduction, in three steps: as built; the secondary referred to the primary; the fundamental only. What is left is one ac network whose voltage transfer is the gain M and whose damping is Q .

- **Refer the secondary to the primary.** The ideal transformer disappears and the load resistance is multiplied by n^2 .
- **Replace the rectifier and the output filter by one resistor.** The rectifier draws a square current wave and the output is stiff, so at the fundamental frequency the whole of it looks like a resistor.
- **Keep only the fundamental of the drive.** The tank is a filter centred near the switching frequency, so the harmonics of the square wave contribute little. This is the *first harmonic approximation*, and it is why every LLC design ends with a simulation or a measurement rather than with the equations.

What is left is one ac network: a voltage divider whose ratio changes with frequency. Two numbers describe it, and both are used on every page from here on:

- **The gain M :** the voltage across R_{ac} divided by the fundamental that drives it. Because C_r and L_r cancel at the series resonance f_r , **$M = 1$ there at any load**. Below f_r the network can step up ($M > 1$); above it, it steps down.
- **The quality factor $Q = Z_0/R_{ac}$,** where $Z_0 = \sqrt{L_r/C_r}$ is the characteristic impedance of the tank. Q says how heavily the tank is loaded: **Q rises with output power**. At $Q = 0$ (no load) the gain curve is tall and peaked; loading flattens the curve and pulls the peak down.

Frequency is always written normalised, $f_n = f_{sw}/f_r$, so that one family of curves serves every tank. $M(f_n, Q)$ always means the gain of the right-hand circuit above.

2.5 The two resonances

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Every LLC has two. While the secondary conducts, the reflected output voltage clamps L_m and it drops out of the circuit, leaving the series resonance

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (1)$$

At no load nothing clamps L_m , and the tank resonates with both inductances:

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}} \quad (2)$$

At f_r the gain is exactly 1 for any load, because the load term disappears. At f_o the no-load gain is unlimited. Between the two the tank can boost; above f_r it cannot. An LLC fed from a fixed dc bus runs close to f_r nearly all the time, and drops into that band only when the bus sags.

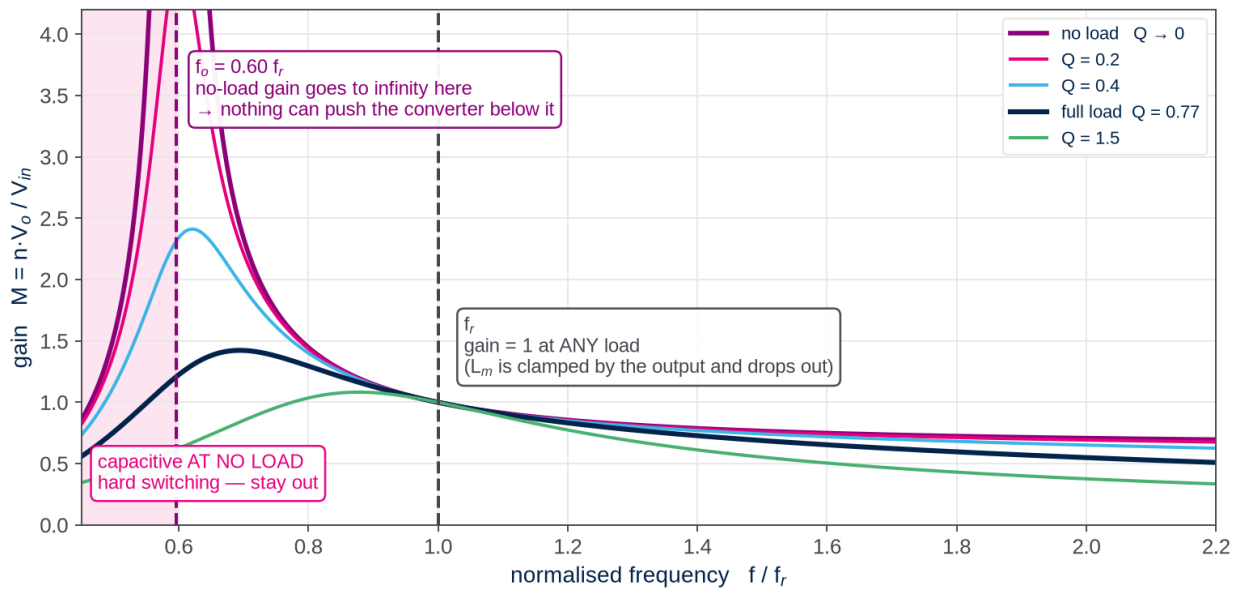


Figure 5: The two resonances, and the band between them in which the tank can boost. The shaded edge is the no-load position of the capacitive boundary; under load it moves up.

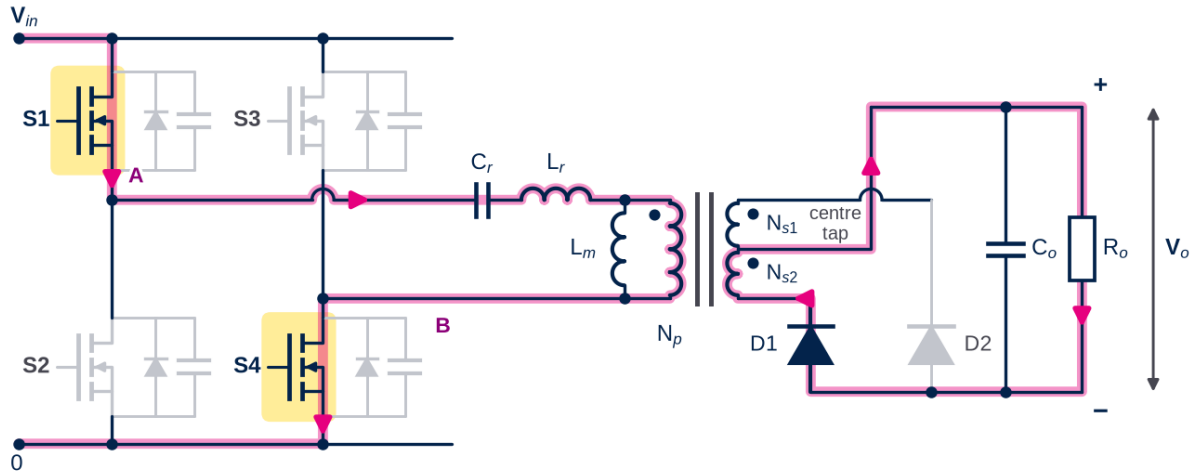
The two expressions are not much use without knowing what each one does to the current, because the current is where the difference shows.

First, the two things the circuit can be doing. Below resonance a half cycle contains both, in this order: power delivery first, then freewheeling, **with the same pair of switches on throughout**. At resonance the half cycle is power delivery only; above resonance the power delivery is cut short.

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1 POWER DELIVERY

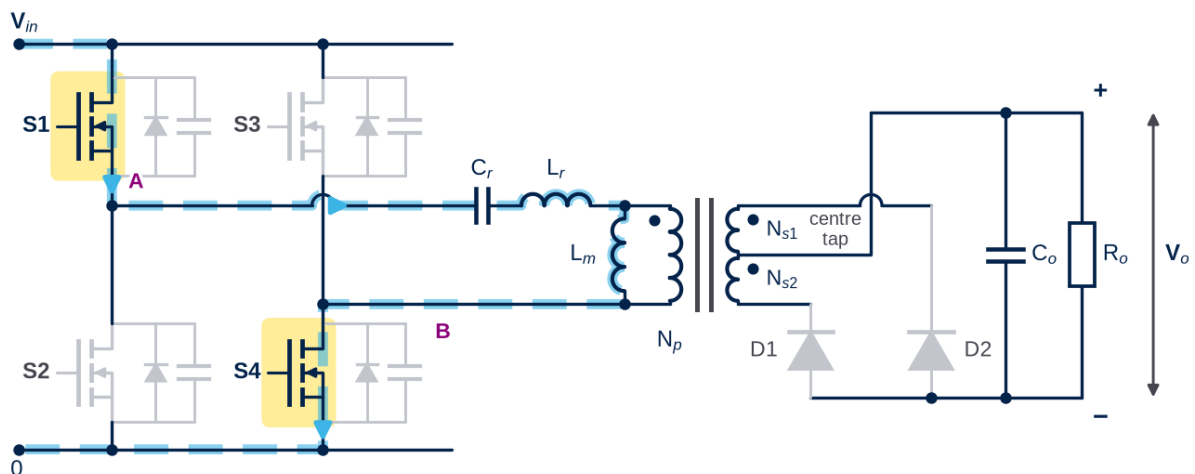
S1, S4 on · D1 conducts · lower half of N_s



L_m is clamped, so i_{Lm} is a straight ramp and $i_{Lr} - i_{Lm}$ is the half sine that crosses.
What is traced is the load component; i_{Lm} flows in L_m too, but it reverses inside this interval.

2 FREEWHEELING

S1, S4 STILL on · secondary off



i_{Lr} has fallen to i_{Lm} . Every rectifier is off, L_m is free again, and the tank rings at f_o .

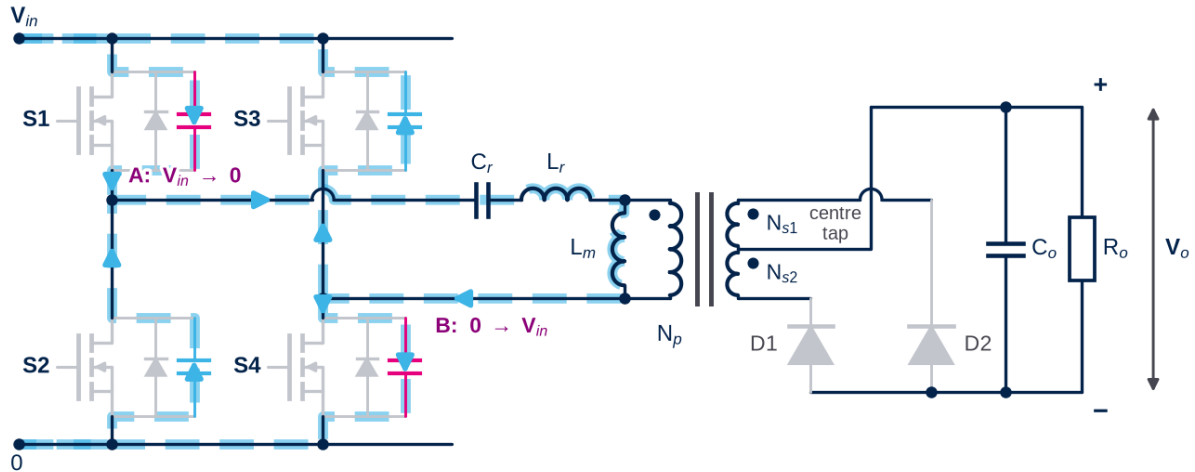
— load current
 - - - magnetising current only
 gated on
 body diode conducting = the ZVS window
— C_{oss} charging, V_{ds} rising
— C_{oss} discharging

Figure 6: One switching period, first half, steps 1 and 2. 1 Power delivery — S_1, S_4 are on, the resonant current exceeds the magnetising current and the difference passes through the transformer to D_1 and the load. L_m is clamped to the reflected output, so this interval resonates at f_r . 2 Freewheeling — the resonant current has fallen to the magnetising current, nothing is left for the secondary, the rectifiers are off and L_m joins the resonance: this interval rings at f_o . The same pair of switches is still on. Each switch is drawn with its body diode and its C_{oss} beside it, greyed while they do nothing.

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3 DEAD TIME (a)

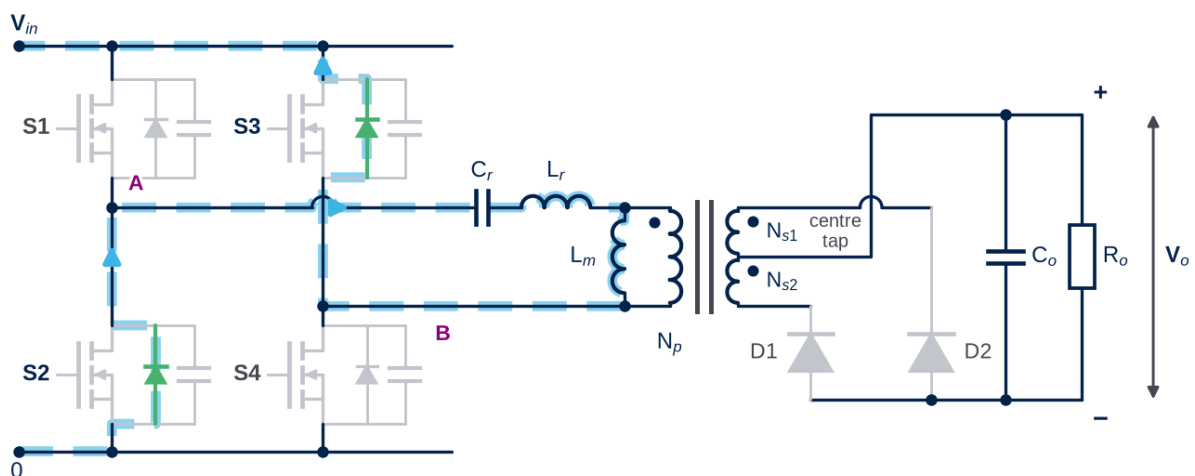
all four off · C_{oss} swaps the midpoints over



All four channels are off, so the ONE tank current can reach the rails only THROUGH the four C_{oss} . It arrives at A from both, so A is pulled down; it leaves B into both, so B is pushed up. That swap IS the transition.

4 DEAD TIME (b) — ZVS

body diodes of S2 and S3 clamp



$V_{ds} = 0$ on S2 and S3 now, because their own body diodes are clamping them. Gate them on HERE.

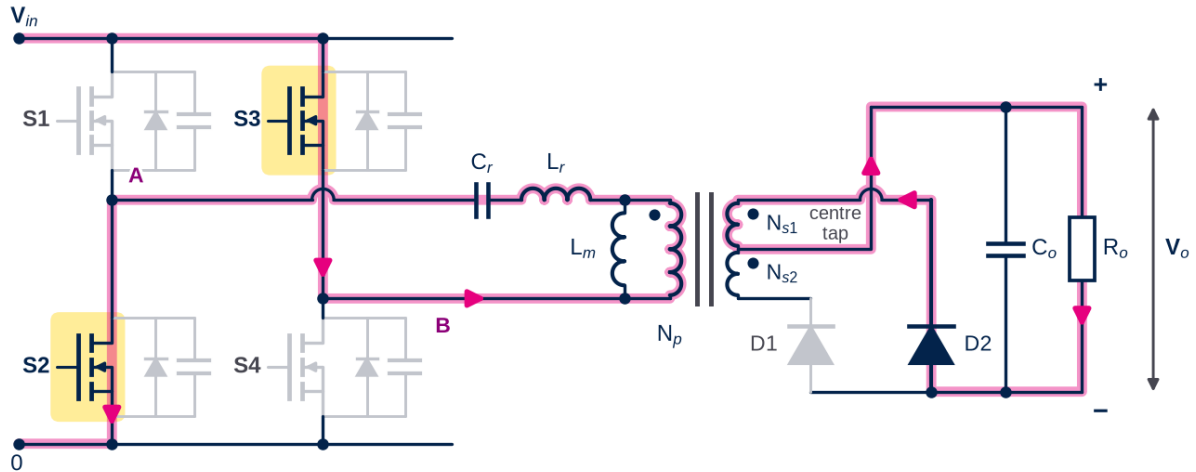
— load current — magnetising current only gated on
 body diode conducting = the ZVS window — C_{oss} charging, V_{ds} rising — C_{oss} discharging

Figure 7: Steps 3 and 4: the dead time. 3 All four switches are off, so the tank current — now the magnetising current alone — can only reach the rails through the four C_{oss} . It arrives at A from both, so A is pulled down, and leaves B into both, so B is pushed up: that swap is the transition. **4** The swing is complete and the body diodes of S_2, S_3 clamp their V_{ds} at zero. This is the ZVS window: the gate that follows has to arrive inside it.

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5 POWER DELIVERY

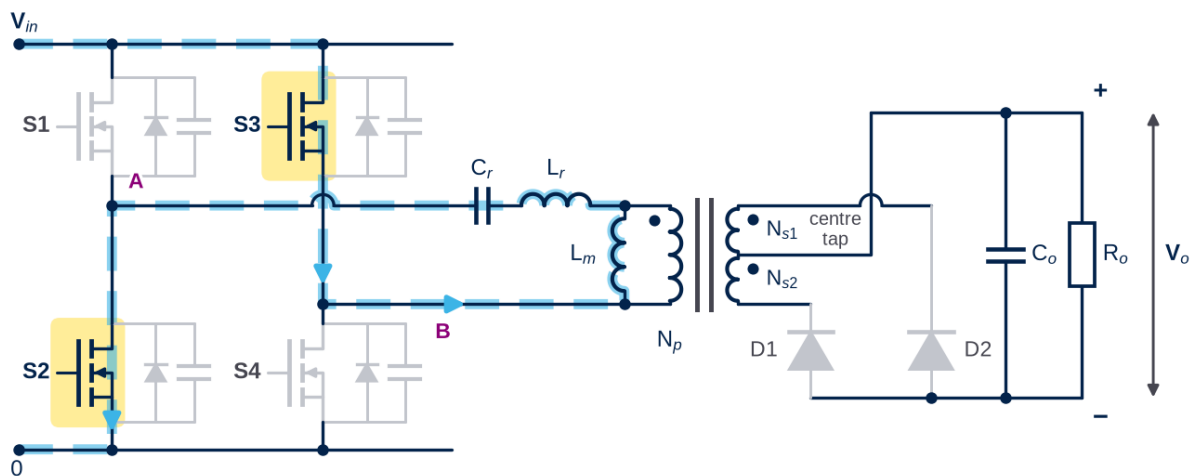
S2, S3 on · D2 conducts · upper half of N_s



The mirror of 1. Just after turn-on the old current still runs back through the two channels. Again only the load component is traced - i_{Lm} reverses inside this interval as well.

6 FREEWHEELING

S2, S3 STILL on · secondary off



The mirror of 2, and the easy one to miss: it ends at the right-hand edge of a waveform plot.

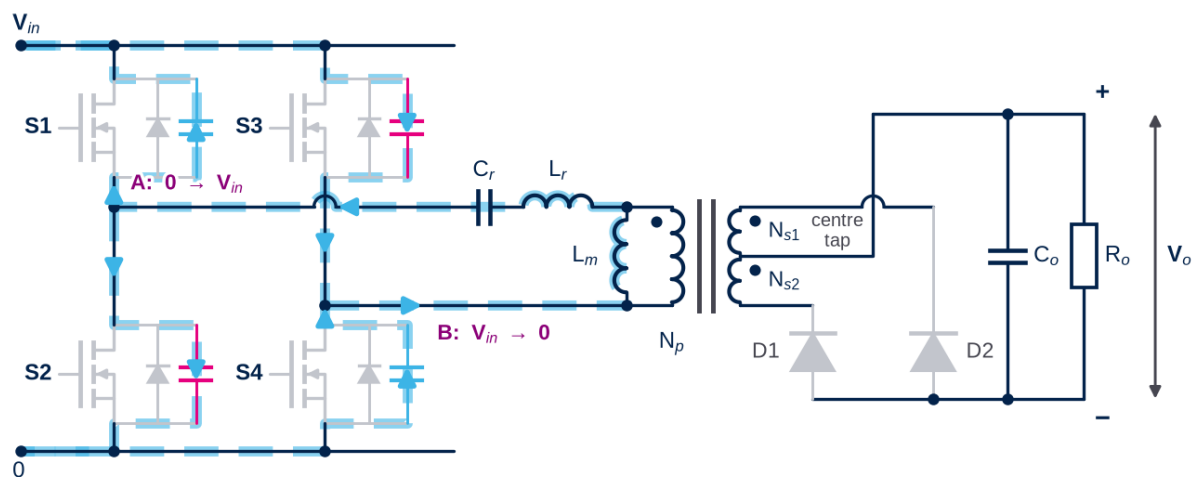
— load current — magnetising current only gated on
 body diode conducting = the ZVS window C_{oss} charging, V_{ds} rising C_{oss} discharging

Figure 8: Second half, steps 5 and 6, the mirror image of 1 and 2. S_2, S_3 carry the power delivery (5, through D_2) and then the freewheeling (6), again with the same pair on throughout.

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7 DEAD TIME (a)

all four off · C_{OSS} swaps the midpoints back

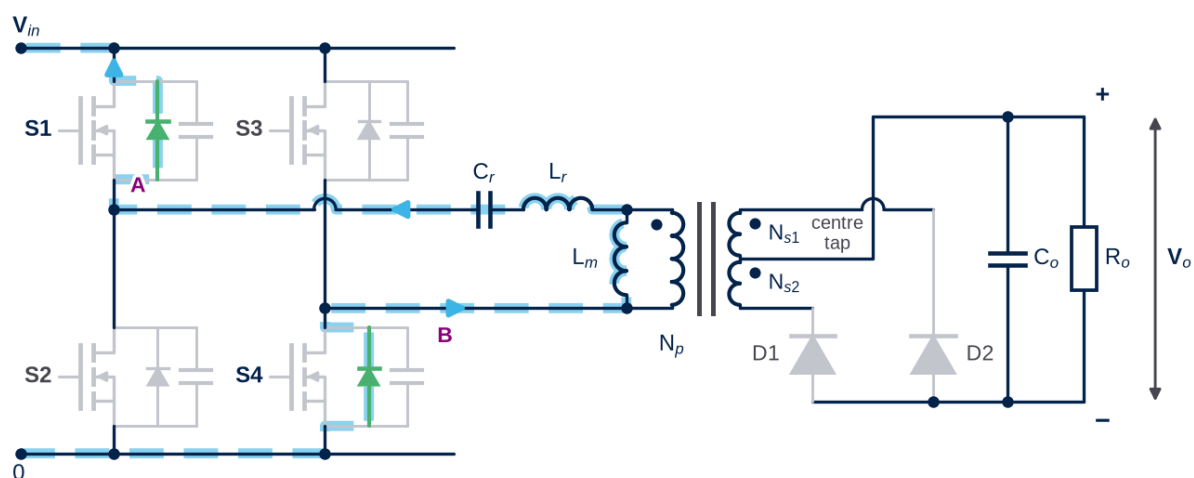


The mirror of 3: the current now arrives at B and leaves A, so B falls and A rises.

Through the capacitances again, never a channel. Magenta: C_{OSS} charging, V_{ds} rising; cyan: discharging.

8 DEAD TIME (b) — ZVS

body diodes of S1 and S4 clamp



The same for S1 and S4, and the period closes. ZVS is in 4 and 8 - and in none of the other six.

█ load current █ magnetising current only █ gated on
█ body diode conducting = the ZVS window █ C_{oss} charging, V_{ds} rising █ C_{oss} discharging

Figure 9: Steps 7 and 8, the second dead time. The mid-points swing back (7) and the body diodes of S_1, S_4 clamp (8), which is where S_1, S_4 then close on zero volts and step 1 begins again. ZVS happens in steps 4 and 8, and in none of the other six.

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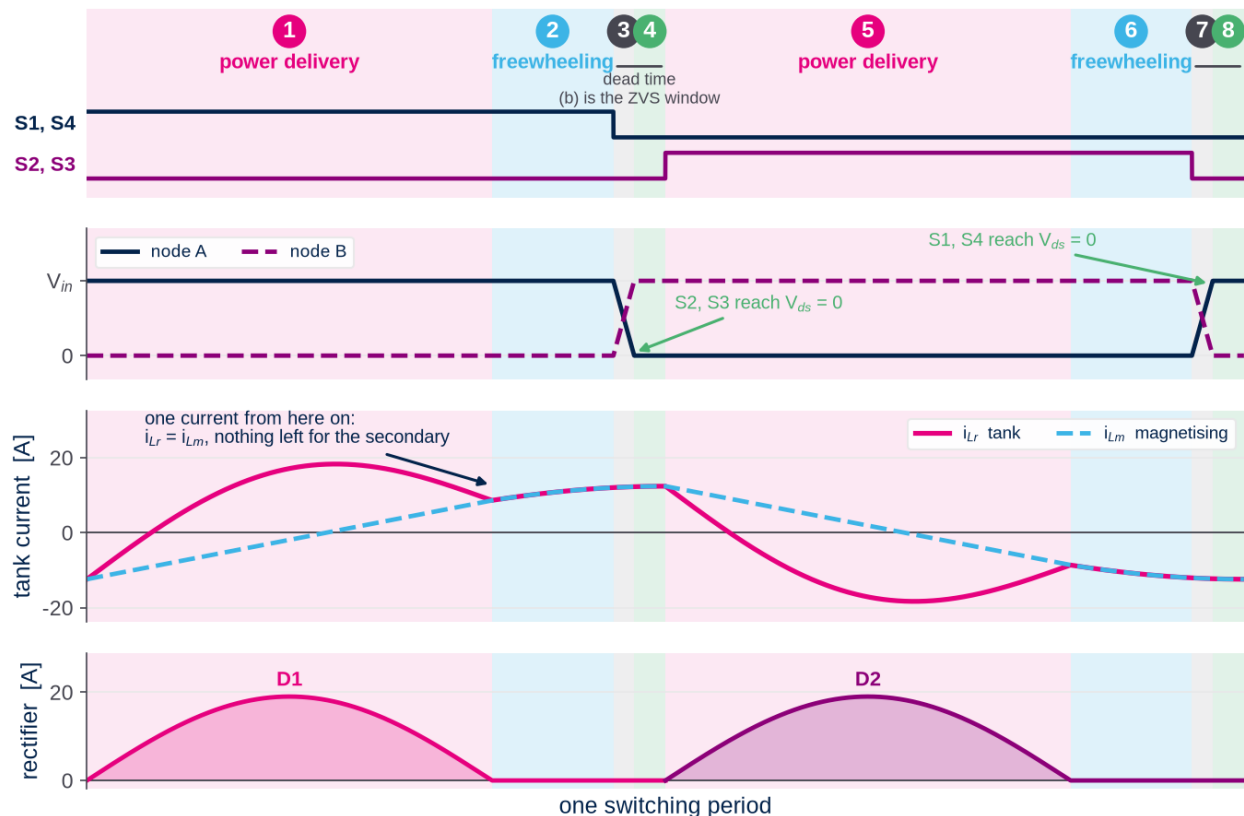


Figure 10: Where each step sits on the waveforms. The numbered bands are the eight panels above, in time order across one switching period: the gates, the two mid-point voltages, the tank and magnetising currents, and the two rectifier currents. Read a panel, then find its band here. Drawn below resonance at $f_{sw} / f_r = 0.70$, with the dead time wider than scale so that steps 3, 4, 7 and 8 have room to be seen; the current amplitudes are this design's own $I_{Lr,pk}$ and $I_{Lm,pk}$ at the low-line corner.

Steps 1 and 2 are the two resonances, and they are the two parts of *one* half cycle: the same pair of switches stays on through both. **ZVS happens in neither**. It happens in the dead time, steps 3–4 and 7–8, when both switches of a leg are off and the tank current carries the mid-point to the other rail — the step in v_d between the gate pulses in Figure 2.

The word “freewheeling” is used for two different intervals. Here it means steps 2 and 6: the secondary is off, L_m has rejoined the resonance, and the same pair of switches is still on. Other documents, among them Toshiba's *Resonant Circuits and Soft Switching* (2019), use the word for the **dead time**, when the current circulates in the body diodes. Both intervals carry current without delivering power, but the dead time exists at every frequency and steps 2 and 6 exist only below resonance. Check which one a source means before comparing numbers.

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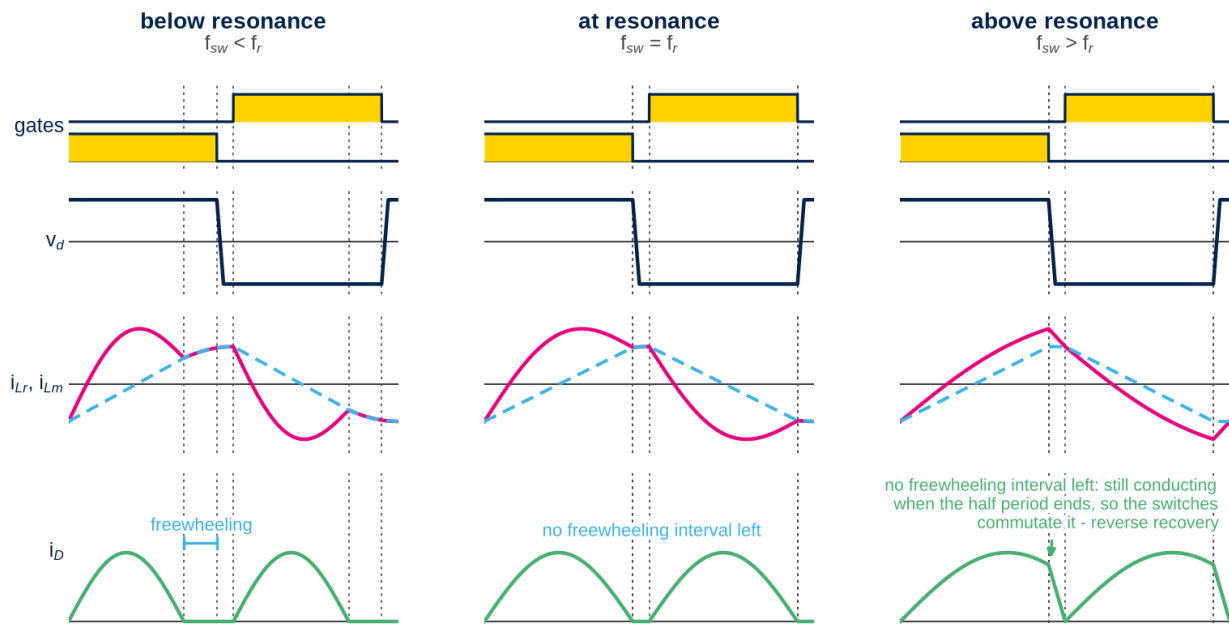


Figure 11: The three cases side by side. Gate signals, the bridge voltage, the resonant current i_{Lr} with the magnetising current i_{Lm} dashed over it, and the rectifier current. What changes across the three columns is how the resonant half period compares with the switching half period. Below resonance there is time left over, and it is spent freewheeling; at resonance the two coincide; above resonance the half period ends first and cuts off the rectifier current while it is still flowing.

- **Below f_r (left).** The resonant half period is shorter than the switching half period. i_{Lr} finishes its half sine early and **lands on i_{Lm}** . From that instant the two are one current and no power crosses to the secondary: that is the freewheeling interval.
- **At f_r (centre).** The two half periods are the same length. i_{Lr} meets i_{Lm} exactly when the bridge switches, and the rectifier current reaches zero at the same instant. This is the most efficient point an LLC has, and it is where a conventional design puts its nominal input.
- **Above f_r (right).** The switching half period ends first, so the resonant half sine is **cut off part way**: i_{Lr} is still well above i_{Lm} when the bridge switches. The primary switches turn off more current, and the rectifier is interrupted while it is still conducting: hard commutation on the secondary.

That freewheeling interval is what f_o is. While it lasts nothing clamps L_m , so the tank is $L_r + L_m$ with C_r , ringing at f_o . It looks flat only because f_o is so much lower than f_r . Lower the switching frequency towards f_o and the interval grows until it fills the whole half period, and the converter delivers no power. So the two frequencies are the two ends of one range: at f_r the flat interval has just disappeared, at f_o it has taken over, and everything useful happens between them.

Symbols are not standard across the literature. Before using a manufacturer's application note alongside this one, check which frequency that document calls f_o . At least one widely used guide gives that name to the series resonance, which is the opposite of the meaning here.

A conventional LLC picks one side and stays there, usually just above f_r at nominal input, where the gain is close to 1 and the circulating current is smallest. Which side the converter designed here runs on is a different question. It is answered in Section 4.5, once the rest of the method is in place.

2.6 The gain function and the three regions

With the usual first-harmonic approximation the tank gain is

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$$M(f_n, Q, \lambda) = \frac{1}{\sqrt{\left(1 + \lambda - \frac{\lambda}{f_n^2}\right)^2 + Q^2 \left(f_n - \frac{1}{f_n}\right)^2}} \quad (3)$$

with f_n , Q and λ as defined in the conventions. Two checks can be made on it by eye before trusting any curve drawn from it. At $f_n = 1$ the second term under the root disappears and the first becomes 1, so **$M = 1$ at any Q** . That is the series resonance, and it is why every gain curve passes through the same point. **At no load** ($Q = 0$, so the second term is gone) raising f_n without limit only brings M down to $1/(1+\lambda)$ and no further; that floor is the subject of Section 4.8. Under load the Q term grows with f_n and the gain keeps falling towards zero.

The curve has three regions. Below the capacitive/inductive boundary the tank is capacitive and the bridge hard-switches, which must never be allowed. Between that boundary and f_r the tank is inductive and boosts; above f_r it is inductive and bucks. In the figure the lower boundary is drawn for the full-load curve. At no load it would sit at f_o ; it moves up as load is added, and the next sections are about that.

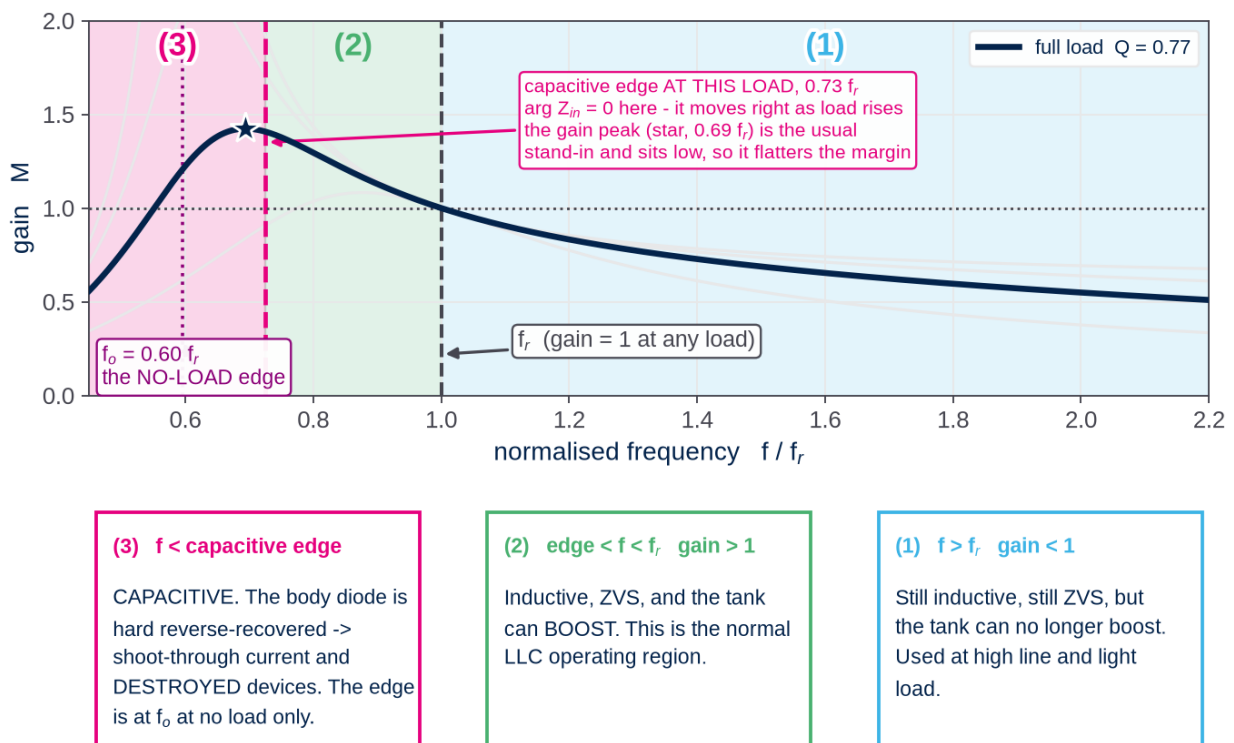


Figure 12: The three operating regions, shaded for the full-load curve drawn. Only the inductive ones are usable, and the boundary between them moves up as the converter is loaded. Region framing after ROHM TechWeb.

2.7 Capacitive and inductive, and why those words

The names have nothing to do with which component dominates. They describe **what the bridge sees**: the phase of the tank input current relative to the square wave that drives it. The tank is a series network. At low frequency C_r dominates and the current *leads*: the bridge is driving something that behaves like a capacitor. At high frequency the inductances dominate and the current *lags*: the bridge is driving something that behaves like an inductor. The word describes the load the bridge works into, not a part.

That phase is the whole of soft switching, which is why the distinction matters more than it sounds:

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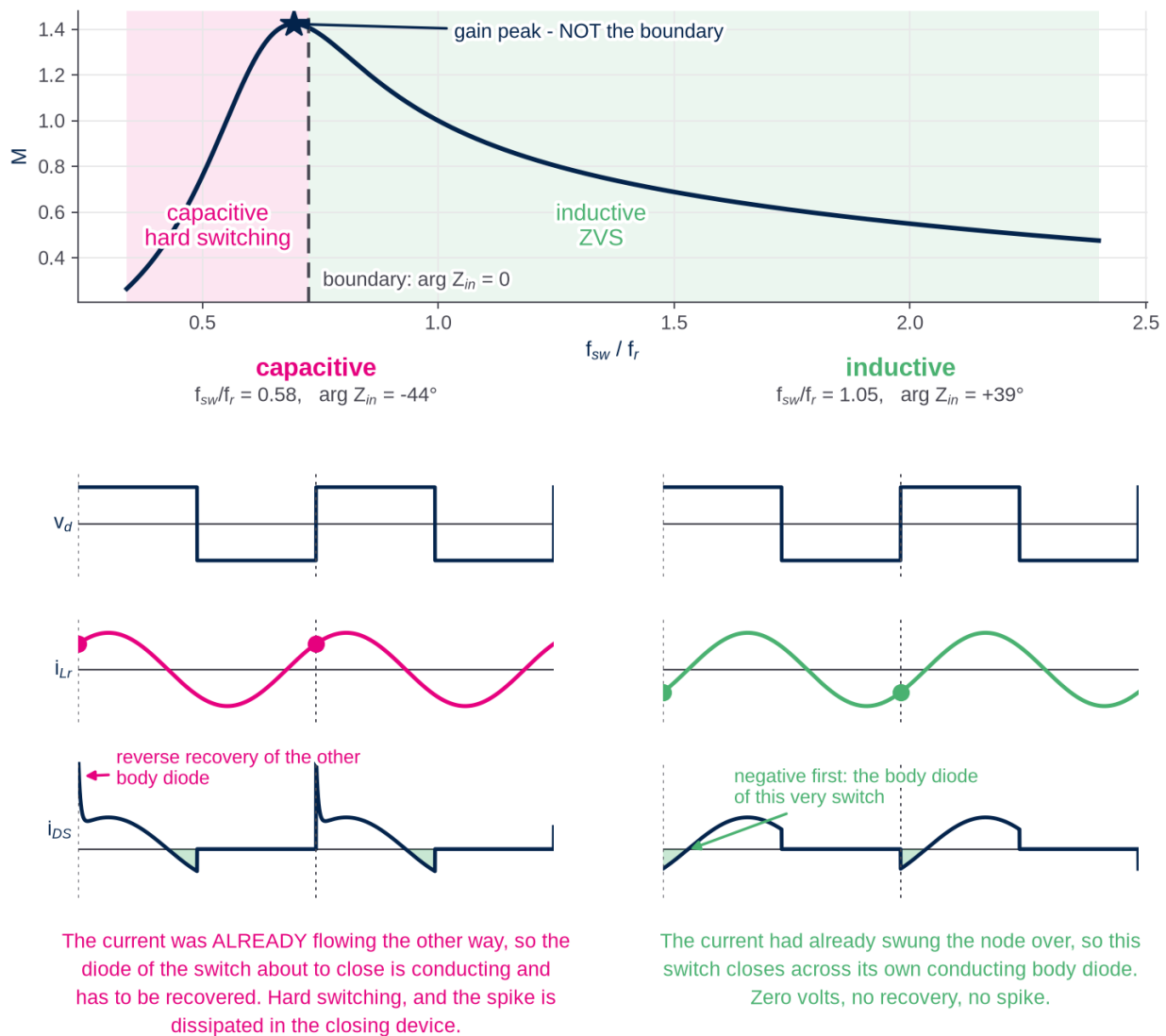


Figure 13: The same converter on either side of the boundary. Above: where the boundary sits on the gain curve. Note that it is not at the peak, but a little above it. Below, for each case: the bridge voltage v_d , the tank current at the phase the tank actually presents there, and the drain current of the switch that is closing. On the left the current leads and is positive at turn-on, and the drain current carries the recovery spike of the opposite body diode. On the right it lags and is still negative, so the switch closes on its own conducting body diode. The two drain currents are drawn to one scale.

- **Inductive: the current lags.** When a switch turns off, the current is still flowing in the direction that pushes the bridge node towards the other rail. The dead time lets it do exactly that, and the next device turns on at zero volts. **ZVS.**
- **Capacitive: the current leads.** By the time the switch turns off, the current has already reversed, so it pushes the node *back* where it came from. The next device then turns on into the full rail voltage. Worse, the body diode of the device that was conducting is forced off with reverse recovery through a low impedance. **Hard switching, and the failure mode that destroys parts.**

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This is why the capacitive region is not a performance trade-off to be balanced against something else. It is a boundary the design stays on one side of, and the controller has a dedicated protection for the case where it does not.

2.8 Why a diode has to recover on one side and not on the other

The drain-current row above is the difference that reaches the hardware, so it is worth following one switching instant through in both cases. Take the leg alone: a high-side switch, a low-side switch, and the tank hanging off the mid-point. Whichever device has just been turned off, **the tank current does not stop** — it is the current in an inductor. The only question is what carries it during the dead time, and the answer decides everything.

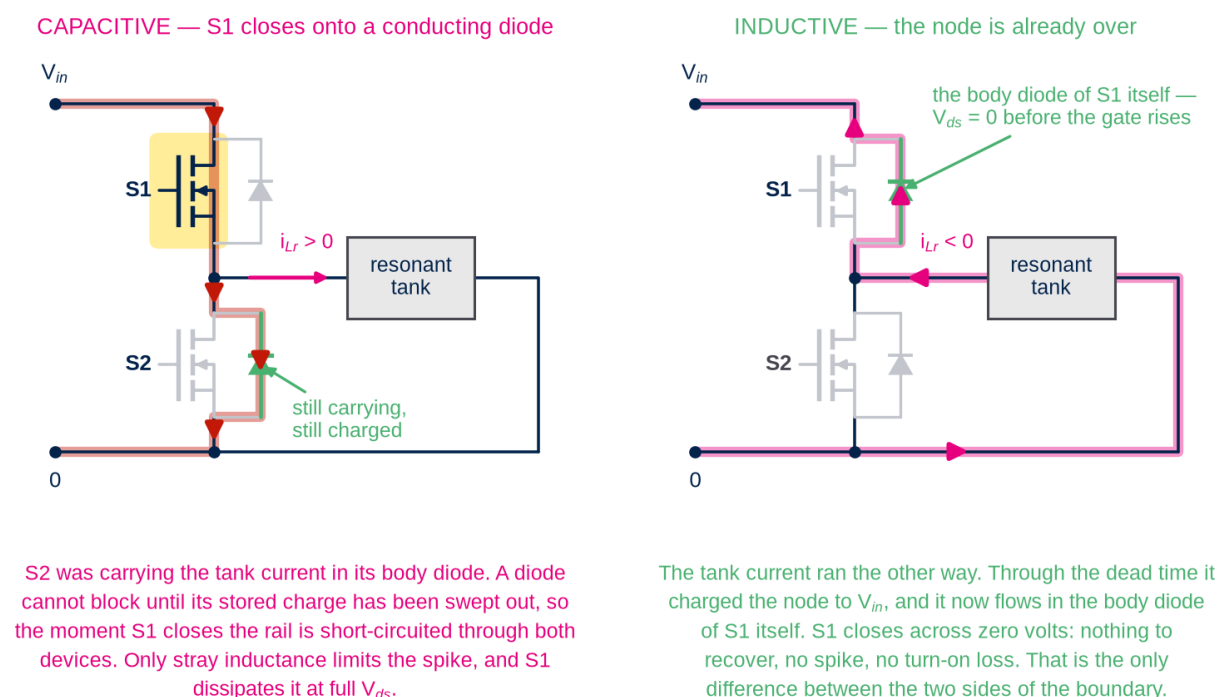


Figure 14: One switching instant in the same leg, either side of the boundary. Left: the tank current is already positive, so it is flowing in the body diode of the low-side device; when the high-side device is gated on, the rail is short-circuited through both until that diode recovers. Right: the tank current is negative, it has carried the mid-point up to the rail during the dead time, and the closing device finds its own body diode already conducting at zero volts. After ON Semiconductor AN-4151.

- **Inductive.** At turn-off the current is flowing in the direction that carries the mid-point towards the *other* rail. It charges one C_{oss} and discharges the other, the node arrives, and the body diode of the incoming device picks the current up and holds the node there. That device is then gated on with V_{ds} already at zero. When its channel takes over, the current in its own body diode falls away with no reverse voltage applied to it, so **there is nothing to recover**.
- **Capacitive.** At turn-off the current has already reversed. It pushes the mid-point back onto the rail it came from, so the body diode of the device that just turned off picks it up and clamps the node there. **The node never moves.** The incoming device is then gated on across the full rail, straight onto a conducting diode.

A diode carrying forward current holds stored charge in its junction, and it cannot block reverse voltage until that charge has been swept out. For as long as that takes, both devices in the leg are conducting and the input rail is short-circuited through them. Three things follow, and all three are bad:

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- The current is limited only by the stray inductance of the loop, so it is a **spike, not a current** — many times the tank current, as the drain-current row shows.
- It flows in the incoming device while that device still stands at the full rail voltage, so it is dissipated there, in a few tens of nanoseconds, every cycle.
- When the diode finally does snap off, the di/dt in the loop inductance appears across it as overshoot, which is what actually breaks the part.

This is the reason the AN asks for a fast-recovery body diode on the primary (Section 6.24). A superjunction device stores a great deal of charge in a body diode that was never designed to be commutated, and in a full bridge that diode conducts before every turn-on even when the converter is behaving. It is also the reason the oscillator floor exists: f_{Min} above f_o is what keeps the converter out of the left-hand picture, and the anti-capacitive protection is what catches it if that fails.

2.9 The two boundaries are not the same boundary

This is the point that is most easily confused, because both pairs of words describe positions on the same frequency axis. They are different lines on it.

	capacitive / inductive	below / above resonance
the boundary	where $\arg Z_{\text{in}} = 0$	f_r , the series resonance
does it move?	yes — with load	no, it is fixed by L_r and C_r
what it decides	ZVS or hard switching, on the primary	boost or buck, and ZCS or not, on the secondary
which side is allowed	inductive only, always	both sides are legitimate

Table 1: Two classifications, two boundaries, two consequences.

The peak-gain frequency sits between f_o and f_r , and it **rises as the converter is loaded**. At no load it coincides with f_o ; at full load it has moved some way up towards f_r . So the capacitive region is not a fixed band below f_o . It grows as load is added, and a frequency that is safely inductive at light load can be capacitive at overload.

The edge is not quite the gain peak. The tank is capacitive while the phase of its input impedance is negative, so the edge is where $\arg Z_{\text{in}} = 0$, and that sits a little *above* the peak of the gain curve. At full load in this design the peak is at 105.4 kHz, but the phase does not reach zero until 110.1 kHz, and at the peak itself the bridge still sees 8.4° of capacitive phase. The peak is the usual stand-in because it needs only the gain curve, but it is optimistic in the dangerous direction: it calls a band inductive while the bridge is still hard-switching. Both are approximations of one real condition, and Section 6.5 measures that condition directly instead: the time the tank current takes to reach zero after the bridge changes state.

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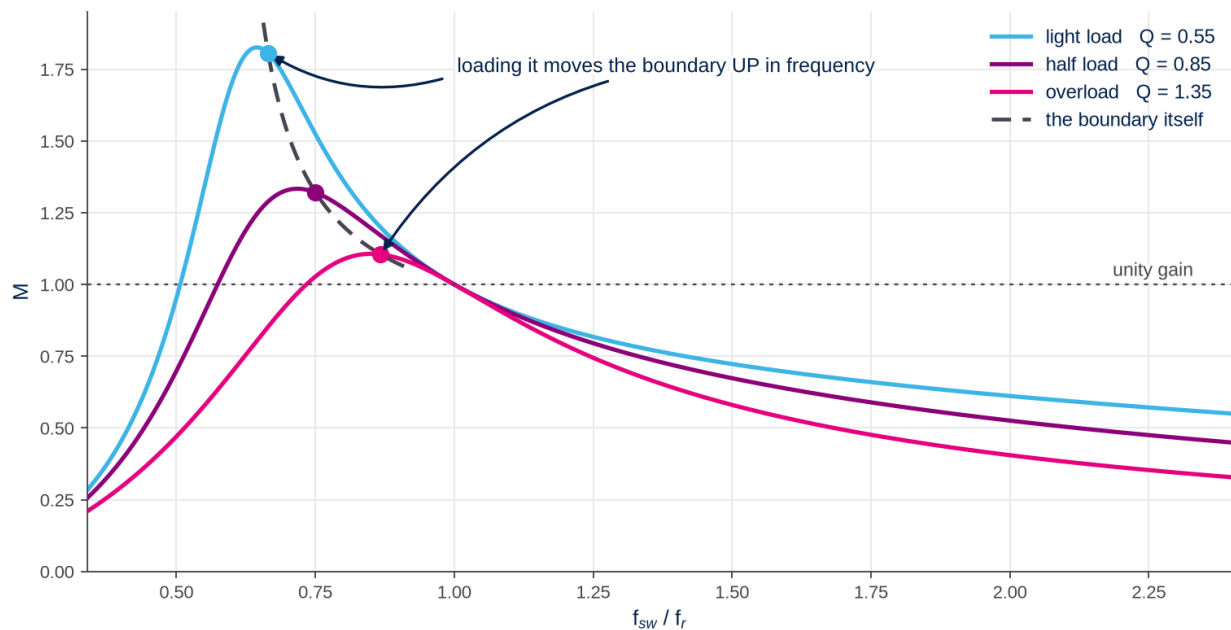


Figure 15: The boundary is not fixed. Loading the converter pushes it to a higher frequency, so a frequency that is inductive at light load can be capacitive at overload. The dashed curve is the boundary itself, traced out as Q varies.

- A converter can be **below f_r and still inductive**. That is the ordinary boosting region, and it is where an LLC normally runs.
- It can be **below f_r and capacitive**, which is the fault. Going below f_r is not the danger; going below the boundary is.
- It cannot be **above f_r and capacitive**, because the boundary is always below f_r . Everything above f_r is inductive at every load.

What this means for the design rule. Keep the oscillator floor f_{Min} above f_o . That is the *no-load* position of the boundary, so it is the weakest form of the requirement: necessary, but not enough on its own. The real check is the ZVS sweep in Section 6.5, which is run under load and at every phase of the line cycle. Behind that, the controller has an anti-capacitive protection.

2.10 ZVS and ZCS are not the same thing

Both are ways of switching a device while one of its two quantities, voltage or current, is zero, so that their product, the loss, is zero. Here they apply to different devices, for different reasons, and only one of them is guaranteed.

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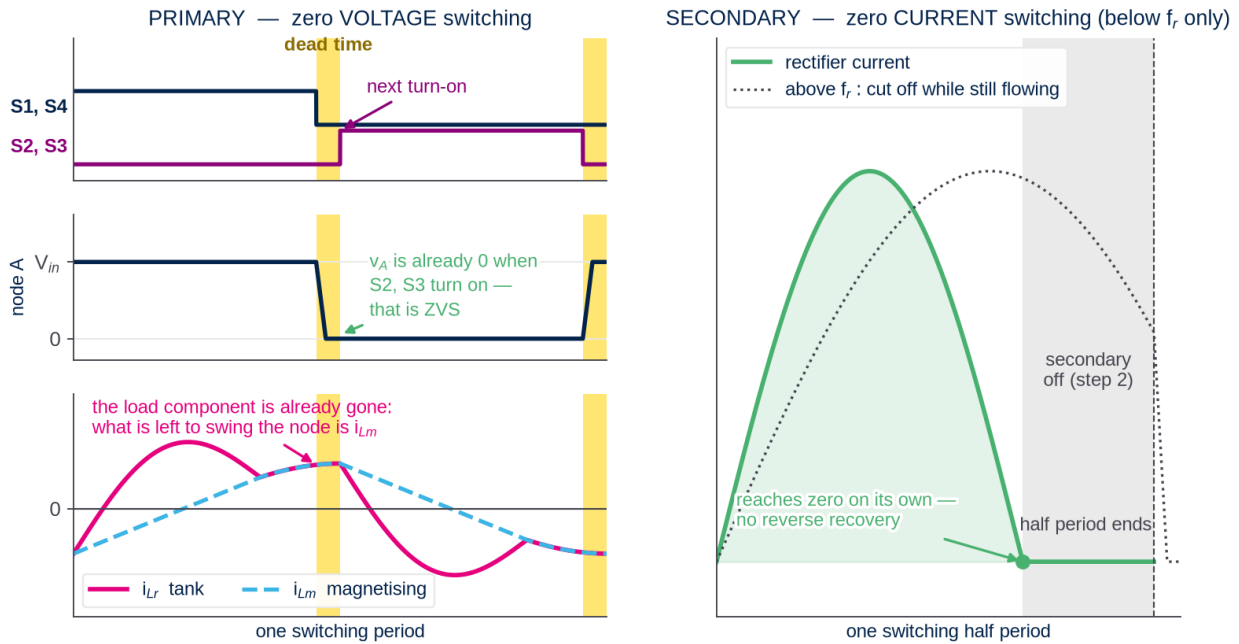


Figure 16: Zero-voltage switching on the primary and zero-current switching on the secondary. Different devices, different mechanisms, different conditions.

- **ZVS, zero-voltage switching, on the primary switches.** The device turns on with no voltage across it. It works because the tank current is still flowing when the previous device turns off, and that current discharges the bridge node during the dead time. It needs **inductive operation**, that is, anywhere above the capacitive/inductive boundary. So it is available on both sides of f_r , and f_o is that limit only at no load. It is required everywhere, and Section 6.5 says how much margin this design has.
- **ZCS, zero-current switching, on the secondary rectifiers.** The device turns off with no current through it, so a diode is not forced out of conduction and a synchronous rectifier has no reverse recovery. It happens because the resonant current reaches zero by itself, which only occurs **below f_r** . Above f_r the rectifier is cut off while it is still conducting, and reverse recovery returns.

So ZCS is not a property of the circuit but of the operating point. Below f_r the secondary rectifiers turn off at zero current and reverse recovery never appears; above f_r they do not, and it does. A converter that stays on one side settles the question once. A converter whose input moves far enough to cross f_r has to be checked on both sides: its secondary body diode and synchronous-rectifier dead time matter whenever it runs above f_r .

2.11 Sizing the dead time so that ZVS actually happens

ZVS is what makes an LLC efficient, and it is a charge problem rather than a voltage problem. During the dead time the magnetising current must move enough charge to swing the bridge node across the rail before the opposite device turns on. The design quantity is T_{ZC} , the time the tank current takes to reach zero after the bridge transition, and the requirement is simply that it must be longer than the dead time:

$$T_{ZC} > t_D \quad (4)$$

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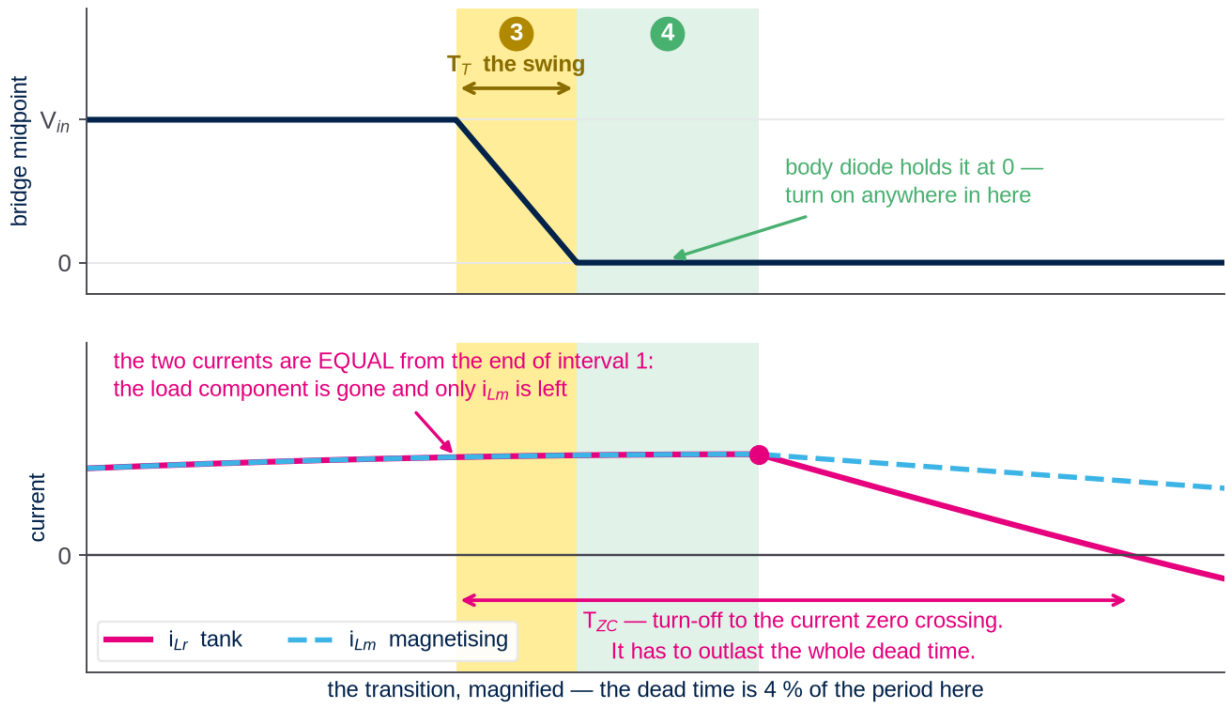


Figure 17: How ZVS happens: the magnetising current, not the load current, discharges the bridge node during the dead time. Two conditions are needed: the current must still be flowing when the gates drop ($T_{ZC} > t_D$), and the swing must finish inside the dead time.

Use the output charge, not the small-signal capacitance. A MOSFET datasheet quotes C_{oss} , $C_{o(er)}$ and $C_{o(tr)}$, and they differ by three to five times. ZVS is a charge question, so $C_{o(tr)}$ (equivalently Q_{oss}) is the one to use. Sizing the dead time from the headline C_{oss} is optimistic by a large factor.

2.12 Not a flyback, and why that changes the core

Almost everybody meets a transformer first in a flyback, and almost every habit formed there is wrong here. Two of them cost money: **why a core area matters at all in a converter that stores none of the energy it delivers**, and **why the saturation test current is nothing like the peak winding current**. Both come from one difference, which is when the two windings conduct.

In a flyback they never conduct together. The secondary polarity dot is inverted, so the rectifier can only conduct while the switch is off. At every instant exactly one winding is carrying current, and therefore at every instant the whole winding current is magnetising current. Energy goes into the core during the on-time and comes out of it during the off-time, and every joule that reaches the output was first stored in the core. The flux follows the *current*, the core is gapped so that it can hold that energy without saturating, and A_e is sized from the peak current.

In an LLC they conduct together. While the secondary is rectifying, both windings are carrying current and their ampere-turns oppose. Only what is left after the subtraction magnetises the core:

$$N_p i_p - N_s i_s = N_p i_\mu \quad (5)$$

The load current passes *through*, by transformer action; it never enters the core as stored energy. What the core does hold is the magnetising energy $\frac{1}{2} L_{\mu} i_\mu^2$, and in this converter that is not a by-product but the mechanism: i_μ is the current that charges and discharges the bridge node during the dead time, which is to say it is what makes ZVS possible (Section 2.11). An LLC transformer is gapped too, but the gap is there to hold L_{μ} at the value the tank

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asked for, not to store what goes to the load.

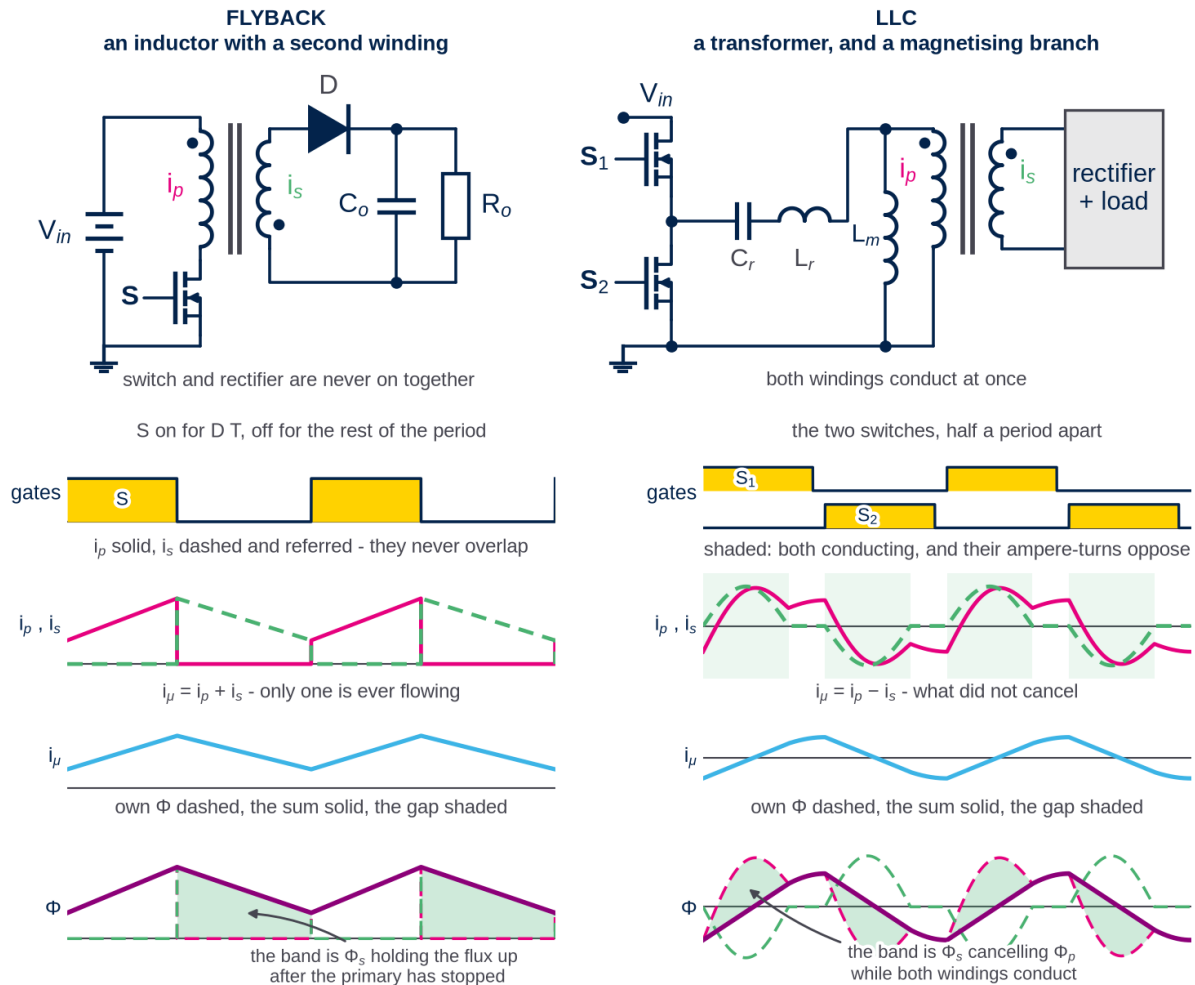


Figure 18: The two transformers from the same four rows. Left: the switch and the rectifier are never on together, so whichever winding conducts carries all of the magnetising current, and the flux is a one-sided ramp that follows it. Right: both windings conduct at once, their ampere-turns oppose, and only the difference $i_\mu = i_p - i_s$ magnetises the core. The i_s traces are referred to the primary, with the sign each polarity dot gives. In the bottom row the dashed curves are what each winding would drive on its own, the solid one is their sum, and the shaded band is what the secondary did to the core: taken away at the same instant on the right, handed over afterwards on the left.

“An LLC transformer stores no energy” is a slogan, not a statement. It stores the magnetising energy, twice every switching period, and the design leans on it. What it does not store is the energy delivered to the load.

So why does A_e come into it at all? Because Faraday's law does not ask whether anything is being stored:

$$B(t) = \frac{1}{NA_e} \int v dt \quad (6)$$

A winding with a voltage across it makes flux; flux in a finite area is a flux density; and ferrite saturates at a flux density. That is equally true of a flyback, an LLC, a mains transformer and a plain choke. **What differs is only the input to the calculation.** In a flyback the volt-seconds are the input voltage times the on-time, which is another way of writing the peak current. In an LLC the conducting secondary clamps the winding to the output for a resonant half period, so the volt-seconds — and with them the flux — are set by the **output voltage alone**.

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Section 6.11 turns that into an equation with neither the input voltage nor the load in it.

Three consequences follow, and they are worth having in mind long before a core is chosen:

- **The flux does not rise with load.** Twice the output current is twice i_s , and the primary has to carry that extra current as well — so both grow together and their difference, i_p , does not move at all. (i_p itself does not simply double: it is i_p plus the reflected load current, and only the second term grows.) A core that is adequate at full load is adequate in overload; what overload threatens is the copper and the semiconductors, not the core.
- **The flux does rise with output voltage.** Anything that lets V_{out} climb raises the flux in proportion, so the ceiling to design against is the over-voltage threshold and not the nominal output.
- **The flux does not fall with switching frequency,** below resonance. The rectifier clamps the winding for a resonant half period $T_r/2$ whatever f_{sw} is doing, which is why f_r and not f_{sw} appears in the flux equation.

The same difference decides what the vendor is asked to test. A DC-overlap measurement is made with the secondary open, which removes the cancellation and puts the transformer back into the flyback condition — all of the test current is magnetising current. Section 6.19 takes that to a number.

2.13 How an LLC is normally designed

The sequence below is the standard one, and every step exists because of something already established above. It is worth having in mind before Section 6, which follows the same order and departs from it in exactly two places.

- **1 Turns ratio from the nominal point.** Choose n so that the converter runs at or near $M = 1$ at nominal input, that is, at f_r . That is where the circulating current is lowest and where the converter should normally run.
- **2 The gain range the tank must cover.** From the input range and the output tolerance, find M_{min} and M_{max} . M_{max} is needed at minimum input and full load, M_{min} at maximum input and minimum load.
- **3 Load resistance referred to the primary, R_{ac} ,** which gives Q its meaning at full load.
- **4 m (equivalently λ) and Q , together, from the peak-gain chart.** They are not independent: for each m there is a curve of attainable peak gain against Q , and the design has to sit under that curve with margin. A larger m gives less circulating current and less peak gain; a smaller m gives the opposite.
- **5 The component values.** Q and R_{ac} give Z_0 ; Z_0 with the chosen f_r gives C_r and L_r ; m then gives L_m .
- **6 Verify.** Check that the gain at the worst corner is still under the peak with margin, that ZVS holds at the worst point rather than at the nominal one, and that the currents and the flux are ones real parts can carry.

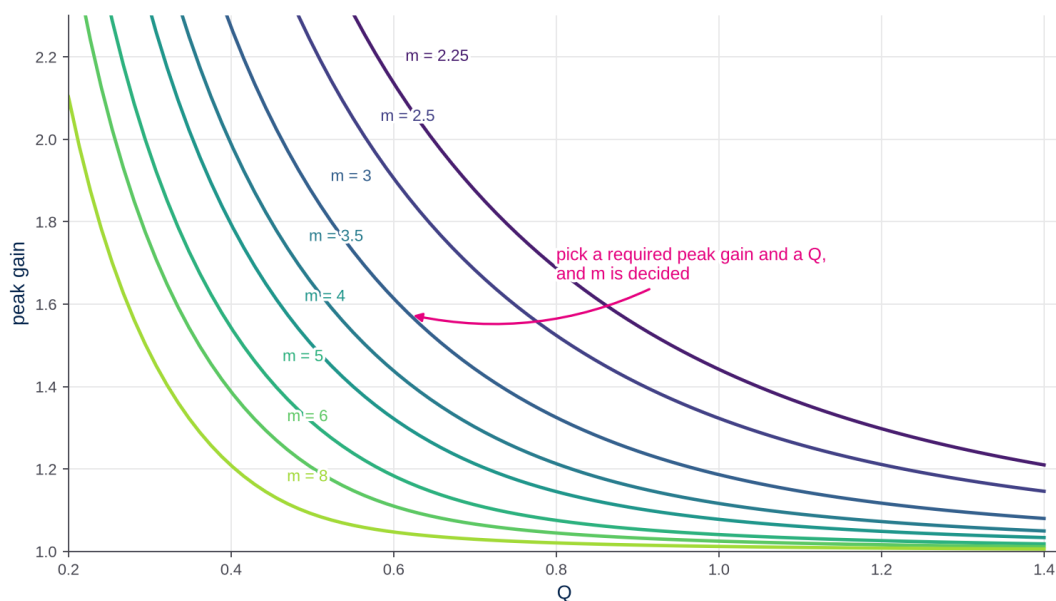


Figure 19: Step 4, and the reason m and Q cannot be chosen separately: the peak gain a tank can reach depends on both. Choosing a required gain and a Q leaves only a band of usable m .

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That is the first half. Everything above is the LLC on its own: a tank, a gain curve, and a frequency that moves the operating point along it. None of it involves the mains. The rest of this section is the other half: what a power factor corrector is for, and how the ordinary one works. It can be read without reference to any of the above. Section 3 is where the two halves are put together.

2.14 What power factor correction is

Start with what happens if nothing is done. A bridge rectifier feeding a capacitor holds the bus near the peak of the mains, so the diodes can only conduct during the short window in which the mains voltage is above the capacitor voltage. All the charge the load will use over the whole cycle has to arrive inside that window.

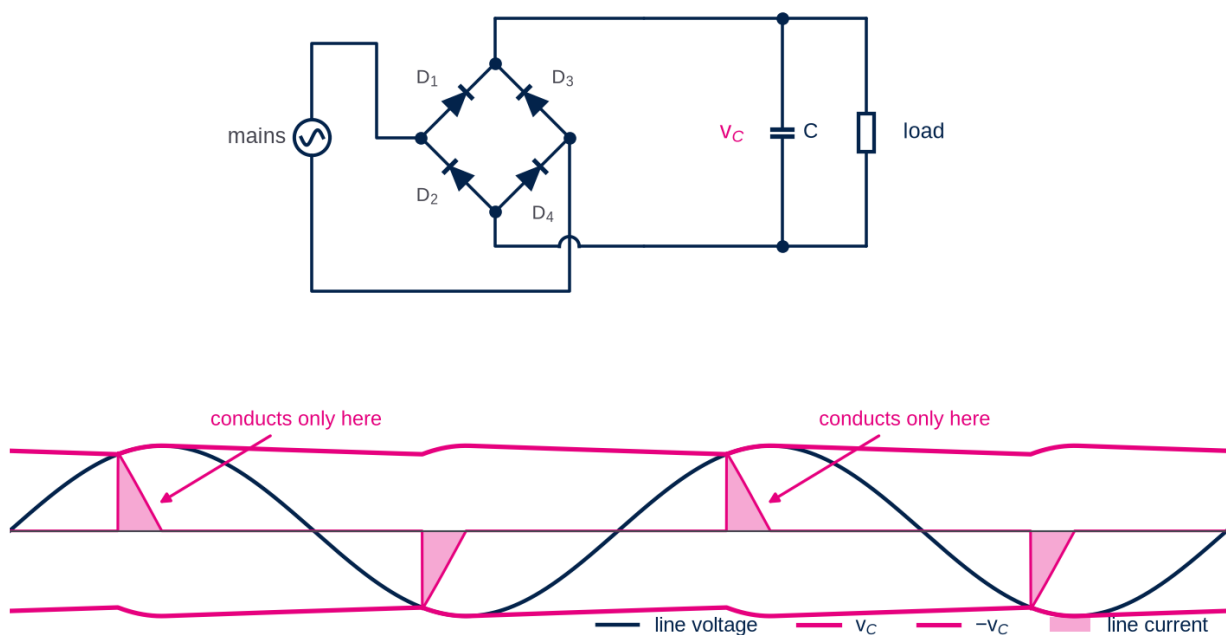


Figure 20: A capacitor-input rectifier and what it draws. v_C stays near the crest, so the diodes conduct only in the narrow windows where the mains is above it, and the current inside them is correspondingly tall.

All the consequences follow from that one fact. The peak current is several times what a resistor of the same average power would draw, so the wiring, the fuse, the bridge and the source all have to be sized for it. And a pulse train that narrow is mostly harmonics.

2.15 Power factor, and why it is not the same as distortion

Power factor is the ratio of real power to apparent power:

$$PF = \frac{P}{V_{rms} I_{rms}} = \cos \varphi_1 \times \frac{I_{1,rms}}{I_{rms}} \quad (7)$$

It has two independent factors, and they fail for different reasons. **Displacement** is the phase between the voltage and the fundamental of the current; it is what a motor gets wrong. **Distortion** is how much of the rms current is at the fundamental at all; it is what a rectifier gets wrong. The current in the figure above is roughly in phase, so its displacement factor is near 1, and it still scores only about 0.6, because the rest of the rms is harmonics.

The related figure of merit is **total harmonic distortion**, the harmonic content as a fraction of the fundamental:

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$$THD = \frac{\sqrt{I_{rms}^2 - I_{1,rms}^2}}{I_{1,rms}}, \quad \frac{I_{1,rms}}{I_{rms}} = \frac{1}{\sqrt{1 + THD^2}} \quad (8)$$

The two are not interchangeable. A converter can hold PF above 0.99 and still fail a harmonic limit, because the standard limits *individual* harmonics in amperes rather than one ratio. Harmonics do not deliver power; they heat the neutral, saturate distribution transformers and disturb other equipment on the same supply, which is why they are regulated. IEC 61000-3-2 sets the limits, and the limit depends on the class. A mains rectifier of this kind is Class D: exempt below 75 W, and held to a milliamp-per-watt line above it. **But Class D only reaches 600 W.** This design draws 685 W at the input, which is above that, so the absolute Class A limits apply instead. Settle the class first.

2.16 How a corrector fixes it

Force the input current to follow the input voltage, and the converter looks like a resistor to the mains: both factors go to one at once. The standard implementation is a **boost converter placed directly after the bridge**, switching fast enough that the mains looks constant within one switching cycle.

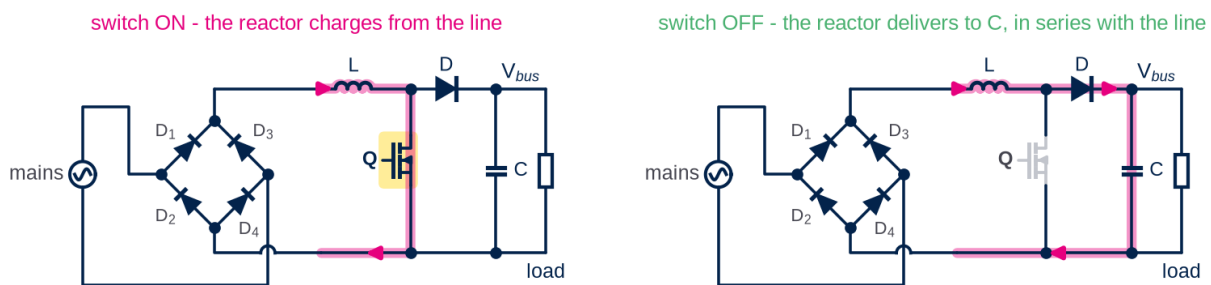


Figure 21: A boost corrector, with the current path traced for each half of the switching period: the reactor charges from the line while the switch is on, and delivers to the bus in series with the line while it is off.

- **Why a boost.** The input is a rectified sine that passes through zero, so the stage must be able to step up by an unlimited ratio near the zero crossing; only a boost can. Its input current is also continuous, which is what makes shaping it possible at all.
- **The inner loop** regulates the inductor current to a reference. It has to be fast compared with the switching frequency.
- **The outer loop** regulates the bus voltage, and it is deliberately made **slower than $2f_i$** . If it were fast, it would fight the ripple on the bus, modulate the current level within the line cycle, and flatten the very shape the corrector exists to make. A slow voltage loop is not a compromise here; it is a requirement.
- **The multiplier** joins them: the shape comes from the mains, the level from the output. That is the whole control law.

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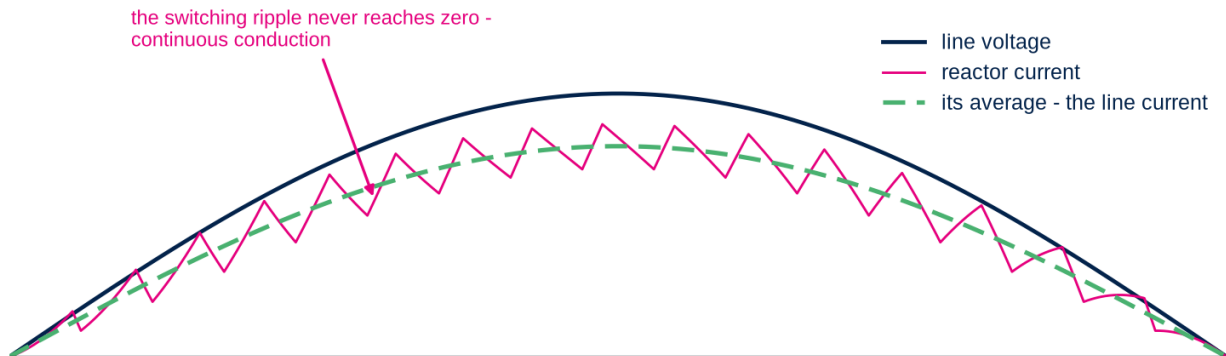


Figure 22: The result, in continuous conduction mode. Switching ripple rides on the reactor current and never brings it to zero; its average follows the input voltage, which is what the two loops were arranged to produce.

This comes back later. Take the boost stage away and the same conflict returns in a sharper form: the outer loop is then the *only* loop, its output is a power command, and any $2f_l$ ripple that gets through it becomes input current distortion directly. That is Section 6.26.

The corrector delivers what was asked for, and creates a new problem in doing so. A current that follows the voltage means an input power that follows $\sin^2\theta$: zero twice per cycle, twice the average at the crests. The load still wants constant power. **Something has to store the difference**, and where that something sits is the subject of the next section.

2.17 Why they are normally two stages

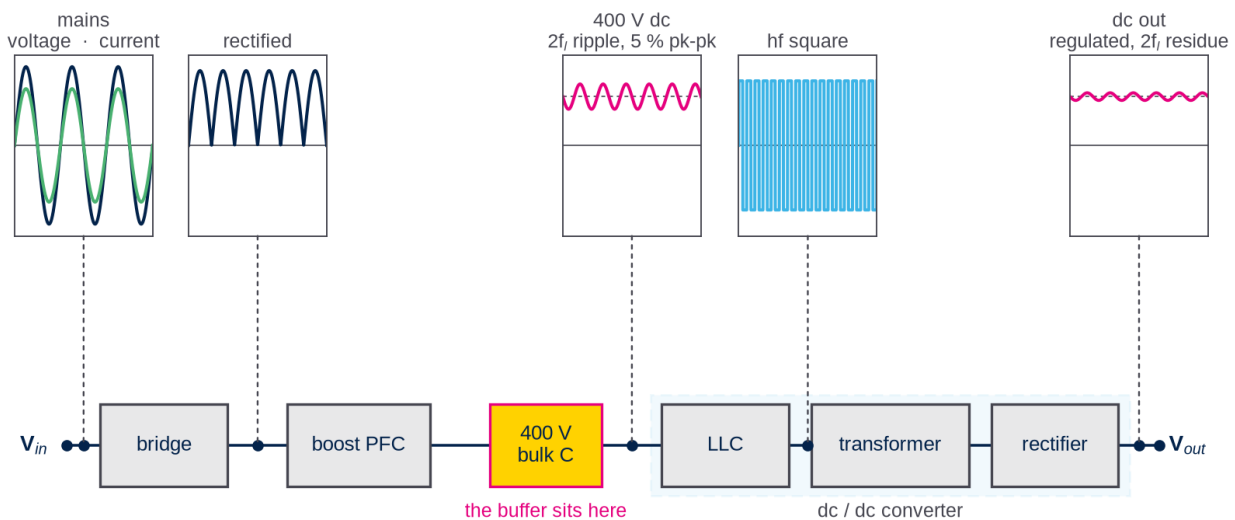


Figure 23: The usual two-stage arrangement, with the waveform at every node. The correction happens in the first block and the LLC works from the dc bus it produces — and the buffer that absorbs the twice-line-frequency pulsation sits between them, at 400 V. The ripple on the dc nodes is drawn larger than it is.

The bus capacitor between a boost PFC and an LLC does two jobs at once, and they are often confused with each other:

- it **buffers** the difference between the pulsating input power and the constant output power;

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- it **isolates** the LLC from the mains waveform, so the tank sees a nearly constant input and has to cover only a narrow gain range.

Removing the boost stage removes both services at once. The buffering has to be done somewhere else, and the tank now has to work directly from a rectified sine that goes to zero a hundred times a second. What that costs, and why it works anyway, is the rest of this note.

3 Why single stage, and what it costs

Two steps of that recipe are the ones this converter breaks. Step 1 has no nominal point to settle at, because the input sweeps a half sine; and step 2 asks for a gain range that no single tank can cover, which is why the bridge itself has to change. Sections 4.1 and 5 are those two departures.

3.1 The energy a unity power factor cannot deliver

Drawing a sinusoidal current in phase with a sinusoidal voltage makes the instantaneous input power a raised sine squared. It swings between zero and twice the average, at twice the line frequency:

$$p_{in}(t) = V_{ac} I_{ac} [1 - \cos(2\omega_l t)] \quad (9)$$

The load wants constant power. The difference has to be stored and returned a few milliseconds later. This requirement does not depend on topology; it follows from asking for unity power factor at all. Figure 24 shows the imbalance that must be buffered.

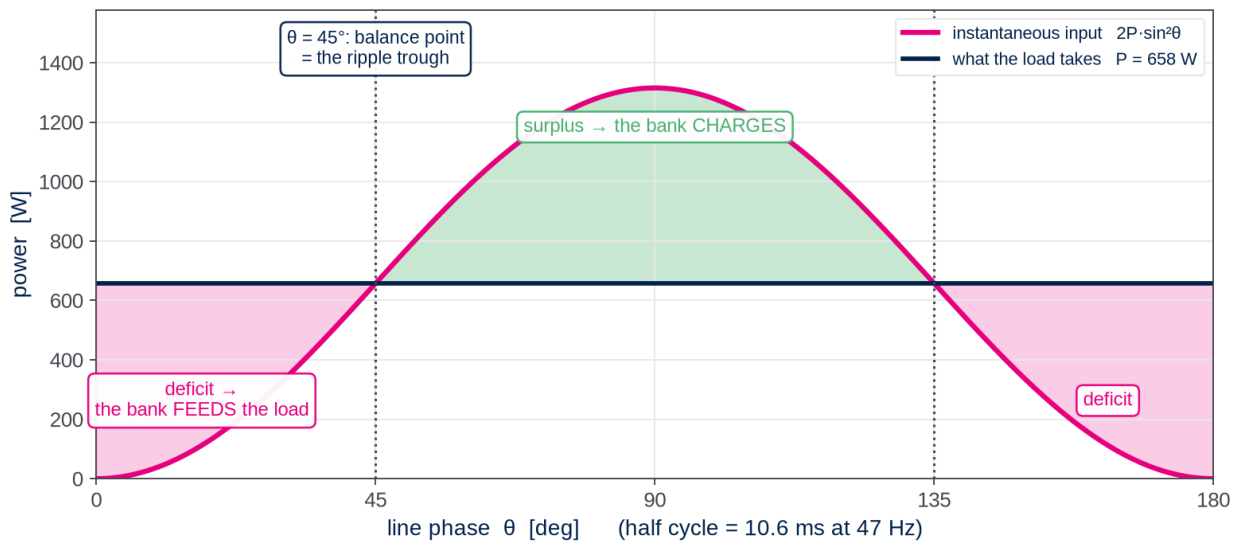


Figure 24: Unity power factor forces a $2f_l$ energy imbalance. The two shaded areas are equal, and they are what a capacitor somewhere must absorb and give back.

3.2 Moving the buffer from 400 V to the output

In the two-stage converter that buffer is the bus capacitor, at 400 V. In the single-stage converter there is no bus, so the buffer moves to the output. **The joules do not change — the voltage they sit at does**, and a capacitor only ever returns the energy between its starting voltage and the lowest the load will accept:

$$E = \frac{1}{2} C (V^2 - V_{min}^2) \quad (10)$$

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Hold-up is the clearest way to see the price, because both architectures must ride out the same 12 ms at full power — 7.9 J either way. On a 400 V bus falling to 320 V that is a 274 μF part. On a 25 V output falling to 19 V it is **67.2 mF**: the same energy, 245 times the capacitance, because the usable voltage window shrank from $400^2 - 320^2$ to $25^2 - 19^2$. That single ratio is the price of the architecture. It should be settled before anything else is designed: if the enclosure cannot house the bank, the output voltage is what has to change.

The bank finally built is larger still. 67.2 mF only satisfies hold-up. Ripple asks for 71.2 mF on the same output and wins, so the design carries **75.2 mF**. Which of the two decides is a question about the specification, and Section 6.23 settles it.

The trade in one line. Removing the boost stage removes a switch, an inductor, a diode and a 400 V electrolytic, and their losses with them. It adds a very large low-voltage capacitor bank, a 2f_r ripple on the output that the load must tolerate, and a resonant tank that has to work over a much wider range. Whether that is a good trade depends almost entirely on how much output ripple the load will accept.

3.3 What the load must tolerate

Because the buffer is on the output, the output carries the 2f_r ripple directly; there is no second stage to reject it. The ripple is set by the bank and the load, not by the control loop, and it cannot be regulated away — attempting that puts distortion on the input current instead (Section 6.26). Here the specification is **5 % peak-to-peak**, which is 1.18 V on a 25 V rail. **How tight that figure has to be is a property of the load, not of the converter**, and it has to come from the specification rather than from a design rule.

4 Operating principle

4.1 Gain is a boundary condition, not a control variable

In a two-stage LLC the controller commands gain: it moves the switching frequency until the output is right. In a single-stage PFC LLC it cannot, because **both ports are voltage sources**. The output is a millifarad-class capacitor whose voltage cannot move within a switching cycle, and the input is the rectified mains, which the converter does not control either. The instantaneous gain is therefore forced:

$$M(\theta) = \frac{n V_{o,eff}}{V_{drive}(\theta)} = \frac{n V_{o,eff}}{\sqrt{2} V_{ac,eq} \sin \theta} \quad (11)$$

with θ the line phase angle. In a two-stage converter the output voltage is free to move — the loop moves it — so commanding a gain is the same as commanding an output. Here both voltages are pinned from outside, so their ratio is fixed moment by moment and the controller cannot change it.

So what does moving the switching frequency do? It moves **Q** — the loading — and Q is power. The tank is given the gain it must produce, and the frequency decides how much current it draws while doing so. **In this converter frequency is a power command, not a voltage command.**

4.2 Two divergences that cancel

Near the mains zero crossing that equation looks impossible to satisfy: as $\theta \rightarrow 0$ the required gain rises as $1/\sin\theta$ without limit. The converter survives because the load disappears at the same time. A unity-power-factor input draws power proportional to $\sin^2\theta$, so

$$Q(\theta) = Q_{pk} \sin^2 \theta \quad (12)$$

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and an unloaded (lossless) LLC has unlimited gain at its lower resonance f_o . The two infinities cancel, and the operating point moves down towards f_o as the mains voltage approaches zero (Figure 25). **f_o is therefore the frequency floor of the entire design**, and putting the oscillator clamp below it is one of the few mistakes that destroys hardware.

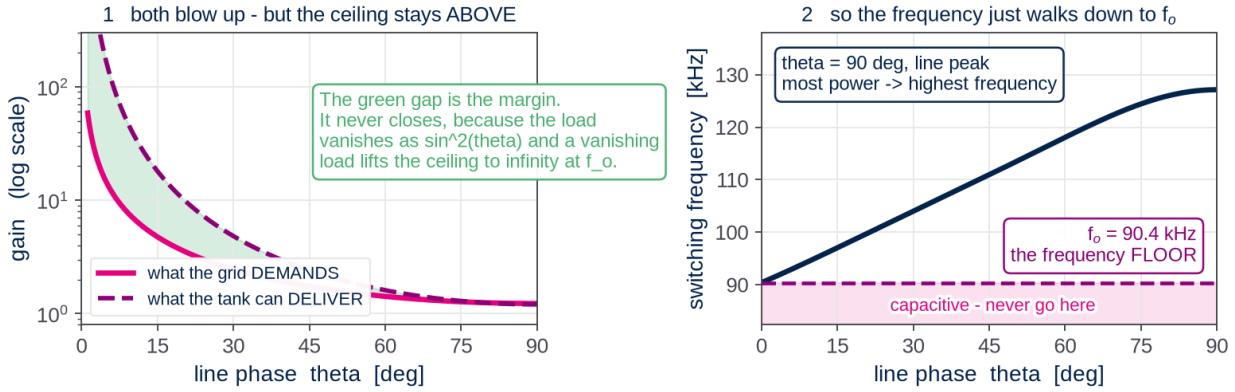


Figure 25: Near the zero crossing the required gain diverges and the load vanishes together, so the operating point converges on f_o .

4.3 Frequency modulation is the power factor correction

Putting these together gives the control law. Over a line half cycle the controller sweeps the switching frequency so the power drawn follows $\sin^2\theta$. There is no current loop, no multiplier and no separate PFC stage: **the frequency profile $f_{sw}(\theta)$ is the power factor correction**. Figure 26 shows it for this design at both ends of the equivalent input range.

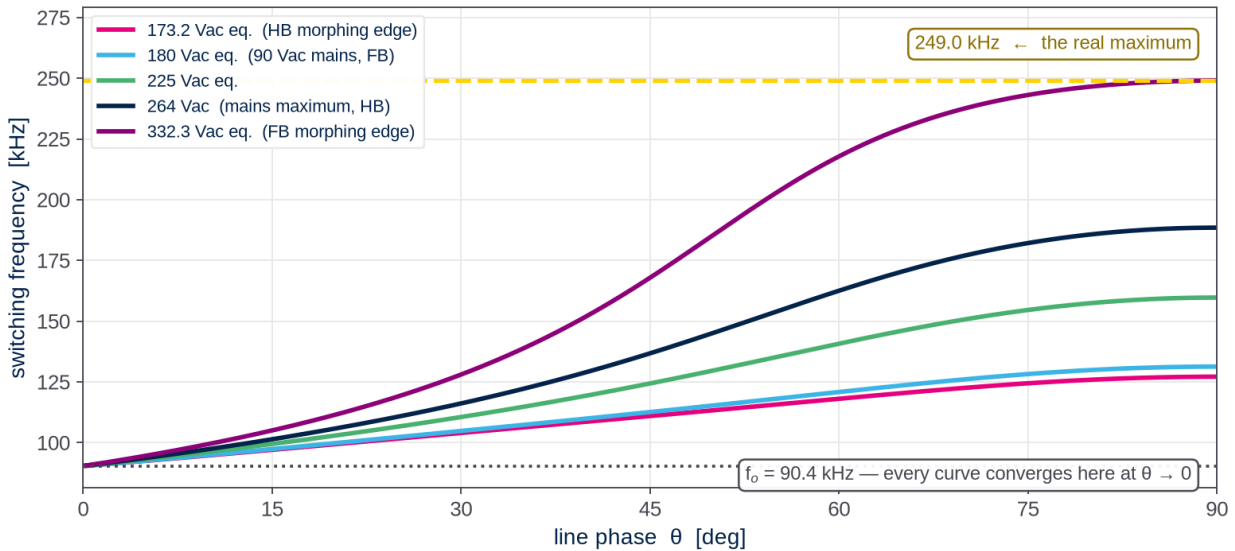


Figure 26: $f_{sw}(\theta)$ over a line half cycle. The excursion is the power factor correction; the floor is f_o . The highest curve is not the maximum mains but the full-bridge morphing edge at 332 V equivalent.

Computing that profile seems to need a numerical root search: at each θ the gain equation has to be solved for f_n , and it has two roots: a capacitive one and an inductive one. **It does not.** Substituting $M_{req} = M_{pk}/\sin\theta$ and $Q = Q_{pk}\sin^2\theta$ and writing $x = 1/f_n^2$ turns the gain equation into a cubic in x :

$$\lambda^2 x^3 + (q - 2\lambda(1 + \lambda))x^2 + ((1 + \lambda)^2 - 2q - \frac{u}{M_{pk}^2})x + q = 0, \quad u = \sin^2 \theta, \quad q = Q_{pk}^2 u^2 \quad (13)$$

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One root is always negative: the cubic equals $+q > 0$ at $x = 0$ and goes to $-\infty$ as x goes to $-\infty$, so it crosses zero somewhere below $x = 0$. The other two roots are where one curve is crossed twice. For $x > 0$ the inverse gain $1/M^2$ of the tank, as a function of x , starts at infinity, falls to a single minimum and rises again. So a required gain that the tank can produce is met at **exactly two** positive values of x : the capacitive crossing and the inductive one. All three roots are then real. The physical, inductive root is the **middle** one, because $x = 1/f_n^2$ decreases as frequency increases. There is no branch to choose: in the trigonometric form of Cardano it is always the $k = 1$ branch. If the required gain is below that minimum there is no positive root at all, which is the no-solution case of Section 4.8 rather than a numerical failure.

Why this matters beyond elegance. A grid search needs an upper bound on f_n , and a bound taken from the ac maximum instead of the morphing corner quietly leaves out the real operating point. The closed form has no grid, no bracket and no starting guess, so that failure cannot happen. The inverse problem — finding the $\sin^2\theta$ that gives a required f_n — is a quadratic and is also exact.

The zero-crossing dead zone. Very near $\theta = 0$ the required gain exceeds anything the tank can produce and the converter simply stops drawing current. This is inherent to the topology and shows up as third-harmonic distortion on the input current. It is one of two mechanisms that set the achievable THD; the other is the voltage loop (Section 6.26).

4.4 The feedback pin is a power command

Combining the burst-mode expression in the datasheet with the multiplier in the block diagram gives the physical meaning of the feedback voltage. It is not an error signal in the usual sense:

$$V_{FB} = V_{os} + \frac{2K_{HV}}{K_M K_{FF}} R_{CS} P_{in} = 0.5 \text{ V} + 0.167 \frac{\text{V}}{\Omega \cdot \text{W}} R_{CS} P_{in} \quad (14)$$

Substituting the 2.8 V feedback span gives $2.8/0.167 = 16.8 \text{ } \Omega \cdot \text{W}$, which is exactly the maximum-power rule in the datasheet. **Maximum power limiting, burst entry and overload detection therefore all sit on one scale**, and that scale is R_{CS} . The outer optocoupler loop commands an *input power*; the internal loop distributes that power over the line cycle as $\sin^2\theta$.

This is why the current-sense pin tolerates no filter and no series resistor. The maximum-power law and the over-current threshold read the same pin, so anything that shifts it shifts both.

4.5 Which side of resonance this converter runs on

Section 2.5 left this question open. Now it can be answered: a single-stage converter does not pick one side. The equivalent input follows a half sine every 10 ms, the required gain follows it, and the operating point crosses f_r and comes back within one line cycle. How much of the cycle it spends on each side depends on the mains voltage.

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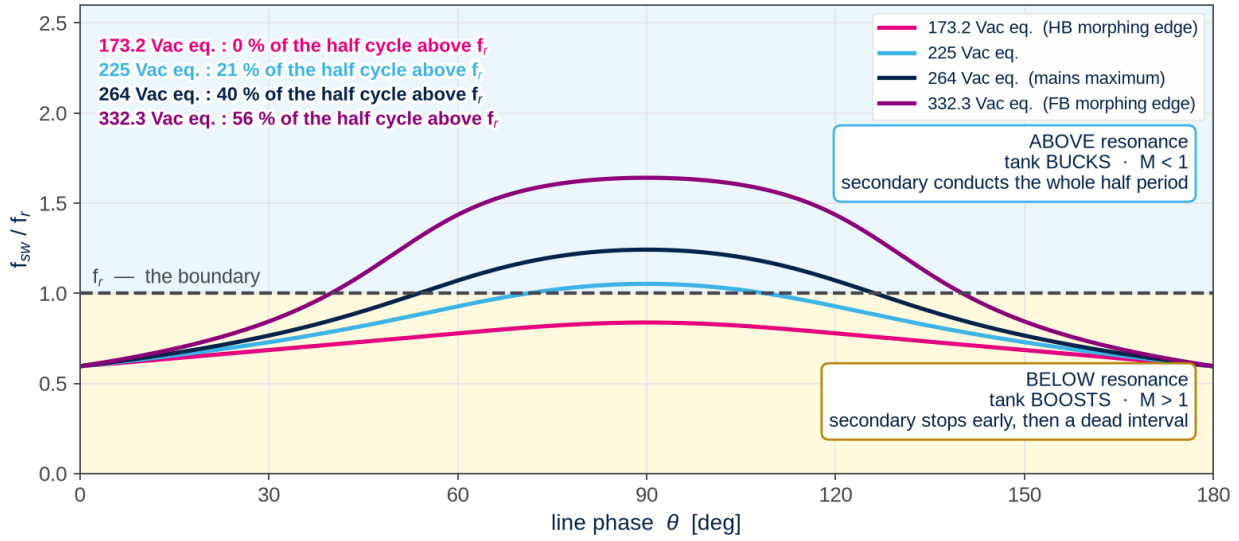


Figure 27: Which side of f_r the converter is on, over a line half cycle, at four equivalent inputs. At the low corner it never leaves the boosting region; at the high corner it spends most of the cycle bucking.

Both sides have to be designed for. The boosting side sets the gain requirement and the ZVS margin. The bucking side sets the top switching frequency, and it takes the secondary out of zero-current switching — so the rectifier body diode and the SR dead time matter here in a way they do not in a conventional LLC. A design checked only at the line peak, or only at low line, has checked only one of the two.

4.6 Why λ must be about 0.5

A classic LLC uses $m = L_p / L_r$ between 5 and 10, that is λ between 0.11 and 0.25. A single-stage converter cannot: it has to produce very high gain near the zero crossing, and peak gain falls as λ falls. This design runs $\lambda = 0.55$, that is $m = 2.82$. The evaluation board ST publishes for this part runs $\lambda = 0.500$, which confirms that this value comes from the topology and not from one particular design.

The price is circulating current. A small L_m means a large magnetising current that carries no power but does carry conduction loss, at every load. **A single-stage PFC LLC is less efficient than a two-stage LLC, and that is inherent rather than a design defect.**

4.7 $f_{sw,max}$ does not check the tank — it computes it

Two of the four candidates for λ carry the specified maximum switching frequency in a denominator:

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad (15)$$

λ_1 is the plain minimum-gain condition. The other two are the same condition with the frequency ceiling folded in: the tank has to reach the lowest required gain *before* it runs out of frequency. Here $f_r / f_{sw,max} = 150/225$, so the squared ratio is 44.4 %, both denominators fall well below one, and the requirement rises from $\lambda_1 = 0.560$ to 1.009 and 1.241. **The number typed as $f_{sw,max}$ roughly doubles the λ the tank is asked for**, and if it were set close to f_r the denominators would go to zero and the required λ to infinity.

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So where does $f_{sw,max}$ come from? The common rule is $1.5 \times f_r$, which is what the 225 kHz here is. It is a **choice**, not a limit — and it sits upstream of the tank, so changing it changes L_m . A maximum in a design sheet usually only checks a result; this one is an input, so a round number typed into it changes the tank without anyone noticing.

It should not be confused with f_{Max} , the oscillator ceiling. That one *is* a limit: R_T , C_T and the idle time fix it in silicon, the control loop cannot push past it however much gain it wants, and it is checked after the fact by k_{ceil} — 320 kHz against an operating maximum of 249 kHz, a factor of 1.28. The two sit at opposite ends of the design: one decides the tank, the other reports whether the oscillator can serve it.

4.8 The other bound on λ , and where it has no solution

The condition above sets a floor under λ . The ceiling comes from the opposite end: the tank must also be able to reach the *lowest* required gain, which happens at the highest equivalent input and no load. That is where the usual design rule quietly breaks down, because the no-load gain of an LLC does not fall without limit. Raising the frequency only brings it down towards a limit (an asymptote):

$$M_\infty = \lim_{f \rightarrow \infty} M_{no\ load} = \frac{1}{1 + \lambda} \quad (16)$$

If the required minimum gain lies below M_∞ , no frequency satisfies the condition at all. A spreadsheet solving for that frequency returns an error rather than a number, and the error is easy to mistake for a broken formula. It is not: the question simply has no answer.

This design is below the asymptote: $M_\infty = 0.6452$ against a requirement of 0.6409. There is no frequency at which the unloaded tank reaches the required gain, and the margin is thin enough — well under one per cent — that no full-load check would ever show it. Raising the turns ratio raises the requirement and brings the solution back.

Losing the solution is not a failure. No load at high line is the region **burst mode** owns, not frequency control. Under load the gain curve falls further and the corner arrives sooner, so every full-load check is unaffected — which is exactly why this can go unnoticed. What matters is to know which of the candidate turns ratios can still regulate by frequency alone at no load, because no full-load number shows it.

5 Topology morphing

A resonant tank has a usable gain range of roughly 1.5:1 at full load. A universal mains asks for 2.93:1. No single tank covers that and keeps ZVS, so the L6790A changes the bridge instead.

Below a mains peak of 235 V the part runs a **full bridge**, which drives the tank with twice the rail. Above 245 V_{pk} it runs a **half bridge**. The tank therefore sees 173.2 to 332.3 Vac equivalent, a range of 1.92:1, and one tank can cover it.

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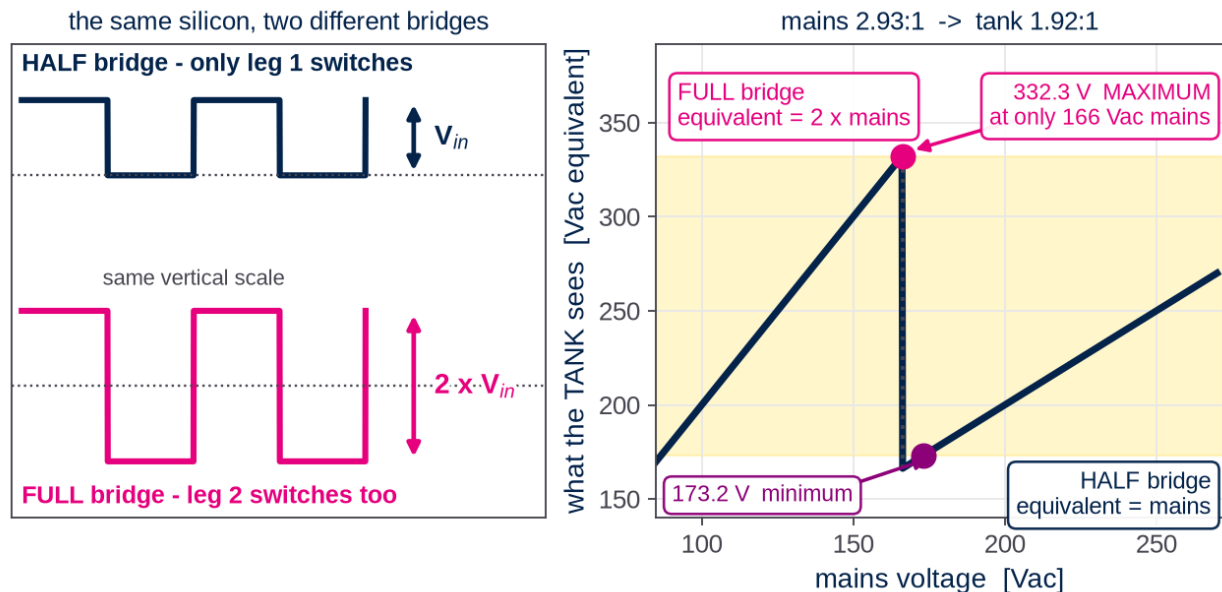


Figure 28: Morphing collapses a 2.93:1 mains range into a 1.92:1 range at the tank. The worst corners are the morphing edges, not the ends of the mains range.

5.1 Full bridge

Diagonal pairs conduct together. S1 and S4 put $+V_{in}$ across the tank, S2 and S3 put $-V_{in}$ across it, so the drive is a square wave of amplitude V_{in} and its fundamental is $(4/\pi)V_{in}$. All four devices switch, and each carries the tank current for half of every period.

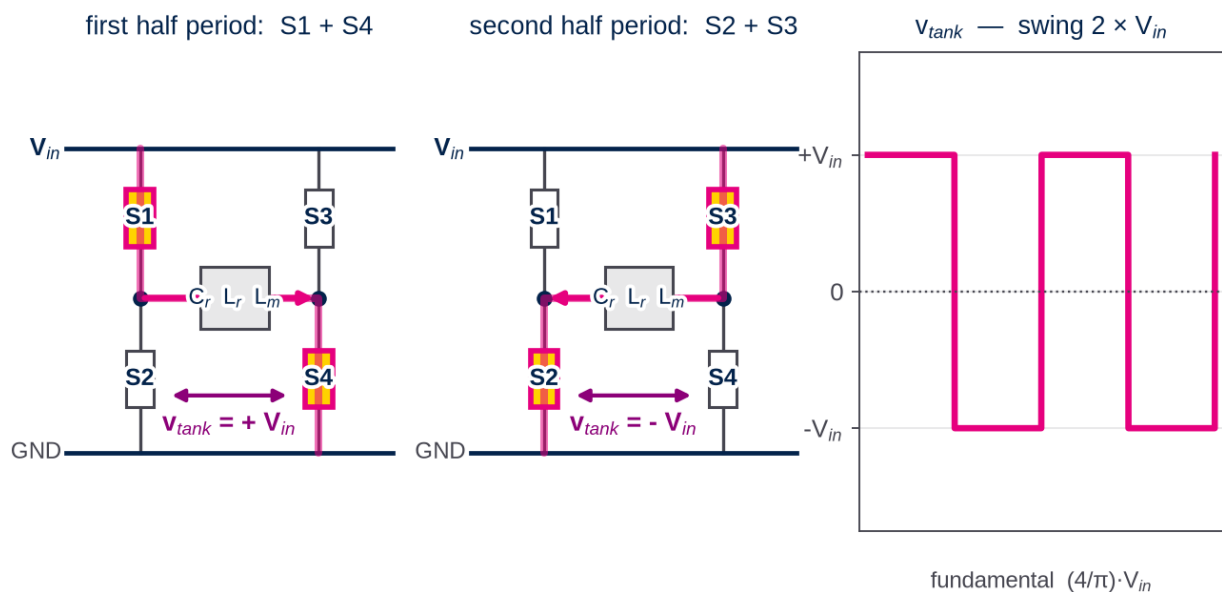


Figure 29: Full bridge: conduction path in each half period, and the resulting tank drive.

5.2 Half bridge

LOUT2 is held statically high. S3 never turns on and S4 never turns off, so the leg-2 midpoint becomes ground and the bridge output swings between 0 and V_{in} . C_r is already in series with the tank, so it blocks the $V_{in}/2$ of dc for free and the tank sees $\pm V_{in}/2$ — a fundamental of $(2/\pi)V_{in}$, exactly half. No extra hardware is involved.

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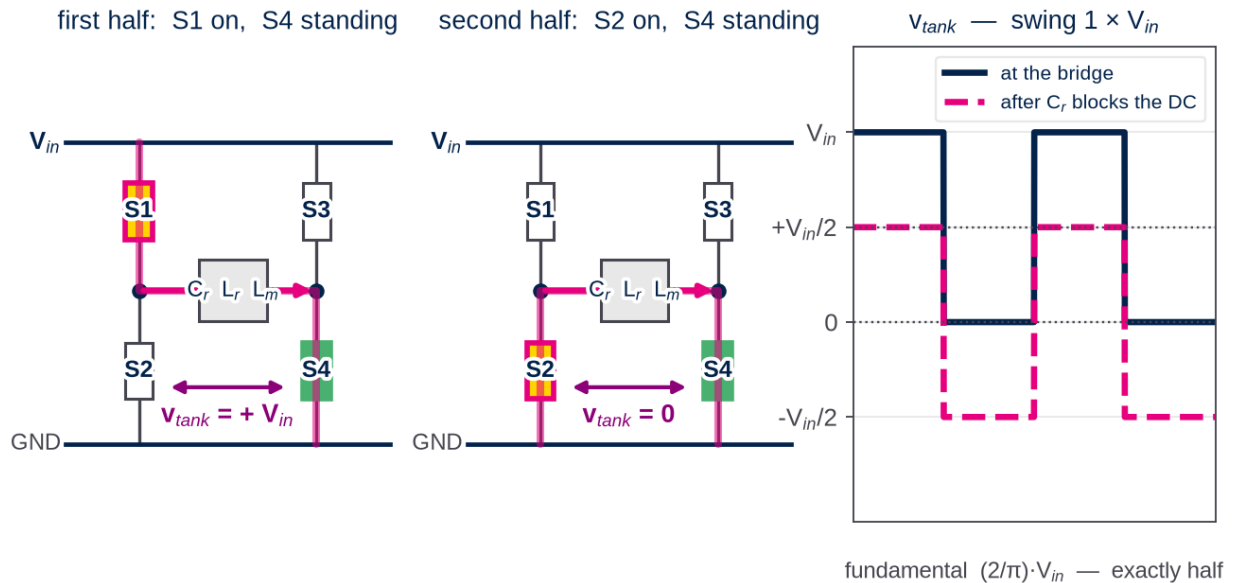


Figure 30: Half bridge: leg 2 stops switching, and C_r removes the dc that the asymmetric drive creates.

The standing device is the hottest one. S4 never switches, but it conducts the full tank current continuously — 10.9 A rms in this design, 9.65 W in one package. It is easy to overlook exactly because it is not switching, and it is the device that sets the heatsink requirement.

5.3 How the controller does it

Only one gate signal changes between the two modes, which is why morphing costs nothing in hardware. The thresholds themselves are **fixed constants inside the IC** and cannot be moved by the designer; R_{CFG} decides only whether morphing is enabled at all and where the brown-out threshold sits.

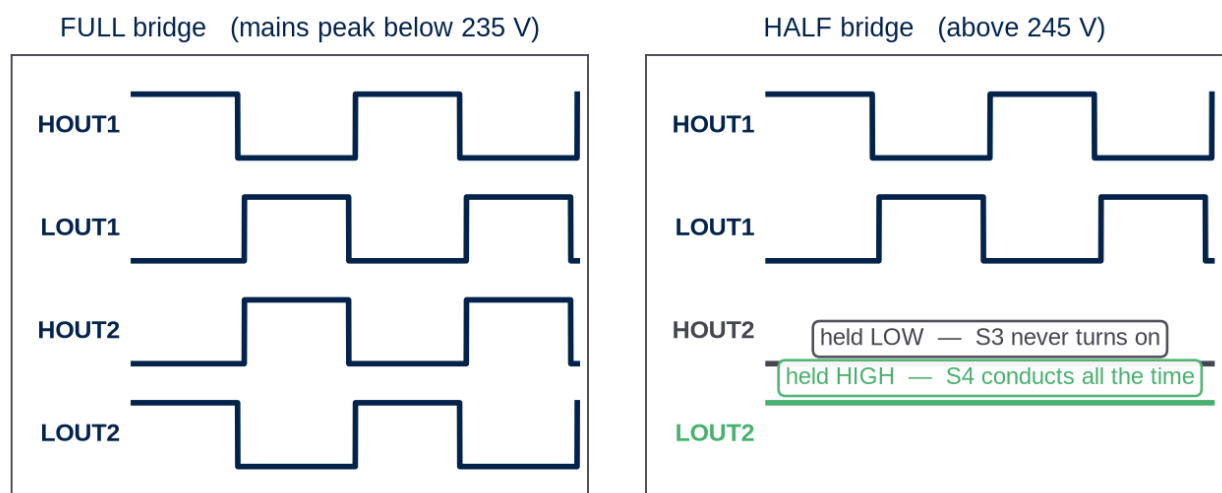


Figure 31: The four bridge drive signals in each mode. LOUT2 held high is the entire mechanism.

5.4 The hysteresis band, and the range it really implies

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The two thresholds are not the same number: the part drops to half bridge at 245 V_{pk} going up and returns to full bridge at 235 V_{pk} coming down. The 10 V of hysteresis stops the mode from toggling back and forth at the threshold, but it also means that **between 235 and 245 V_{pk} the mains voltage alone does not determine the mode** — the direction of travel does.

Taking each threshold on the side where its mode is guaranteed gives the design range 173.2 to 332.3 Vac ($1.92:1$). Including the band gives **166.2 to 346.5 Vac ($2.08:1$)**. This design passes every check at the wider range as well, but a design with less margin would not, and the narrower figure must not be used when tightening a specification.

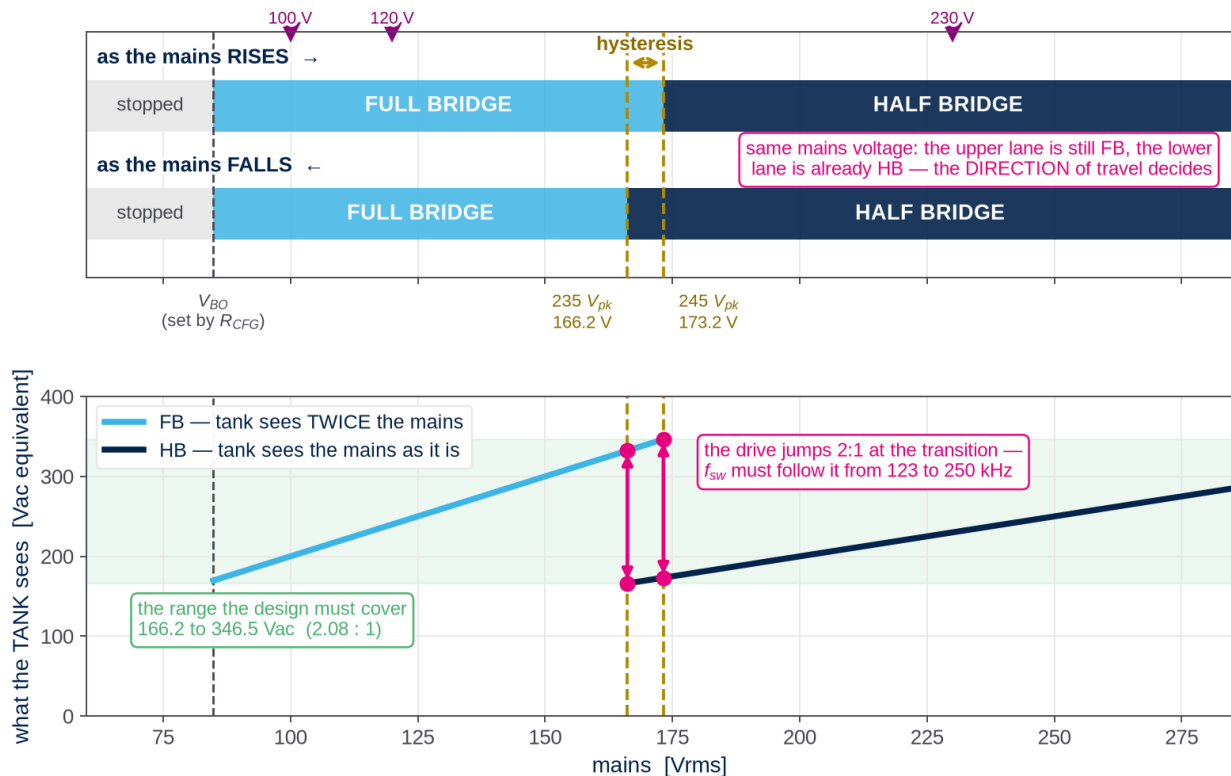


Figure 32: Where each mode applies. Inside the band the mode depends on which direction the mains arrived from.

Where you will actually meet the band. Real mains never sits at 166 to 173 Vrms , so in service the mode is settled at start-up and stays. The band is met on a programmable ac source and in line dip and surge testing. Test it with a **step** across the threshold, not a ramp — a ramp gives the voltage loop time to follow, and a step does not.

Transition transient. Crossing a threshold changes the tank drive $2:1$ within one line cycle, and f_{sw} must follow from 127 to 249 kHz against a voltage loop that crosses at 17.6 Hz . The output droops until the loop recovers. This event is **less severe than the hold-up case**, which removes power entirely for 12 ms and is already met, so it is not a reason to enlarge C_{out} or speed up the loop. That comparison has not yet been checked in the time domain, by simulation or by measurement.

6 Design procedure

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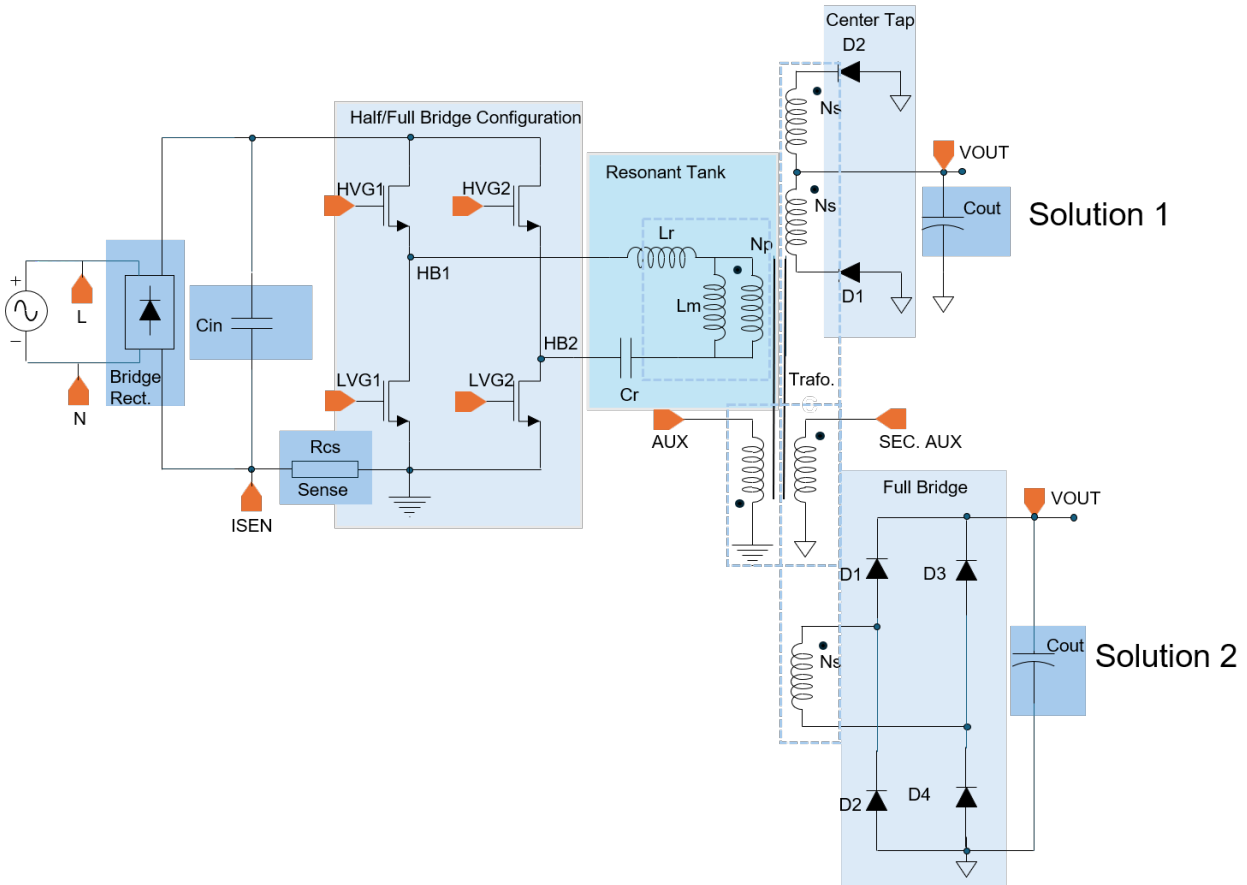


Figure 33: The complete power stage. The bridge rectifier feeds the half/full-bridge leg pair directly — there is no bulk capacitor and no boost stage. The secondary is drawn both ways: centre tap (Solution 1) and full bridge (Solution 2).

Pick the secondary before the tank: it sets $N_{rect'}$ and N_{rect} sets the reflected voltage. A centre tap puts one device in the conduction path, a full bridge puts two, so at a low output voltage the centre tap halves the secondary conduction loss. The price is twice the reverse voltage on each device, and a secondary winding that works only half of the time. This design uses the centre tap.

The sequence below is the order in which the quantities actually depend on one another. Working out of order will produce a tank that has to be redone.

6.1 The specification

The worked example in Section 7.1 states the specification in full, sorted by what kind of number each one is; the procedure below refers to it but does not repeat it. Two qualifiers matter to every step here: **the worst line frequency is the lowest**, so everything that depends on f_l uses 47 Hz, and **hold-up is specified at the worst line phase**, the ripple trough.

6.2 The equivalent input range

This is the step that is most often done wrong. With morphing the tank does not see 90 Vac at the bottom and 264 Vac at the top. It sees an **equivalent** input given by the bridge mode:

$$V_{ac,eq} = 2 V_{ac} \quad (\text{full bridge}) \quad V_{ac,eq} = V_{ac} \quad (\text{half bridge}) \quad (17)$$

At the half-bridge edge the mains is 245 V_{pk} and the tank sees 173.2 Vac equivalent; at the full-bridge edge the mains is 235 V_{pk} in full bridge and the tank sees 332.3 Vac equivalent. **These two edges are the design corners**,

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not the ends of the mains range: the low edge is the worst case for gain and ZVS, the high edge for switching frequency.

6.3 Turns ratio and reflected voltage

The turns ratio only ever appears multiplied by the effective output voltage, so what matters is the **reflected voltage**:

$$V_{refl} = n V_{o, eff} = n(V_{out} + N_{rect} V_f) \quad (18)$$

Two ratios have to be kept apart: n is the equivalent-model ratio in the gain equation, n_T the physical turns ratio with the leakage folded into L_r . At $\lambda \approx 0.5$ they differ by more than 20 %; Section 6.9 has the relation, and the transformer drawing must carry open- and short-circuit inductance as well as turns.

6.4 The resonant tank

The equivalent ac load resistance for a centre-tapped secondary is

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o, eff}^2}{P_{in, LLC}} \quad (19)$$

with $P_{in, LLC}$ the power into the resonant stage. The 4 is the usual 8 with the *line-peak* power $2P$ in the denominator rather than the average — see the note below.

The gain requirement at the low equivalent corner and the ZVS requirement together cap the quality factor at Q_{ZVS} , and $R_{ac} Q_{ZVS}$ is then the impedance the tank is *sized* from: it fixes C_r through equation 1, and C_r with the target f_r fixes L_r . Three rules govern what happens next, and they matter more than the arithmetic:

- The three *calculated* figures are not one tank. C_r comes from the design impedance; L_r is then whatever pairs with the **selected** C_r , which is why it is not $R_{ac} Q_{ZVS}/2\pi f_r$; and the L_m figure is L_r divided by the λ that *no load at the high corner* asks for — a condition a design may knowingly not meet (Section 4.8), so it is not a value to round towards.
- **Never round L_m up.** That lowers λ and takes the ZVS margin with it. C_r above its calculated value and L_m below have the opposite effect: Q_{pk} falls and λ is held.
- **Stay below the Q_{ZVS} cap, not on it.** That gap is where most of a design's ZVS margin comes from, and it is a deliberate choice, not an accident of rounding. L_m is chosen together with n so that $n_T = n\sqrt{1+\lambda_{act}}$ lands on a ratio that can actually be wound.

The same expression, evaluated at the line peak. Every LLC text writes $R_{ac} = (8/\pi^2)n^2 V_o^2/P$ with the one output power a two-stage converter has. Here the drawn power is $p(\theta) = 2P\sin^2\theta$, so R_{ac} has to be pinned to one instant. Pinned to the **line peak**, where $p = 2P$ and the tank is most heavily loaded, the 8 becomes a 4. It is not a different formula and it has nothing to do with half against full bridge — the bridge factor belongs to the equivalent input voltage. Q then means the quality factor at the line peak, with $Q(\theta) = Q_{pk} \sin^2\theta$ elsewhere. Using the two-stage value understates the loading by two, and every tank value that follows is wrong.

6.5 ZVS verification

The closed-form ZVS estimate in most design guides is a fitted approximation evaluated at the design Q_{ZVS} , not at the Q the converter runs at. Against a full sweep it errs in **either** direction by tens of per cent, and the sign depends on the tank; optimistic is the dangerous direction, because it reports ZVS margin that is not there.

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And it has to be told which λ to use. The shortcut is written with a bare λ , and it means the *design* λ — the largest of the four candidates — together with the design Q_{ZVS} . Read the same expression with λ_{act} , which is what the symbol means everywhere after the tank is chosen, and the phase can come out **negative** — a capacitive answer for a tank that is well inside the inductive region. The sweep carries no such ambiguity: it uses λ_{act} and the Q of the moment, which is what the hardware has.

The reliable procedure is to sweep θ , input voltage and load with the **selected** tank and take the minimum. Section 7.5 runs it for this design, with numbers on both directions of the closed-form error, and the worst point is not where one would expect.

6.6 Which side of resonance the converter runs on

Below f_r the secondary current is a truncated sine with a dead interval and the conduction ratio $d = f_{sw}/f_r$ is less than one; above f_r the current is continuous and d clamps at one. The rms figures that ratings and losses are built on contain d , so the difference is real, not a matter of words.

Which side of resonance the converter runs on is not a separate choice. The turns ratio decides it. The demand is $M_{req} = 2nV_{o,eff}/(\sqrt{2} V_{eq})$, so a larger n asks for more gain, and more gain means further below resonance. Two candidate transformers on the same tank can end up on opposite sides of f_r at the same mains voltage.

Worth settling before the transformer is ordered. Comparing candidate turns ratios on gain margin alone hides it, and the two sides do not stress the secondary the same way: below resonance the rectifiers turn off at zero current, above it they do not.

6.7 Currents over the line cycle

Ratings come from the worst switching cycle; losses come from the line-cycle rms. They are different numbers and must not be confused. Two further rules apply to a single-stage converter in particular.

- The **composite** tank current is what the primary devices and the sense resistor see. It is the sum of the reflected load current and the magnetising current, and its peak is neither the sum nor the larger of the two peaks, because they occur at different instants.
- Below resonance the secondary conducts for only $d = f_{sw}/f_r$ of each half period. Omitting that conduction ratio over-estimates the rms, and the loss by the square of that.

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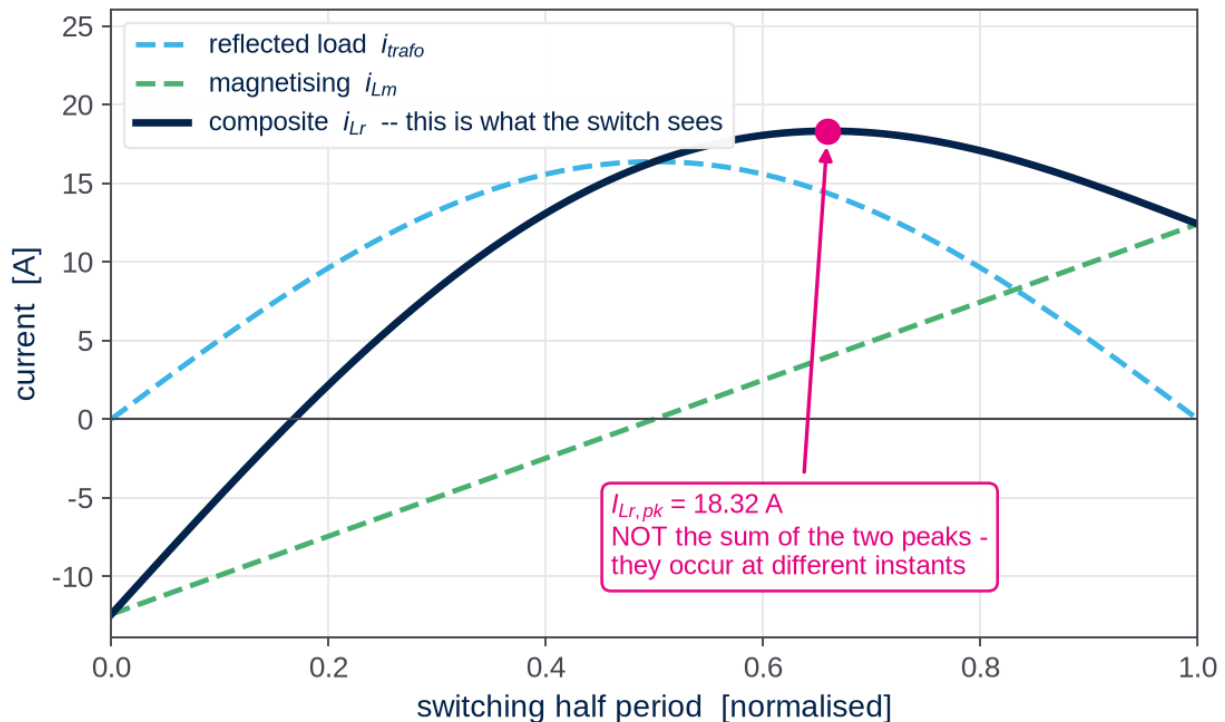


Figure 34: The composite tank current, and why its peak is not the sum of the two component peaks.

6.8 The transformer

The tank has fixed L_r , L_m and a ratio; the transformer has turns, an open-circuit inductance and a leakage. Getting from one to the other is where most of the errors in a resonant design are made, because **two different turns ratios and two different inductances carry almost the same names**. Section 2.12 has already set out why this transformer is sized the way it is; this section turns that into the numbers a supplier can measure.

Every number on the drawing is read off one of three waveforms — the primary winding current, the current in each secondary winding, and the secondary winding voltage — or off one bench test. Figure 35 marks where, and Table 2 says what each mark sets, over which interval it is taken and what it comes to in the worked design. The sections that follow derive the entries; they are easier to read with the figure open.

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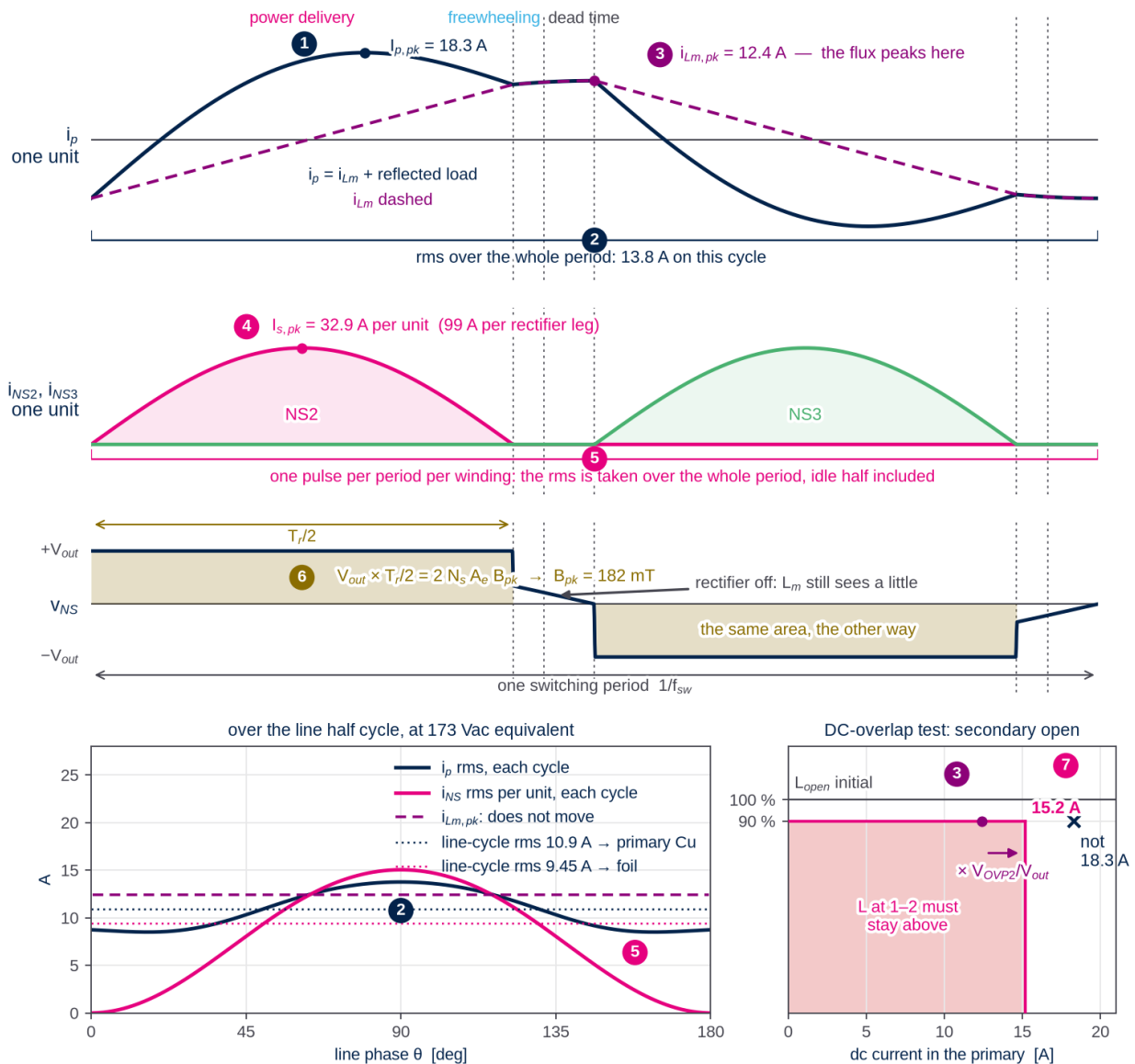


Figure 35: Where each transformer specification is read. Top: the primary winding current of one unit, the current in its two secondary windings and the secondary winding voltage over one switching period, at the worst cycle of the design (line peak, 173 Vac equivalent, full load). Lower left: the rms of the two winding currents over the line half cycle, and the line-cycle values the copper is sized on. Lower right: the DC-overlap test, which is a bench condition and not an operating waveform.

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Mark	Item	Read from — and over what	Worked design	What it sets
1	Primary peak current	Top of i_p : the half sine of load current riding on the magnetising ramp. Worst switching cycle: line peak, minimum equivalent input, full load.	18.3 A	Primary switch and OCP1 headroom. Not the saturation test.
2	Primary rms current	i_p^2 averaged over the whole switching period, then over the line half cycle (lower left). Every unit carries the same current: the primaries are in series.	13.8 A worst cycle, 10.9 A line cycle	Primary copper: 10.9 A / 4.5 A/mm ² = 2.43 mm ² per unit
3	Magnetising peak	The dashed i_{Lm} at the switching instant. Below resonance it depends on neither f_{sw} nor load, only on V_{out} , so it is the same on every cycle (the flat line, lower left).	12.4 A	The peak flux; and, scaled to V_{OVP2} , the DC-overlap test current (mark 7)
4	Secondary peak current	Top of the pulse in either secondary winding, same worst cycle. Per unit it is $1/N_x$ of the rectifier-leg value: the secondaries are in parallel.	32.9 A per unit (99 A per leg)	Rectifier peak rating; the foil termination
5	Secondary rms current, each winding	One pulse per period in each winding, so the rms is taken over the whole period with the idle half counted, then over the line half cycle.	9.45 A per unit (28.3 A per leg)	Secondary copper: 2.10 mm ² → foil 0.20 × 10.5 mm
6	Winding volt-seconds	The secondary winding voltage is V_{out} for the resonant half period $T_r/2$ while the rectifier conducts. That area is the flux swing $2N_s A_e B_{pk}$; f_r is the worst case because the interval never gets longer than $T_r/2$.	$B_{pk} = 182$ mT on A_e $= 113$ mm ²	Core area and N_s : $A_e \geq 103$ mm ² at 0.20 T
7	DC-overlap (saturation) test	Not a waveform. Secondaries open, dc current in the primary, inductance at the primary read against current. The test current is mark 3 scaled to the highest output the controller allows: $V_{OVP2}/V_{out} = 1.224$.	15.2 A, $L_{open} \geq 90$ % of initial	Core saturation margin; the number the supplier tests to
—	L_{open} , L_{short}	LCR meter at the primary with the secondaries open, then shorted. Terminal quantities, so a supplier can test them.	10.33 / 3.67 μ H per unit	$L_m + L_r$ and L_r : the tank itself

Table 2: What each mark in the figure sets. Currents per unit of the assembly; the primaries are in series and the secondaries in parallel.

6.9 Two ratios, two inductances

When all of the leakage is referred to the primary, the tank sees an ideal transformer of ratio n , a series L_r and a shunt L_m . That n is **not** the wound ratio. With k the coupling coefficient,

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}} \quad (20)$$

so $n_T = n \sqrt{1 + \lambda}$. At $\lambda \approx 0.5$ the two differ by more than 20 %, which is far more than any gain margin.

The same split appears in the inductances. The tank model uses L_r and L_m ; the physical transformer is described by a magnetising inductance L_μ with leakage on both sides:

$$L_\mu = \sqrt{L_m (L_m + L_r)}, \quad L_{L1} = L_m + L_r - L_\mu, \quad L_{L2} = \frac{L_{L1}}{n_T^2} \quad (21)$$

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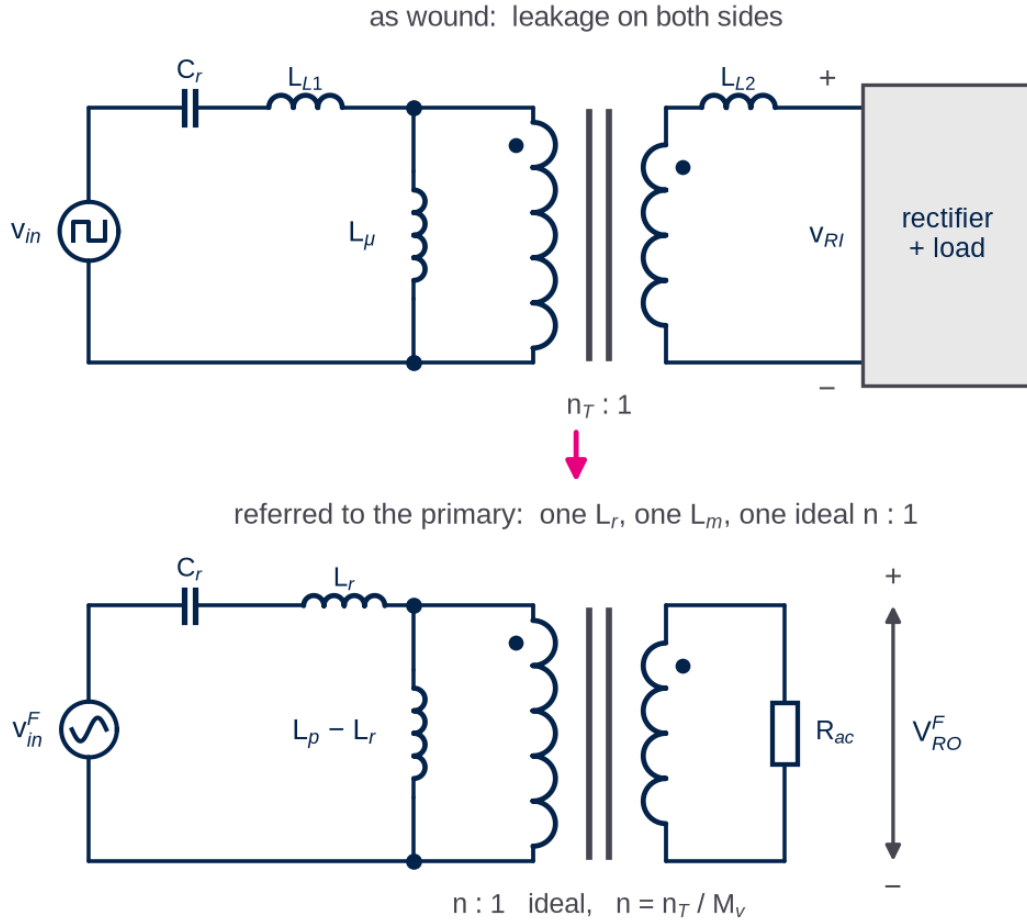


Figure 36: The same transformer drawn both ways. Above, as wound: the leakage split between the two sides, the physical magnetising inductance L_μ in shunt, and the wound ratio $n_T : 1$. Below, referred to the primary: one L_r , one L_m , and an ideal ratio $n : 1$ that is no longer the wound one. The two shunt elements are not the same number — $L_\mu = \sqrt{(L_m(L_m + L_r))}$ — and neither are the two ratios.

That three-element form assumes the leakage splits evenly between the two sides. The real split is set by the winding arrangement rather than by the design, and it does not matter: the tank sees only the two quantities a meter can read.

6.10 What the transformer specification must say

Only two inductances are measurable at terminals, and they are exactly the two the tank needs:

$$L_{open} = L_\mu + L_{L1} = L_m + L_r, \quad L_{short} = L_{L1} + \frac{L_\mu n_T^2 L_{L2}}{L_\mu + n_T^2 L_{L2}} = L_r \quad (22)$$

L_{open} is read at the primary with every other winding open, L_{short} with the secondary shorted. **Specify those two and the turns, never L_μ** — no terminal pair exposes L_μ , so a supplier cannot test it, and a part that meets it can still miss the tank. Also state the turns as numbers: with one or two secondary turns, the inductance of that winding is dominated by the lead-out loop, so $\sqrt{(L_1/L_2)}$ does not give the turns ratio.

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6.11 Peak flux is set by the secondary, not the primary

Mark **6** of Figure 35: the secondary winding voltage is a rectangle of height V_{out} and width $T_r/2$, and the flux swings from $-B_{pk}$ to $+B_{pk}$ across it. The equation is that area divided by $2N_s A_e$, and nothing else enters.

Below resonance the secondary conducts for a resonant half period and the output voltage stands across the secondary winding for all of it. Above resonance it conducts for a shorter switching half period, so f_r is the worst case everywhere on the line cycle:

$$B_{pk} = \frac{V_{out, eff}}{4f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4f_r N_s B_{max}} \quad (23)$$

N_p does not appear. Adding primary turns does not reduce the flux; only adding secondary turns does. That is an uncomfortable result for a low-voltage, high-current output, where the secondary wants to be a single turn: halving N_s doubles the core area needed. That is why a single-turn secondary is expensive in core area even though it is the obvious choice for the current.

Computing the same flux from an inductance is correct only with the **physical** L_μ : $B_{pk} = L_\mu i_{\mu, pk} / (N_p A_e)$ reduces exactly to the expression above. Using the tank model L_m with the real N_p understates the flux by $\sqrt{1+\lambda}$ — about a quarter at $\lambda \approx 0.5$ — because the L_m branch carries $n V_{out}$ while the winding reflects $n_T V_{out}$.

6.12 Choosing the core: two areas, and the one that usually decides

A core has to satisfy two independent conditions, and they have nothing to do with each other. The magnetic one is the equation above: enough cross-section A_e to carry the flux at the turns chosen. The electrical one is the winding window A_w : enough room for the copper the currents need. Catalogues quote both, and their product is what handbooks call the **area product**. Either can be the binding one.

Read the flux check against A_{min} , not A_e . A_e is the effective area for inductance; A_{min} is the narrowest section the flux actually has to pass. On a PQ core the two differ by ten to twenty per cent, and it is A_{min} that saturates first: the real peak is $B_{pk} A_e / A_{min}$. Both numbers are in the datasheet, and only one of them is usually used.

In a **single-stage PF LLC** there is a **third condition, and it is normally the one that decides**: the leakage inductance has to come out at L_r . Written as a fraction of what a meter reads at the primary,

$$\frac{L_{short}}{L_{open}} = \frac{L_r}{L_m + L_r} = \frac{\lambda}{1 + \lambda} \quad (24)$$

At $\lambda = 0.55$ that is **35 %**. A conventional transformer, wound to couple as well as it can, leaks a few per cent; this one has to leak an order of magnitude more, *on purpose*. Leakage that large is not a winding tolerance, it is a winding **geometry** — and geometry costs window area. A core that passes on A_e and on copper can still fail here.

6.13 The winding arrangement is the leakage

Leakage is the flux that links one winding and not the other, so it is set by how far apart the two windings are and by how much of the window separates them. That gives the designer one knob and a well-known ordering:

- **Interleaved** — primary, secondary, primary, all concentric. Lowest leakage, typically one or two per cent. This is what a conventional forward or flyback transformer wants, and it is the opposite of what is needed here.
- **Concentric, not interleaved** — the whole primary, then the whole secondary, over it. A few per cent. Still far short.
- **Side by side** — the primary on one half of the winding width, the secondary on the other. Tens of per cent, and adjustable by the gap left between them. **This is the arrangement a single-stage tank asks for**, and it is why the window has to be generous: half of it goes to each winding, and part of what is left is deliberately empty.

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- **A magnetic shunt** — a ferrite bar in the window between the two windings, which carries leakage flux on purpose. It buys leakage without spending winding width, at the cost of a part and of a tolerance that is harder to hold.

This inverts the usual advice. Every transformer text says to interleave, because in every other topology leakage is loss and overshoot. Here L_r is a design value that the tank needs, and a supplier who "improves" the coupling has broken the converter — f_r moves, λ collapses and the gain curve goes with it. Say so on the drawing, in words, next to L_{short} .

6.14 The gap, and why the drawing must not name it

L_m is two orders below what an ungapped core of this size gives, so the core is gapped. The inductance factor that follows from the turns is

$$A_L = \frac{L_{\text{open}}/N_x}{N_p^2} \quad g \approx \frac{\mu_0 A_e}{A_L} \quad (25)$$

with N_x the number of units the assembly is built from. The second expression is only an estimate: it assumes the gap carries the whole reluctance and it ignores fringing, which always makes the real gap larger than it predicts. **Specify A_L , or better L_{open} itself, and leave the gap to the supplier** — that is the number they grind to, and it is the number a meter can check.

6.15 The window: wire, current density and what actually fits

Marks **2** and **5** of Figure 35. The copper is sized on the **line-cycle** rms: the rms of each switching period (the bracket under the trace), averaged in I^2 over the line half cycle (the lower-left panel). The secondary value is per winding and per unit — each winding carries one pulse per period, and its idle half is inside the average.

Each winding needs a conductor cross-section set by its own rms current and the current density the design can cool:

$$A_{cu} = \frac{I_{rms}}{J} \quad \sum_w N_w A_{cu, w} \leq k_u A_N \quad (26)$$

J is an assumption, not a constant: 4 to 5 A/mm² is usual for a transformer of this size in free air, less if it is enclosed. k_u is the window utilisation, and it is brutal — round wire in a round bundle, insulation, the bobbin wall, the margins and the layer-to-layer tape leave **0.3 or less** of the window as copper when the wire is Litz. Sizing on bare copper alone overstates what fits by three times.

The primary current does not divide between units. With the primaries in series every unit carries the whole primary current, and only the secondaries share. So the primary copper is the same in each unit as it would be in one big transformer, while the secondary copper is divided — which is exactly why splitting helps a low-voltage, high-current output.

6.16 Skin depth, and why the wire is not a wire

At the switching frequency current does not fill a conductor. It crowds into a surface layer of depth

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}} \quad (27)$$

which at $f_r = 152$ kHz and 100 °C is **0.19 mm** in copper. A conductor thicker than about 2δ carries no more current than one of 2δ — it only adds weight. Worse, the field from the *other* turns drives circulating current in each

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conductor (the proximity effect), and in a side-by-side winding, where the two windings sit in each other's leakage field, that term can dominate the skin term.

- Use **Litz** on the primary: many strands, each well under 2δ , individually insulated and transposed. Strand diameter around 0.2 mm is a usual choice at this frequency.
- On a two-turn, high-current secondary, **copper foil** is usually better than wire: the thickness can be held near δ while the width fills the window, and it terminates well into a centre tap.
- Whatever is chosen, the ac resistance is what heats the winding. The dc resistance $\rho l_N N/A_{cu}$ is a floor, not an answer.

6.17 Isolation, margins and what they cost in window

The transformer is the isolation barrier of the whole supply, so the winding that looks like a copper problem is also a safety one. Two arrangements are common, and they spend the window differently. **Margin tape** leaves a creepage margin at each end of the bobbin — typically 3 mm each side for reinforced isolation from a universal mains — so it takes 6 mm off the usable winding width and costs nothing in material. **Triple-insulated wire** needs no margin, so the winding uses the full width, but the wire is thicker for the same copper and it costs more.

On a side-by-side winding the margin is taken twice over: once for the isolation, and once again for the gap that sets the leakage. Work the window budget with both in it, or the part that is ordered will not be the part that was calculated.

6.18 Loss, and the temperature the drawing has to survive

- **Core loss** comes off the material curve at the operating flux and frequency, not off the headline figure. A datasheet quotes P_v at one point — 100 or 200 mT at 100 kHz — and a design at a different flux and a different frequency has to be read from the curve.
- **Copper loss** is $I^2 R_{ac}$, and R_{ac} is the number the previous section is about. Both windings count, and in a side-by-side arrangement the proximity term is not small.
- **The temperature rise** is what actually limits the design, and it depends on the surface area and the airflow, neither of which is in any of the equations above. Treat the computed loss as an input to a thermal measurement, not as an answer.

Flux and loss do not move together here. B_{pk} is set at f_r by the secondary volt-seconds, so it does not change when the converter runs faster — but core loss does, and in a single-stage converter the frequency sweeps over the line cycle. The core sees its worst flux at the bottom of the sweep and its worst loss per cycle count at the top.

6.19 The saturation test is not the peak winding current

Mark 3 and the bench panel of Figure 35. The current to ask for is the peak of the dashed magnetising trace at the switching instant, scaled to the over-voltage ceiling — not the peak of the winding current at mark 1.

A DC-overlap test leaves every other winding open, so none of the primary ampere-turns is cancelled by the secondary and the same current makes far more flux than it does in operation. It is the flyback condition of Section 2.12, reached deliberately. Ask for the current that reproduces the operating flux in that test:

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_\mu} \quad (28)$$

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That denominator is L_μ , not L_{open} . In the open-circuit test the primary current is entirely magnetising current, so what links the core is the L_μ part of the flux linkage and the rest is primary leakage, which goes round no core at all. Dividing by L_{open} understates the current by $1/\sqrt{1+\lambda}$ — about a fifth at $\lambda \approx 0.5$. There is a check that costs nothing: the answer **must** come out equal to $i_{\mu,pk}$, because in that test they are the same current.

Giving the supplier the peak tank current instead asks for a flux density no ferrite reaches, and the supplier will either oversize the core or ask why. Nor is an arbitrary margin wanted on top of I_{eq} : the flux ceiling the core has to survive is already defined, by the highest output voltage the controller allows before it shuts down. Scale I_{eq} by that ratio and there is no number left to invent.

$$I_{sat} = I_{eq} \frac{V_{OVP2}}{V_{out, eff}} \quad (29)$$

6.20 The specification, and what it is not allowed to leave out

The open-circuit inductance tolerance cannot simply be the usual round number. f_o goes as $1/\sqrt{L_{open}}$, so a low part raises the frequency floor towards the oscillator clamp. The fall a tank can tolerate follows from k_{floor} and is typically a few per cent, so the usual $\pm 10\%$ can allow a part that meets its specification to hard switch. Section 6.25 derives the figure and gives the two ways out. Ask the supplier for a tighter low-side limit, or first make room in the oscillator setting and then ask for $\pm 10\%$.

Core, bobbin, wire and winding order are the supplier choice, and insulation and creepage follow the applicable safety standard. Nothing above constrains them; everything above is measurable at the terminals.

6.21 If one transformer is not practical

At a low output voltage and high current the secondary is a single heavy turn and the core comes out large, which may not fit a height or footprint limit. The transformer can then be built as N_x identical units with **primaries in series and secondaries in parallel**. Series primaries carry the same current, so the paralleled secondaries share automatically without balancing resistors.

- Open-circuit and leakage inductance divide by N_x , and so do the secondary currents. **The primary current does not divide.**
- The DC-overlap test current does not change: flux and primary ampere-turns scale together.
- The wound ratio of the assembly becomes $N_x N_p / N_s$, so **the achievable ratios are quantised in steps of N_x / N_s** . That has to be respected when the tank is chosen, not discovered afterwards.

6.22 The input capacitor

With no bulk capacitor, C_{in} is a film capacitor that absorbs switching ripple and nothing else. A large value is not the safe choice: it holds charge across the zero crossing and distorts the line current, so it is a THD term, not a filter term.

$$C_{in} = 3 \frac{nF}{W} \times P_{in} \quad (30)$$

That is a **lower** bound, so round up to the next standard value and rate it for the full line voltage.

6.23 The output capacitor bank

Two conditions apply, and the larger wins. The ripple condition is

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$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2} \quad (31)$$

and the hold-up condition is

$$C_{out} \geq \frac{2P_{out}T_{hold}}{(V_{out} - \frac{1}{2}\Delta v_{pp})^2 - V_{o,min}^2} \quad (32)$$

Hold-up starts at the worst line phase, and that belongs in the sizing. If the mains disappears at the trough of the $2f_l$ ripple the bank is already half a ripple below nominal when it starts, which is why V_{out} above carries the $-\frac{1}{2}\Delta v$ term. The difference is not small: it costs about a tenth of the ride-out time. Note also that Δv_{pp} there is the ripple the bank actually produces, not the allowance. The condition therefore depends on its own answer, which is why it is checked once the bank is chosen.

Which one wins is a question about the specification and not about the output voltage: both conditions scale as $1/V_{out}^2$, so V_{out} cancels when they are divided. Writing $k = V_{o,min}/V_{out}$, ripple dominates when

$$(1 - \frac{\Delta v}{2})^2 - k^2 > 4\pi f_l \Delta v T_{hold} \quad (33)$$

This screening form uses the *allowed* Δv rather than the achieved ripple, because it is asked before the bank exists. On the specification below the two come out close, and loosening Δv to 10 % would make hold-up the deciding condition instead.

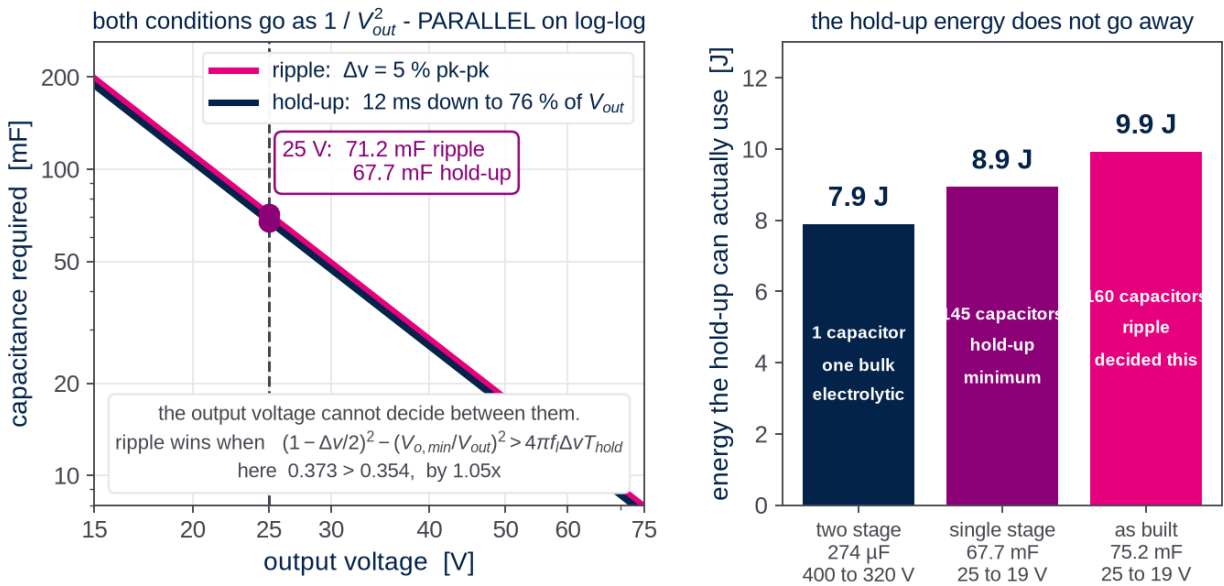


Figure 37: Both sizing conditions fall as $1/V_{out}^2$, so only the specification separates them. Right: hold-up asks for the same energy either side of the boost stage — what changes is the capacitance that holds it.

- Include the $2f_l$ component in the ripple current. The switching-frequency part alone is far too small: the $2f_l$ envelope carries a comparable rms of its own, and the two are orthogonal so they add in quadrature.
- With a centre-tapped secondary, judge the capacitor ripple current at the **output node**, not per winding.
- Use the ESR at the **switching frequency**. An electrolytic $\tan\delta$ is quoted at 120 Hz and is five to ten times larger there.

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A risk this architecture creates. The controller start-up window is finite, and a bank of this size is a far heavier start-up load than the few millifarads a conventional design presents. Cold start into the full bank should be one of the first things measured on hardware.

6.24 Semiconductor requirements

State requirements rather than pick parts: what to pick depends on package, thermal design and cost. Four rules decide whether the requirement is even stated correctly.

- Rate the primary switches on the **composite tank peak**, not on the reflected load current. The load component alone under-rates the device by more than a tenth here, because the magnetising current is missing from it.
- Divide currents by the number of devices actually in parallel before judging stress. Tabulated currents are **per switch position** on the primary and **per leg** on the secondary; a centre-tapped leg carries the whole secondary current, not half of it.
- Compute loss from $R_{DS(on)}$ at $T_{j,max}$, not at 25 °C. The datasheet maximum is process spread at 25 °C; temperature is a separate multiplier, roughly 1.8 for a 600 V superjunction device and 1.9 for a low-voltage synchronous rectifier. Using the 25 °C value to size a heatsink is optimistic by about a factor of two.
- A curve plotted against T_a may still be read as T_j **if it is a pulse test** — the die does not self-heat during the pulse. For any non-pulsed curve this is not valid.

The standing device in half-bridge morphing is the one to watch (Section 5.2). It never switches and carries the whole tank current, and it is where the conduction-loss budget is met or knowingly missed — Section 7.11 shows which happened here.

6.25 Controller network

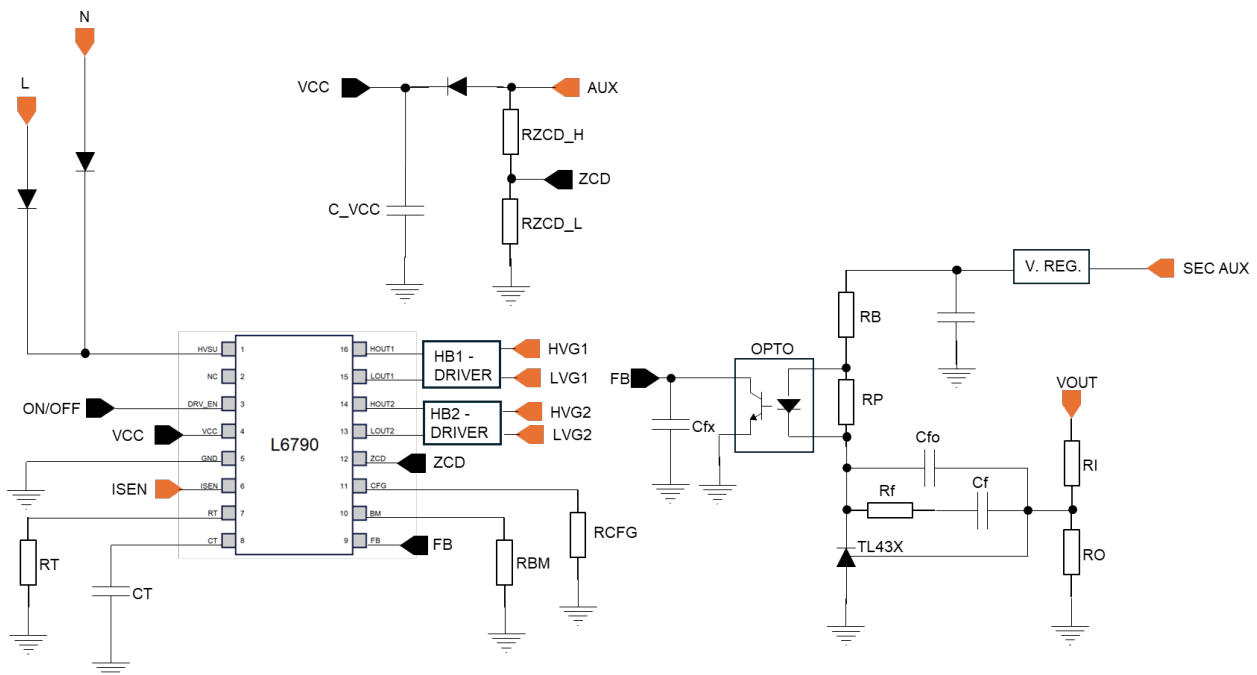


Figure 38: The controller and its passive network. HVSU is fed from the AC side of the bridge; the auxiliary winding drives ZCD through a divider; R_T and C_T set the oscillator limits; R_{CFG} and R_{BM} are read at power-up.

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Decide first what the auxiliary winding is for. When it also supplies V_{CC} , the rectified auxiliary voltage at OVP2 must stay under the V_{CC} rating, and that caps its turns ratio. Here V_{CC} comes from elsewhere, so the winding only senses, the only constraint is that the turns come out whole, and the datasheet ratio check is not applicable (Section 7.19).

Each of these parts follows from something already fixed, and Section 7.14 sizes them one at a time in the order they have to be done. The one that deserves the most attention throughout is R_T , because it sets the oscillator floor and that floor must stay above f_o .

That floor is usually the thinnest margin in the design. f_{Min} must sit above f_o , or nothing but the anti-capacitive protection stands between the converter and the capacitive region. It also depends on the oscillator idle time, which the draft datasheet states inconsistently, so whatever it computes to has to be confirmed by measuring $f_{sw}(\theta)$ on hardware.

That ratio is also what the transformer tolerance has to fit inside, and this is easy to miss because the two numbers are in different documents. Since $f_o = 1/(2\pi\sqrt{L_{open}C_r})$, a **low** open-circuit inductance raises f_o . The fall that can be tolerated is

$$\frac{\Delta L}{L} = 1 - \frac{1}{k_{floor}^2} \quad (34)$$

A floor margin of a few per cent allows a fall of a few per cent in inductance. A transformer drawing asking for the usual $\pm 10\%$ does **not** fit inside that: at the bottom of the tolerance the converter hard switches. There are two ways out — tighten the tolerance on the low side, or lower R_T to lift f_{Min} and make the room. **Check this before the transformer is ordered**, not after.

6.26 Voltage loop and compensation

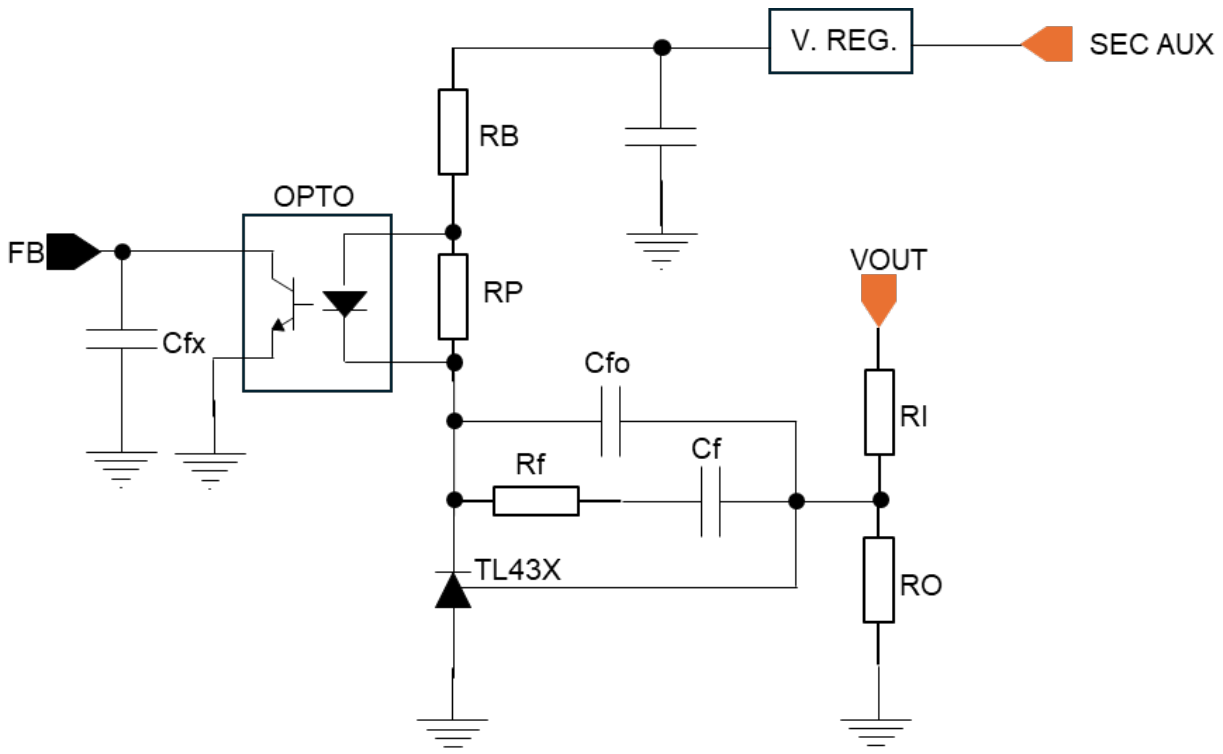


Figure 39: The secondary-side error amplifier and the optocoupler. R_L and R_O set the regulated output, $R_f C_f$ and C_{fo} shape the compensator, and R_p and R_B bias the shunt regulator and the optocoupler diode.

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The output is a capacitor fed by a constant-power source, so the control-to-output transfer is a **pure integrator**: -90° at every frequency. That makes the loop easy to stabilise but hard to place, because the crossover frequency is set by a constraint that has nothing to do with stability.

Any $2f_l$ ripple that survives the loop and reaches the feedback pin becomes a power command that varies over the line cycle, and therefore input current distortion — mostly third harmonic. Raising the crossover reduces output ripple and **increases** distortion. The two constraints pull in opposite directions, and the practical answer is a crossover between 15 and 20 Hz.

The compensator is a Type II network placed by the Venable K-factor method. With α_v the ratio between the error-amplifier gain at $2f_l$ and the gain the loop needs there,

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right] \quad (35)$$

K_v puts the zero and the pole the same factor below and above the crossover. That places the phase boost where it is needed:

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v E A_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v} \quad (36)$$

f_{MB} is the point they sit around: the zero goes below it and the pole above it, by the same factor. A third pole f_{px} is added near a kilohertz to roll off switching noise.

6.27 Gain margin, and why it needs checking

Phase margin alone does not prove stability. Above f_{px} this network has one zero against two poles, so the phase heads for -270° and **-180° is crossed at a finite frequency**. That crossing needs no root search either: setting the zero and pole phase contributions equal and taking the tangent of both sides leaves one square root,

$$f_{180} = \sqrt{f_p f_{px} - f_z (f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})| \quad (37)$$

The value under the root is positive whenever the zero sits well below both poles, which is what the K-factor placement gives. If it ever came out negative the loop would not be extra safe: the two poles would pull the phase below -180° before the zero could lift it, at every frequency, so the phase margin at any crossover would be negative. Judge the gain margin against 6 dB as a floor and 10 dB as a comfortable target. With a crossover in the tens of hertz the margin is usually large in this topology — which is a reason to compute it, not a reason to assume it.

6.28 Feedback ripple against the burst threshold

A crossover this low leaves $2f_l$ ripple on the feedback pin. Because that pin is a power command, the ripple moves the commanded power up and down every half line cycle; if it crosses the burst-entry threshold the converter chatters in and out of burst. The ripple amplitude at the burst point is

$$\Delta V_{FB} = \frac{P_{in,BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l) \quad (38)$$

Lowering R_{BM} by $\Delta R_{BM} = \Delta V_{FB} / (2 \times 0.01 \text{ V/k}\Omega)$ moves the burst entry point below that ripple. **Raising the crossover would shrink the ripple but worsen the distortion**, so trading it against R_{BM} is the practical answer rather than speeding up the loop.

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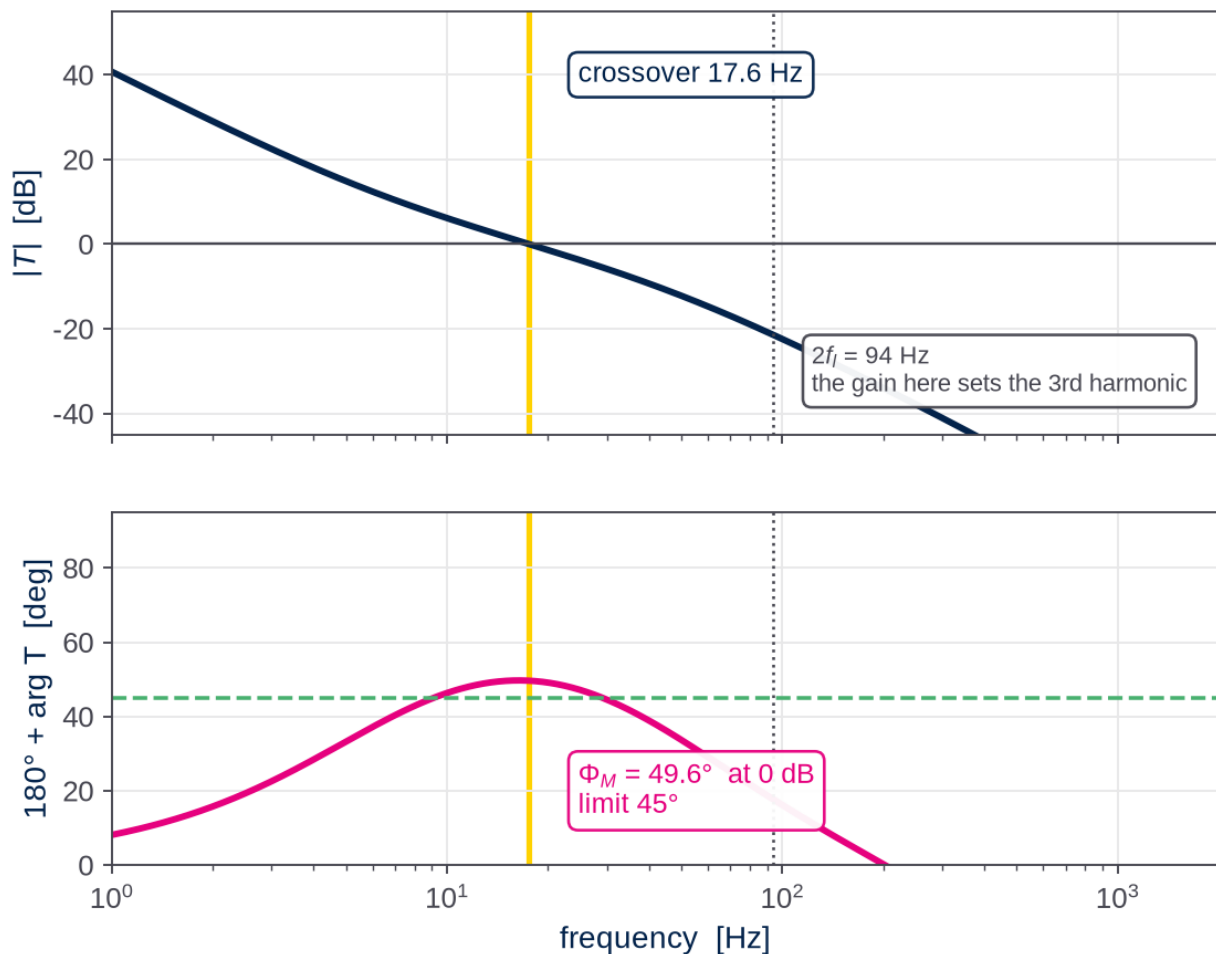


Figure 40: Open-loop gain of the selected compensation network, and $180^\circ + \arg T$. That distance is the phase margin only where $|T|$ crosses 0 dB.

A loop this slow cannot recover from a large load step. The controller's anti-saturation circuit does that instead, which is why the optocoupler bias resistor must sit inside its permitted window. Checking the window is not optional.

7 Design example

Everything so far is method: what each quantity is, which equation gives it, and what goes wrong if it is worked out the wrong way.

Here are the numbers. One design runs from the specification down to component values. Each value names the equation it came from and shows the inputs put into it, so the arithmetic can be checked. The specification is **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, centre-tapped synchronous rectification, no bulk capacitor and no boost stage.

7.1 The specification, sorted by what kind of number it is

A specification sheet usually runs all of its numbers together in one table, and that is where much later trouble starts. Four different kinds of quantity are in there. Some are imposed from outside and cannot be traded. Some were chosen by the designer and could have been chosen otherwise. Some are assumptions standing in for a measurement nobody has made yet. And some are simply what the first three produce. They are kept apart here,

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because the only useful question about a number — can I move it? — has a different answer in each group.

Item	Symbol	Value
Mains input	V_{ac}, f_l	90 to 264 Vac, 47 to 63 Hz
Output	V_{out}, I_{out}	25 V, 26.3 A (657.5 W)
Output ripple, $2f_l$ pk-pk	Δv	$\leq 5\%$ of V_{out}
Hold-up	$T_{hold}, V_{o,min}$	12 ms down to 19 V, at 100 Vac, worst line phase

Table 3: Set by the load and the mains. Not the designer's to change.

Two of these four have a condition attached that is easy to forget and costly to forget. **The worst line frequency is the lowest**, because the longest line period gives the largest ripple and the shortest hold-up, so everything that depends on f_l uses 47 Hz here, not 50 or 60. And **hold-up is specified at the worst line phase**, which means the bank starts half a ripple below nominal rather than at V_{out} — worth 11 % of the ride-out time here, as Section 7.10 shows.

Item	Symbol	Value	Consequence
Secondary rectifier	N_{rect}	1 (centre tap, SR)	halves the secondary conduction loss; doubles the device reverse voltage
Target series resonance	f_r	150 kHz	sets the magnetics size and, with C_r , fixes L_r
Specified maximum f_{sw}	$f_{sw,max}$	225 kHz	a $1.5 \times f_l$ convention, and it is what sets λ — Section 4.7
Bridge dead time	t_D	220 ns	the time the ZVS sweep has to beat
V_{CC} source	—	external 12 V rail	removes the turns-ratio ceiling on the auxiliary winding

Table 4: Chosen here. Each could have been chosen differently; the section named is where that shows up.

Item	Symbol	Assumed	If it is wrong
LLC stage efficiency	η_{HB}	98 %	R_{CS} and R_{ac} move by about 3 % for a 95 % outcome — nothing downstream is sensitive
Rectifier forward drop	$N_{rect} V_f$	0 V	a synchronous rectifier drops $IR_{DS(on)}$, carried in the loss budget rather than in the reflected voltage
Oscillator idle time	T_{idle}	250 ns	the draft datasheet also says 700 ns, which would close the floor margin entirely — the first thing to measure
$R_{DS(on)}$ temperature factor	k_T	1.8 / 1.9	primary / secondary, read off datasheet curves at $T_{j,max}$; a wrong factor mis-sizes the heatsink directly
Burst entry point	r_{BM}	75 W (11 % of rated)	sets R_{BM} , which then has to be checked against the feedback ripple

Table 5: Assumed, in place of a measurement. Replace each one from the first prototype.

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Item	Symbol	Value	From
Output ripple in volts	Δv_{pp}	1.18 V	5 % of 25 V
Reflected output voltage	$V_{o,eff}$	25.0 V	$V_{out} + N_{rect} V_r$, with $V_r = 0$
Rated input power	P_{in}	684.8 W	P_{out} through the bridge, EMI and LLC loss budgets
Overall efficiency	η_{tot}	96.02 %	P_{out}/P_{in}
Equivalent input range	$V_{ac,eq}$	173.2 to 332.3 Vac	the morphing edges, not the mains range — next section

Table 6: Follows from the three tables above — arithmetic, not input.

7.2 What the design came out as

Block	Values
Tank	C_r 100 nF, L_r 11 μ H, L_m 20 μ H, λ 0.55, f_r 151.7 kHz, f_o 90.4 kHz
Transformer	$N_p:N_s$ 15:2, n 6.02, n_T 7.50, L_{open} 31.0 μ H, L_{short} 11.0 μ H, $A_e \geq 103$ mm ² at B_{pk} 0.20 T
Frequency	f_{sw} 127.1 kHz at the low corner, 249.0 kHz at the high corner
Output	470 μ F $\times$ 160 = 75.2 mF, ripple 1.18 V (4.74 %), hold-up 13.42 ms
Controller	R_T 11 k Ω , C_T 470 pF, R_{CS} 24.0 m Ω , R_{CFG} 30 k Ω , R_{BM} 30 k Ω
Loop	f_{cross} 17.57 Hz, Φ_M 49.64°, third harmonic 4.23 %

Table 7: Principal values.

7.3 What a verification margin is

Every check has the same shape. Take what the design gets, X_{act} , divide it by what the design needs, X_{req} , and call the answer k :

$$k = \frac{X_{act}}{X_{req}} \implies \text{pass when } k > 1 \quad (39)$$

$k = 1.05$ is five per cent of room, $k = 0.9$ is ten per cent short. Every check is forced into this shape because the raw quantities do not agree on which way is good — T_{ZC} should be large, f_o small, a loss small, a hold-up long — while ratios all point the same way and have no units. So the smallest k in the table is the thinnest part of the design, whatever it happens to be about.

Check	$k = \text{has} / \text{needs}$	Substituted	k
ZVS at the worst point of the sweep	$T_{ZC,min} / t_D$	401 ns / 220 ns	1.824
Oscillator floor clears the lower resonance	f_{Min} / f_o	92.25 kHz / 90.39 kHz	1.021
Oscillator ceiling clears the operating maximum	$f_{Max} / f_{sw,max}$	319.61 kHz / 249.03 kHz	1.283
Over-current threshold clears the tank peak	$I_{OCP1} / I_{Lr,pk}$	22.92 A / 18.32 A	1.251
Hold-up achieved against hold-up required	t_{hold} / T_{hold}	13.42 ms / 12 ms	1.119
Secondary loss against its budget	$P_{budget} / P_{SR,leg}$	3.00 W / 2.83 W	1.063
Primary loss against its budget	$P_{budget} / P_{mos,dc}$	3.00 W / 9.65 W	0.311

Table 8: The verification margins, and the ratio each one is.

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k below 1 is a result, not a failed calculation. $k_{\text{Ploss}} = 0.311$ says the standing primary device spends 9.65 W against a 3 W budget. The budget is itself a choice, so the ratio could be made to pass by writing 10 W into the numerator — at which point the check stops checking anything. It is left as it is, and Section 6.24 says why the design goes ahead anyway.

Do not read two of these as percentages. $k_{\text{floor}} = 1.021$ looks like a safe 2 %. But two things move it — the transformer open-circuit tolerance and the oscillator idle time — and neither is known to 2 % yet. k_{zvs} is the same: its numerator came from a sweep, not a formula. The rest of the table is arithmetic on values that are already fixed.

7.4 Following the numbers through

The table above is where the design ends. This is how it got there: every quantity in the order it was actually computed, with what it was computed from. Read down the “from” column and it is a chain — nothing in it appears before the things it needs, which is the whole reason the design procedure is in that order.

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#	Quantity	Value	Eq.	From
1	Input power P_{in}	684.8 W	—	P_{out} 657.5 W plus the bridge, EMI and LLC loss budgets
2	Power into the tank $P_{in,LLC}$	670.9 W	—	P_{in} less the bridge and EMI budgets
3	Equivalent input, low corner	173.2 Vac	17	$245 V_{pk}$ in half bridge, $\div\sqrt{2}$
4	Equivalent input, high corner	332.3 Vac	17	$235 V_{pk}$ in full bridge, $\times 2 \div \sqrt{2}$
5	Turns ratio n	6.024	20	chosen with L_m so that n_T lands on a windable ratio — the plain calculation gives 6.046
6	Reflected voltage	150.6 V	18	$n V_{o,eff}$
7	Gain demanded at the low corner	1.2294	11	$2 n V_{o,eff} / (\sqrt{2} \times 173.2)$
8	Gain demanded at the high corner	0.6409	11	the same expression at 332.3 Vac
9	AC load resistance R_{ac}	13.70 Ω	19	$(4/\pi^2) n^2 V_{o,eff}^2 / P_{in,LLC}$ — the 4 is the line peak
10	Quality-factor cap Q_{ZVS}	1.7019	4	the smaller of the gain and ZVS limits at the low corner
11	Design impedance	23.32 Ω	—	$R_{ac} Q_{ZVS}$
12	C_r	45.50 $\rightarrow$ 100 nF	1	$1/(2\pi f_r \times \text{design impedance})$, then then rounded up on purpose, to gain Q margin
13	L_r	11.26 $\rightarrow$ 11 μ H	1	whatever pairs with the selected C_r at $f_r = 150$ kHz
14	λ required	1.241	15	the largest of four candidates — two of them carry $f_{sw,max}$ in a denominator
15	L_m	8.87 $\rightarrow$ 20 μ H	—	not that ratio: chosen with n , knowingly missing the no-load condition
16	Realised λ_{act}	0.550	—	L_r/L_m of the selected parts
17	Realised f_r, f_o	151.7 / 90.4 kHz	1, 2	from C_r, L_r and L_r+L_m
18	Realised Q_{pk}	0.766	3	$\sqrt{(L_r/C_r)} = 10.49 \Omega$, divided by R_{ac} — well under the cap
19	Physical turns ratio n_T	7.500	20	$n \sqrt{(1+\lambda_{act})}$, landing on 15:2

Table 9: The design, step by step.

Six of those rows carry the real decisions, so they are written out in full. Each one is the equation named beside it, with this design's numbers put in:

- **Row 5, the turns ratio.** Equation 20 is inverted: $n = n_T / \sqrt{(1+\lambda_{act})} = 7.500 / \sqrt{(1+0.550)} = 7.500 / 1.2450 = \mathbf{6.024}$. The wound ratio is the input here, not the output, because 15:2 is what can be built.
- **Row 7, the gain the tank is asked for.** Equation 11 at $\theta = 90^\circ$: $M = 2nV_{o,eff} / (\sqrt{2}V_{ac,eq}) = 2 \times 6.024 \times 25.0 / (1.4142 \times 173.24) = 301.2 / 245.0 = \mathbf{1.2294}$.
- **Row 9, the load the tank sees.** Equation 19: $R_{ac} = (4/\pi^2) n^2 V_{o,eff}^2 / P_{in,LLC} = 0.4053 \times 36.29 \times 625 / 670.9 = \mathbf{13.70 \Omega}$. Note the 4, not the textbook 8: $P_{in,LLC}$ is the line-peak power.
- **Row 12, the resonant capacitor.** The design impedance and the target f_r give it through equation 1 rearranged: $C_r = 1/(2\pi f_r R_{ac} Q_{ZVS}) = 1/(2\pi \times 150 \text{ kHz} \times 23.32 \Omega) = 45.50 \text{ nF}$, taken to **100 nF**.
- **Row 13, the resonant inductor.** Same equation, now solved for L with the selected C_r : $L_r = 1/((2\pi f_r)^2 C_r) = 1/((2\pi \times 150 \text{ kHz})^2 \times 100 \text{ nF}) = 11.26 \mu\text{H}$, taken to **11 μ H**. This is why L_r is not $R_{ac} Q_{ZVS} / 2\pi f_r$: C_r moved first.

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- **Row 18, the quality factor the tank actually runs at.** $Z_0 = \sqrt{L_r/C_r} = \sqrt{(11 \mu\text{H} / 100 \text{ nF})} = 10.49 \Omega$, and $Q_{pk} = Z_0/R_{ac} = 10.49/13.70 = \mathbf{0.766}$, against the 1.702 cap of row 10.

And the rest of the chain, in the same form, so that nothing in the table is a number that appeared without being worked:

- **Rows 1 and 2, the power the tank has to pass.** The output is 657.5 W. The input bridge is allowed 5.52 W and the EMI filter 8.34 W, and the LLC stage itself 13.42 W at $\eta_{HB} = 98 \%$. So $P_{in} = 657.5 + 13.42 + 8.34 + 5.52 = \mathbf{684.8 \text{ W}}$, and what reaches the tank is P_{in} less the first two: $684.8 - 5.52 - 8.34 = \mathbf{670.9 \text{ W}}$.
- **Rows 3 and 4, the corners.** Equation 17, at the two morphing thresholds and not at the mains limits: in half bridge $245 \text{ V}_{pk}/\sqrt{2} = \mathbf{173.2 \text{ Vac}}$, and in full bridge $2 \times 235 \text{ V}_{pk}/\sqrt{2} = \mathbf{332.3 \text{ Vac}}$. Everything downstream is designed between those two numbers.
- **Row 10, the cap on the quality factor.** Two limits are evaluated at the low corner and the smaller is kept: the Q at which the tank can still reach the gain of row 7, and the Q at which ZVS still completes inside t_D . Here that is $Q_{ZVS} = \mathbf{1.7019}$, and multiplying it by R_{ac} gives the design impedance of row 11: $13.70 \times 1.7019 = \mathbf{23.32 \Omega}$.
- **Row 14, the λ the tank is asked for.** Four candidates are computed and the largest is binding. The plain minimum-gain condition gives $\lambda_1 = 0.560$; folding in the frequency ceiling through equation 15 raises it to 1.009, and the dead-time form to $\mathbf{1.241}$. That last one is the requirement.
- **Rows 15 and 16, where the design departs from the arithmetic.** L_r/λ_{req} would be $11/1.241 = 8.87 \mu\text{H}$. That number is the *no-load* condition at the high corner, which this design knowingly does not meet, so it is not rounded to. L_m is chosen at $\mathbf{20 \mu\text{H}}$ together with n so that n_T lands on a windable ratio, and the λ that results is $\lambda_{act} = L_r/L_m = 11/20 = \mathbf{0.550}$.
- **Row 17, the two resonances the selected parts actually make.** $f_r = 1/(2\pi\sqrt{L_r C_r}) = 1/(2\pi\sqrt{(11 \mu\text{H} \times 100 \text{ nF})}) = \mathbf{151.7 \text{ kHz}}$, and the same expression with $L_r + L_m = 31 \mu\text{H}$ gives $f_o = \mathbf{90.4 \text{ kHz}}$. Those two, not the targets, are what every later check is measured against.
- **Row 19, the ratio that is actually wound.** $n_T = n\sqrt{(1+\lambda_{act})} = 6.024 \times \sqrt{(1+0.550)} = 6.024 \times 1.2450 = \mathbf{7.500}$, which is 15 : 2 across the assembly and 5 : 2 in each of the 3 units.

Steps 12, 15 and 18 are the judgement calls. Everything else in that table is arithmetic that any two engineers would reproduce identically. C_r was taken to 100 nF against a calculated 45.5 nF and L_m to 20 μH against a calculated 8.9 μH , and the two together are why the tank runs at $Q_{pk} = 0.766$ instead of at the 1.702 cap. That gap is most of the ZVS margin this design has.

Load	173.2 Vac eq.	180.0 Vac eq.	225.0 Vac eq.	264.0 Vac eq.	332.3 Vac eq.
100 %	460	517	671	674	603
75 %	757	776	791	720	561
50 %	1084	1079	971	808	501
25 %	1470	1445	1231	974	401

Table 10: T_{ZC} [ns] over the operating space, selected tank. The dead time is 220 ns.

7.5 ZVS over the whole operating space

The closed form of Section 6.5 reads 292 ns on this tank against a swept 460 ns, so it understates the margin by 36 %; on another tank in the same family it overstates by about 23 %. Read with $\lambda_{act} = 0.550$ instead of the design $\lambda = 1.241$ it returns -471 ns — capacitive, for a tank that is well inside the inductive region. So this design is judged by the sweep, not by the closed form.

The worst point is not the corner the tank was designed at. Full load at the low equivalent input is where the *gain* requirement is set, and it gives $T_{ZC} = 460 \text{ ns}$, a margin of 2.09. The smallest number in the table is **401 ns at 25 % load and the 332.3 Vac equivalent corner**, a margin of 1.82 against $t_D = 220 \text{ ns}$. That is the number this design is held to.

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Why light load at high line can be the ZVS corner. The charge that swings the bridge node is carried by the magnetising current, and its peak goes as $V_{in}/(4f_{sw}L_m)$. Shed load and the controller raises f_{sw} to cut the gain, so that current falls even though the node still has the same capacitance to move. Whether this beats the full-load corner depends on the tank: it does not on a high turns ratio, where the design already sits deep below resonance, and it does on a lower one. **Sweeping only the design corner will therefore report a margin that is not there**, which is the same trap as trusting the closed form.

7.6 Which side of resonance this design runs on

Section 6.6 says the turns ratio decides this. With $n_T = 7.50$ the answer is in the table below. At all but the lowest line conditions the tank spends part of every line cycle above f_r . There the secondary loses zero-current turn-off, so the rectifier body diode and the SR dead time both need checking.

Line condition	V_{eq}	peak f_{sw}	side of f_r
minimum line	180 V	131.3 kHz	below
tank minimum	173 V	127.1 kHz	below
110 Vac	220 V	156.4 kHz	above
nominal	225 V	159.7 kHz	above
230 Vac	230 V	163.2 kHz	above
maximum line	264 V	188.5 kHz	above
FB threshold	332 V	249.0 kHz	above

Table 11: Peak f_{sw} over the half cycle against $f_r = 151.7$ kHz. 5 of the seven line conditions cross into above-resonance operation for part of the cycle.

7.7 The currents this design has to carry

Ratings come from the worst switching cycle and losses from the line-cycle rms, so both are listed and they are different numbers. The row to watch is the composite tank peak: it is what the primary devices and R_{CS} actually see, and it is neither the sum of the two component peaks nor the larger of them.

Quantity	Value	Where it is used
f_{sw}	127.1 kHz	oscillator floor check
$I_{sec,pk}$	98.6 A	secondary rectifier peak
$I_{trafo,pk}$	16.37 A	reflected load component
$I_{Lm,pk}$	12.41 A	magnetising component
composite tank peak	18.32 A	primary device rating, OCP margin
$I_{pri,rms}$ (worst cycle)	13.76 A	primary rms rating
I_{pri} (line-cycle rms)	10.93 A	primary conduction loss
I_{rect} per leg (line-cycle rms)	28.34 A	secondary conduction loss
I_{Cout} (line-cycle rms)	30.24 A	output bank ripple current

Table 12: Currents at the low equivalent corner, full load, $\theta = 90^\circ$.

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Quantity	Calculated	Selected	Direction
C_r	45.50 nF	100 nF	rounded up
L_r	11.26 μ H	11 μ H	rounded down
L_m	8.87 μ H	20 μ H	not a rounding — see below
$\lambda_{act} / f_r / f_o$	—	0.550 / 151.7 kHz / 90.4 kHz	from the selected parts

Table 13: Tank: calculated against selected.

7.8 The transformer, as built

The tank asks for $L_r = 11 \mu\text{H}$, $L_m = 20 \mu\text{H}$ and $n_T = 7.50$. At 26.3 A out the secondary wants to be one or two heavy turns, and a single core that can carry the flux at that few turns comes out too large for the enclosure. This design therefore uses 3 identical units, primaries in series and secondaries in parallel, as Section 6.21 describes.

Quantity	Per unit	Assembly	Note
Turns	5 : 2	15 : 2	primaries in series, secondaries in parallel
Open-circuit inductance	10.3 μ H	31.0 μ H	$L_r + L_m$, measured with every other winding open
Short-circuit inductance	3.67 μ H	11.0 μ H	this is L_r : no separate resonant inductor
A_L	413 nH	—	open-circuit inductance / N_p^2
Peak flux density	182 mT	—	$V_{o,eff} / (4 f_r N_s A_e)$ — N_p does not enter
Core area	$\geq 103 \text{ mm}^2$	—	to hold B_{pk} at 0.20 T; EE6405 offers 113 mm^2 , a factor of 1.098
Saturation test current	15.2 A	—	the flux-equivalent 12.4 A at the V_{OVP2} ceiling, not the 18.3 A tank peak

Table 14: The transformer as it is actually wound and specified.

Three of those rows are computed rather than specified, and this is the arithmetic:

- **Peak flux.** Equation 23, at f_r because that is the worst case anywhere on the line cycle: $B_{pk} = V_{o,eff} / (4 f_r N_s A_e) = 25.0 \text{ V} / (4 \times 151.75 \text{ kHz} \times 2 \times 113.1 \text{ mm}^2) = \mathbf{182 \text{ mT}}$. N_p is absent, which is the whole point of that equation.
- **Core area needed.** The same equation solved for A_e at $B_{max} = 0.20 \text{ T}$: $A_e \geq 25.0 \text{ V} / (4 \times 151.75 \text{ kHz} \times 2 \times 0.20 \text{ T}) = \mathbf{103 \text{ mm}^2}$. EE6405 gives 113.1, so the margin is 1.098 — and halving N_s would double the requirement to 206 mm^2 , which EE6405 does not meet.
- **Saturation test current.** Equation 28, per unit and over L : $I_{\mu,eq} = B_{pk} N_p A_e / L_{\mu} = 0.1821 \text{ T} \times 5 \times 113.1 \text{ mm}^2 / 8.30 \mu\text{H} = 12.41 \text{ A}$, which is $i_{\mu,pk}$ to the last digit, as it has to be. Raised to the over-voltage ceiling by equation 29, $30.61 \text{ V} / 25.0 \text{ V} = 1.2244$, it is specified as **15.2 A**. The tank peak is 18.3 A and handing that over instead would ask the supplier for 1.48 times the flux the core ever sees.

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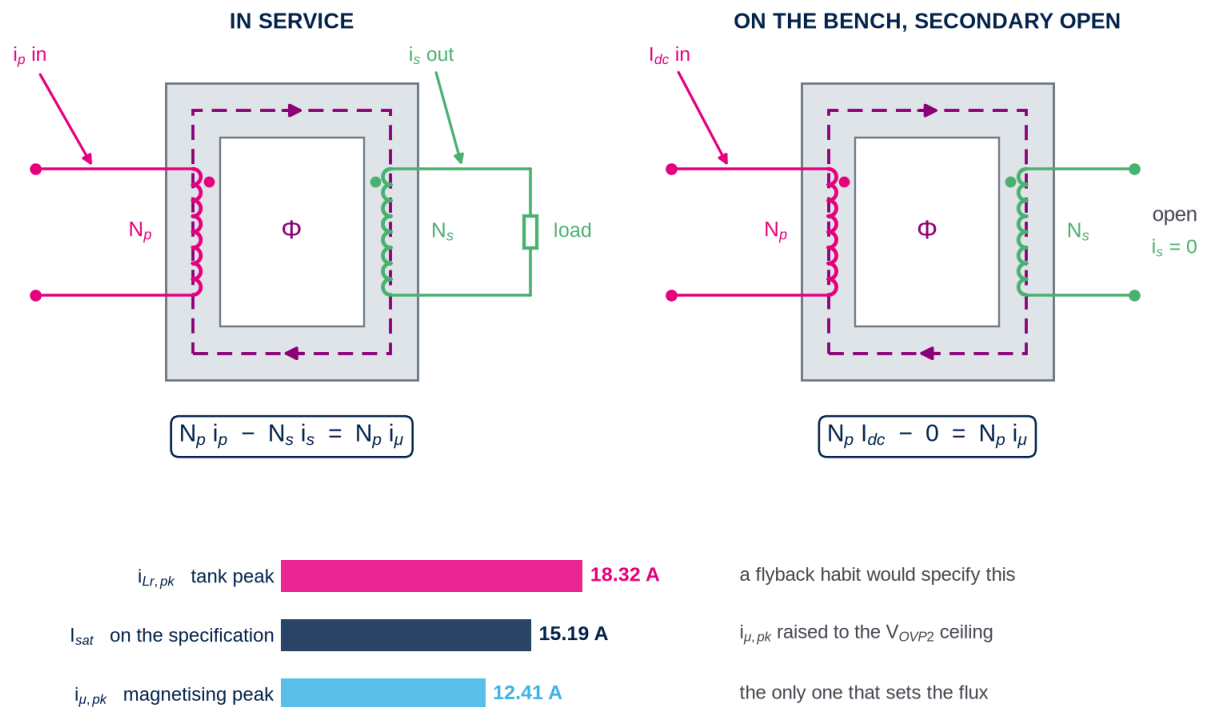


Figure 41: Why that test current is the one to ask for. In service the secondary's ampere-turns cancel most of the primary's, and only the difference magnetises the core; on the bench, with the secondary open, there is nothing to cancel and the whole of the test current is magnetising current. So the bench reaches the design flux at 12.41 A — the magnetising peak — and not at the 18.32 A the primary really carries. The polarity dots, not the drawn winding sense, define the sign in the equation under each panel.

The quantisation is a constraint on the tank, not a detail of the build. With 3 units in series the assembly ratio is $3N_p/N_s$, so the only ratios available are multiples of $3/N_s$. $n_T = 7.50$ exists because $15:2$ exists; a tank that had asked for 7.90 could not have been wound this way at all. That has to be checked while the tank is being chosen.

7.9 Choosing the core, and building the winding on it

The tank has fixed what the transformer must be: $L_{open} = 10.3 \mu H$ and $L_{short} = 3.67 \mu H$ per unit, 5 : 2 turns, and a leakage that is 35 % of the open-circuit inductance. Now a core has to be picked that can do all three. Two standard PQ sets are worked through below; both are TDK catalogue parts and every figure in the left-hand columns is from their datasheets.

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	PQ 32/30	PQ 40/40	what it has to beat
A_e	153.8 mm ²	189.0 mm ²	≥ 103 mm ² at 0.20 T
A_{min}	127.5 mm ²	174.0 mm ²	the section that really saturates
B_{pk}	134 mT	109 mT	at A_{min} : 162 and 118 mT
N_s needed	1.34 $\rightarrow$ 2	1.09 $\rightarrow$ 2	a whole number, so 2 on either
window A_N	104 mm ²	309 mm ²	≥ 68 mm ² for the copper alone
window used	66 %	22 %	before margins and the separation
mean turn l_N	62 mm	87 mm	copper length, so copper loss
gap for $A_L = 413$ nH	0.47 mm	0.57 mm	first estimate, before fringing

Table 15: Two candidate cores against the three conditions. Datasheet figures on the left, this design on the right.

Both cores pass the flux condition, so the flux does not decide. A_e clears the requirement by 1.49 on the smaller core and 1.83 on the larger, and either takes the same two secondary turns. What separates them is the window. The copper alone asks for 68 mm² at 4.5 A/mm² and $k_u = 0.30$, which is 66 % of the smaller former and 22 % of the larger — *and that is before any of the window has been given to the margins and to the separation that the 35 % leakage needs.* On the smaller core there is nothing left to give.

So the core is chosen by the leakage, not by the flux. That is the single-stage result in one line, and it is worth stating because it is the opposite of the usual order: in a conventional converter the core comes from A_e and the window follows, and here the window decides. **PQ 40/40 is taken** — N95 or an equivalent low-loss grade, gapped to A_L . The price is a mean turn 40 % longer than the smaller core and a set that weighs 57 % more.

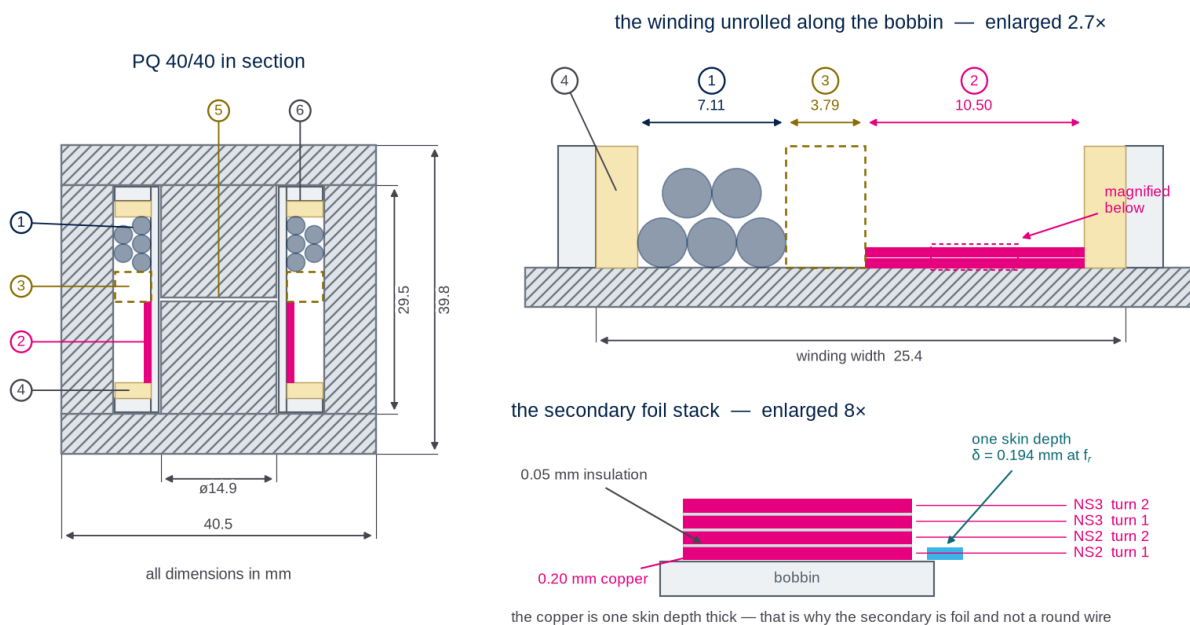


Figure 42: The chosen core in section, the winding unrolled along the bobbin, and the secondary foil stack, all to scale from the TDK drawings. Primary and secondary sit side by side, and the 3.79 mm left between them is the deliberate separation that makes L_{short} come out at 3.67 μ H per unit. Table 17 is where every number on it comes from.

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Mark	Item	Value	Note
1	Primary NP1, pins 1–2	5 T	Litz 310 × Ø0.10 mm, bundle Ø2.37 mm, 2 layers (3 + 2); conducts every half period
2	Secondary NS2, pins 3–5, and NS3, pins 4–6	2 + 2 T	foil 0.20 × 10.5 mm, 4 layers; NS2 conducts in the first half period, NS3 in the second
3	Separation	3.79 mm	the width left over; it is the leakage, $L_{\text{short}}/L_{\text{open}} = 35\%$
4	Margin tape	2.0 mm each flange	creepage to the flange; assumed
5	Centre-leg gap	≈ 0.57 mm	ground to $A_L = 413$ nH; the estimate ignores fringing
6	Window A_N	309 mm ²	20.5 mm ² of bare copper in it
—	Auxiliary NAUX, pins a–b	1 T	any wire; ZCD sense only, no load

Table 16: Legend of the figure, and the winding list of one unit.

Why the secondary is "2 + 2". The rectifier is centre-tapped, so the secondary is two identical windings, NS2 and NS3, of $N_s = 2$ turns each, joined at the tap. Each conducts on alternate half periods (Figure 35, second trace), so the bobbin carries 4 foil turns of which two are carrying current at any moment. That is what the drawing means by 2 + 2, and why the foil stack has 4 layers.

Step	Primary NP1	Secondary NS2 (and NS3)
Line-cycle rms current per unit (marks 2 and 5)	10.93 A — the whole primary current, the primaries are in series	28.34 A / 3 units = 9.45 A
Copper at $J = 4.5$ A/mm ²	10.93 / 4.5 = 2.43 mm²	9.45 / 4.5 = 2.10 mm²
Conductor ($2\delta = 0.39$ mm at f_r)	Litz: 2.43 mm ² / ($\pi \times 0.10^2/4$) = 310 strands of Ø0.10 mm; served bundle Ø2.37 mm at 55 % fill	Foil 0.20 mm thick (one skin depth, all of it conducts): 2.10 / 0.20 = 10.5 mm wide
Turns	5, in 2 layers (3 + 2)	2 + 2: two windings of 2, one per half period
Width taken on the bobbin	3 × 2.37 = 7.11 mm	the foil width, 10.5 mm
Radial build	2 × 2.37 = 4.74 mm	4 × (0.20 + 0.05 insulation) = 1.00 mm
Left for the separation	25.4 – 2×2.0 – 7.11 – 10.5 = 3.79 mm (mark 3 in the figure)	
Bare copper in the window	5 × 2.43 + 4 × 2.10 = 20.5 mm² ; at $k_u = 0.30$ that is 68 mm ² , 22 % of A_N	

Table 17: From current to copper: how the numbers in the figure come about. One unit; the two secondary windings are identical.

Reduced to what a supplier can measure at the terminals, this gives the specification sheet below. Nothing on it is a construction instruction, and L_μ is deliberately absent — Section 6.10 says why.

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Item	Value	Condition
Turns ratio	$N_p : N_s = 15 : 2$	two secondary windings, centre tap outside the part
Open-circuit inductance	31.0 μH , -4.1 % at worst	primary, all other windings open
Leakage inductance	11.0 $\mu\text{H} \pm 10\%$	primary, secondary shorted
DC overlap	$\geq 90\%$ of initial inductance	test current 15 A (mark 7)
Primary current	10.9 A rms / 18.3 A pk	line-cycle rms (mark 2), composite peak (mark 1)
Secondary current, each winding	28.3 A rms / 99 A pk	line-cycle rms (mark 5), peak (mark 4)
Core area	$A_e \geq 103 \text{ mm}^2$	holds B_{pk} at or below 0.20 T (mark 6)
Switching frequency	127 to 249 kHz	at full load

Table 18: Transformer specification for the worked design.

7.10 The output bank, as sized

Both conditions of Section 6.23, worked out on the specification above.

Condition	Asks for	Driven by
$2f_r$ ripple $\leq 5\%$	71.25 mF	I_{out} , $f_l = 47 \text{ Hz}$ and the allowed Δv — this one wins
Hold-up 12 ms to 19 V	67.22 mF	the energy 7.89 J between the starting voltage and 19 V, starting at the ripple trough
Selected	75.2 mF	470 $\mu\text{F} \times 160$, the next assembly up from 71.2 mF

Table 19: The two sizing conditions, evaluated.

- **The ripple condition**, equation 31: $C_{out} \geq P_{out} / (2\pi f_{l,min} \Delta v_{out}^2) = 657.5 \text{ W} / (2\pi \times 47 \text{ Hz} \times 0.05 \times 625 \text{ V}^2) = \mathbf{71.25 \text{ mF}}$, with Δv as a fraction and V_{out} in volts.
- **The hold-up condition**, equation 32, starting half a ripple low: $C_{out} \geq 2P_{out} T_{hold} / ((V_{out}^{-1/2} \Delta v_{pp})^2 - V_{o,min}^2) = 2 \times 657.5 \text{ W} \times 0.012 \text{ s} / (24.41^2 - 19^2) = \mathbf{67.22 \text{ mF}}$. Starting at V_{out} instead would have asked for only 59.77 mF.

The two are only 1.060 apart. That is close, so the screening inequality is worth carrying. At $k = V_{o,min} / V_{out} = 0.760$ it reads 0.373 against 0.354: the left side is larger, so ripple wins. Relax Δv to 10 % and the two sides swap.

Quantity	Value	Against
Achieved ripple	1.18 V (4.74 %)	5 % allowed
Achieved hold-up	13.42 ms	12 ms required — and 15.10 ms if it had been started at V_{out} instead of at the trough
Ripple current, switching component	23.8 A rms	—
Ripple current, $2f_r$ component	18.6 A rms	orthogonal to the above
Ripple current, total	30.24 A rms	0.19 A in each of 160 parts
Bank ESR	0.1875 m Ω	30 m Ω each, 160 in parallel
Ripple and noise	17.4 mV	with the 10 μF ceramic bypass

Table 20: What the selected bank then delivers, and what it has to survive.

7.11 Where the power goes

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The specification allowed 27.3 W of loss to get from $P_{in} = 684.8$ W to $P_{out} = 657.5$ W. That allowance was split three ways before any component existed, and the itemised device losses are computed afterwards. The two do not have to agree, and here they do not.

Item	Value	Note
Budget — input bridge	5.52 W	synchronous bridge assumed
Budget — EMI filter	8.34 W	—
Budget — LLC stage	13.42 W	$\eta_{HB} = 98\%$ of $P_{in,LLC}$
Primary conduction, standing device	9.65 W	the half-bridge leg that never switches — the single largest item
Primary switching, three positions	4.83 W	not a problem
Secondary rectifiers	5.65 W	1.411 W per device, 2 in parallel per leg
Current-sense resistors	2.86 W	4.54 W at the worst switching cycle
Output capacitor ESR	0.162 W	—
Itemised total	23.15 W	and the magnetics are not in it

Table 21: The budget as assumed, and the device losses as computed. The second block is not a breakdown of the first.

That disagreement shows that the efficiency assumption is optimistic. The itemised device losses already come to 23.2 W against the 13.4 W that $\eta_{HB} = 98\%$ allows the LLC stage, and the transformer is not even in the list. So the assumption is optimistic, which is why it is filed as an assumption rather than as a figure. Nothing important depends on it: taking 95 % instead moves R_{ac} and R_{CS} by about 3 % and nothing downstream is sensitive to that. What it does change is the thermal design, and that is settled by measurement rather than by either number.

7.12 What the semiconductors have to be

Requirements, not part numbers. What to pick depends on package, thermal design and cost. The four rules of Section 6.24 decide whether these are even stated correctly.

Item	Requirement
Primary drain-source voltage	≥ 485 V, so a 600 V class part
Primary body diode	fast recovery: in full bridge the body diode conducts before turn-on
Primary $R_{DS(on)}$ per device	≤ 25.2 m Ω hot to meet the loss budget
Primary $C_{o(tr)}$	small enough that the tank can swing the node inside 220 ns
Secondary drain-source voltage	≥ 80 V including the centre-tap doubling
Secondary rectifier current	28.34 A rms per leg, 98.6 A peak
Secondary package	check the lead and clip rating, not only the die

Table 22: Semiconductor requirements for the worked design.

7.13 The voltage loop, as built

Section 6.26 gives the method. These are the numbers it produces on this converter, in the order they are found.

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#	Quantity	Value	From
1	Output divider	R_1 100 k Ω / R_o 11 k Ω	sets the regulated output against the 2.495 V reference
2	Plant gain G_o	130.06 rad/s	a pure integrator — the output is a capacitor fed by a power source
3	Error-amplifier gain needed at $2f_l$	0.4541	what the loop needs there, against what the amplifier has
4	K factor K_v	2.7218	places the zero and pole symmetrically
5	Geometric mean f_{MB}	16.921 Hz	the point they are placed about
6	Zero / pole	6.217 / 46.057 Hz	f_{MB}/K_v and $K_v f_{MB}$
7	High-frequency pole f_{px}	999.4 Hz	added to roll off switching noise
8	Components	C_{Fo} 82 nF, C_F 560 nF, R_F 47 k Ω , C_{fx} 1.55 nF	the nearest standard values to the pole and zero above
9	Optocoupler bias	R_P 1.3 k Ω , R_B 6.2 k Ω	R_B must land inside 5.76 to 6.48 k Ω or the anti-saturation circuit cannot act

Table 23: The compensator, from the plant to the parts.

Quantity	Value	Against
Crossover f_{cross}	17.57 Hz	the 15 to 20 Hz this topology lands in
Phase margin	49.64°	45° as a floor
Gain margin	34.18 dB at 202 Hz	6 dB floor, 10 dB comfortable
Third harmonic on the input current	4.23 %	the price of crossing this low
Feedback ripple at the burst point	52 mV	move R_{BM} down 2.60 k Ω to clear it

Table 24: And what that loop then measures as.

7.14 The parts around the controller

These are the results. The sections after this one derive each value in the order it has to be done, because every one of them is sized against something already fixed.

Part	Value	Result
C_T / R_T	470 pF / 11 k Ω	f_{Min} 92.2 kHz, f_{Max} 319.6 kHz
R_{CS}	24.0 m Ω (120 m Ω $\times$ 5 in parallel)	OCP1 at 1.251 $\times$ the composite peak
R_{CFG}	30 k Ω	V_{BO} 84.8 V, morphing enabled
R_{BM}	30 k Ω	burst-mode entry point
ZCD divider	330 k Ω / 19 k Ω	OVP1 28.16 V, OVP2 30.61 V
C_{in}	2200 nF film	about 3.2 nF/W — there is no bulk capacitor
Compensation	82 nF / 560 nF / 47 k Ω / 1.55 nF	f_{cross} 17.57 Hz, ϕ_M 49.64°

Table 25: Controller network for the worked design.

7.15 The oscillator: C_T first, then R_T

The VCO charges C_T from the sum of a fixed current V_{ref}/R_T and the error-amplifier current I_{EA} , up to $V_{ref} = 1.5$ V, then adds a fixed idle time:

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$$\frac{T_{sw}}{2} = \frac{C_T V_{ref}}{\frac{V_{ref}}{R_T} + I_{EA}} + T_{idle} \quad (40)$$

I_{EA} is the only control input in the converter. More current means a higher frequency, which means less power. Everything the control loop does, it does through this one term.

C_T is chosen first because it sets the *span*, and R_T after it because it sets the *floor*:

$$C_{T,max} = \frac{I_{EA,max}}{2V_{ref}} \cdot \frac{(1 - 2T_{idle}f_{sw,max})(1 - 2T_{idle}f_{sw,min})}{f_{sw,max} - f_{sw,min}}, \quad C_{T,min} = \frac{1}{R_{T,max}} \left(\frac{1}{2f_{sw,min}} - T_{idle} \right) \quad (41)$$

$$R_{T,ceil} = \frac{1}{C_T} \left(\frac{1}{2f_{sw,min}} - T_{idle} \right) \quad (42)$$

with $f_{sw,min}$ set to f_o — the frequency floor of the whole design, for the reason in Section 4.2. Here that window is 176 to 703 pF, the datasheet allows 270 to 1000 pF, and the part is taken from the middle of the overlap rather than from either edge: $C_T = 470$ pF, COG. Then $R_{T,ceil} = 11.24$ kΩ gives $R_T = 11$ kΩ.

$R_{T,ceil}$ is a maximum, not a minimum, whatever the name suggests. $f_{Min} = 1/[2(C_T R_T + T_{idle})]$ falls as R_T grows. Round this number up and the VCO clamp drops below f_o , leaving only the anti-capacitive protection between the converter and the capacitive region. Round it down.

The three frequencies the selected pair actually produce are

$$f_{Min} = \frac{1}{2(C_T R_T + T_{idle})}, \quad f_{Max} = \frac{1}{2\left(\frac{C_T}{\frac{I_{EA,max}}{V_{ref}} + \frac{1}{R_T}} + T_{idle}\right)} \quad (43)$$

Putting the selected pair into the first of those, with $T_{idle} = 250$ ns: $f_{Min} = 1/[2(470 \text{ pF} \times 11 \text{ kΩ} + 0.25 \text{ μs})] = 1/(2 \times 5.420 \text{ μs}) = 92.25 \text{ kHz}$, which is what has to clear f_o .

Quantity	Value	Limit	Margin
f_{Min}	92.2 kHz	above $f_o = 90.4$ kHz	1.021
f_{Max}	319.6 kHz	above $f_{sw,max} = 249.0$ kHz	1.283
f_{SU}	503.5 kHz	below the 675 kHz silicon ceiling	1.341
$R_T C_T$	5.17 μs	2.5 to 12 μs	2.068 / 2.321
R_T	11 kΩ	5 to 30 kΩ	2.200 / 2.727
C_T	470 pF	270 to 1000 pF	1.741 / 2.128

Table 26: Oscillator: what the two parts produce, and the limits each result has to clear.

T_{idle} is the weakest number in this chapter. The draft datasheet gives 700 ns in the text and a table that back-solves to 250 ns; this design is worked at 250 ns. At 700 ns the same R_T and C_T put f_{Min} essentially on top of f_o and the floor margin disappears. It is not a precision question, it is whether the two clamps still bracket the operating range, and it is the first thing to measure on hardware.

7.16 The current-sense resistor, which sets three things at once

R_{CS} is asked for by two conditions and the smaller one wins:

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$$R_{CS, max1} = \frac{16.8 \Omega \cdot W}{P_{in}} \quad R_{CS, max2} = \frac{0.55 V}{I_{Lr, pk}} \quad (44)$$

The first is the maximum-power law of Section 4.4; the second is the OCP1 trip point. Opened out, equation 44 is two divisions — $16.8/P_{in}$ and $0.55/I_{Lr, pk}$ — and they give 24.53 and 30.03 mΩ, so the power law is the limiting one. Dissipation then fixes how many resistors that has to be split over. At the worst switching cycle that is 4.54 W, so **5 in parallel**; 120 mΩ each gives **$R_{CS} = 24.0 \text{ m}\Omega$** .

The denominator of $R_{CS, max2}$ is the composite tank peak, not the reflected load current. R_{CS} sits in the bridge return, so it carries the magnetising component too. Using $I_{trafo, pk} = 16.37 \text{ A}$ instead of $I_{Lr, pk} = 18.32 \text{ A}$ loosens the limit by about 12 % and the over-current protection with it.

Result	Value	Against
Maximum input power	700.0 W	$P_{in} = 684.8 \text{ W}$
OCP1 trip	22.92 A	composite peak 18.32 A, margin 1.251
OCP2 trip	31.25 A	immediate stop, 50 μs, restart at f_{SU}
Sense dissipation	4.54 W total, 0.909 W each	1 W parts, margin 1.100

Table 27: What the selected R_{CS} then fixes.

7.17 Burst mode: one resistor, read once at power-up

$$R_{BM} = 16.7 \frac{\text{k}\Omega}{V^2} R_{CS} P_{in, BM}, \quad V_{BM, eq} = \frac{R_{BM}}{100 \text{ k}\Omega} + 0.5 V \quad (45)$$

Entering burst at 75 W of input power — about 11 % of rated — asks for 30.06 kΩ, so **$R_{BM} = 30 \text{ k}\Omega$** and the equivalent threshold on the FB pin is 0.800 V. The valid range is 15 to 140 kΩ. Tie the pin to ground and burst mode is off altogether; deep burst stays active either way.

This threshold and the feedback ripple have to be checked against each other, or the converter chatters in and out of burst once per half line cycle — Section 6.28.

7.18 Brown-out and bridge configuration: the CFG pin

One resistor does two unrelated jobs, and both are read once at power-up:

$$V_{BO, pk} = R_{CFG} \times 4 \frac{V}{\text{k}\Omega}, \quad V_{BO, rms} = \frac{V_{BO, pk}}{\sqrt{2}} \quad (46)$$

The brown-out threshold is compared against the *peak* of the mains, so it has to be converted before it is compared with an rms specification — a $\sqrt{2}$ that is easy to drop. Asking for brown-out just below the minimum line gives $R_{CFG, max} = 31.82 \text{ k}\Omega$; **this is a maximum, so it rounds down, to 30 kΩ**, and $V_{BO} = 84.85 \text{ V}_{rms}$ against a 90 Vac minimum — margin 1.061. Rounding up instead would stop the converter starting at low line.

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R_{CFG}	LOUT2	Configuration
15 k Ω	open	morphing, fixed brown-out (60 V / 70 V peak)
15 to 47 k Ω	open	morphing, adjustable brown-out — this design
47 to 100 k Ω	open	fixed full bridge
15 k Ω	to GND	fixed half bridge, split C_r , fixed brown-out
15 to 100 k Ω	to GND	fixed half bridge, split C_r , adjustable brown-out

Table 28: What R_{CFG} and LOUT2 select together. The 235 and 245 V thresholds are fixed inside the IC and cannot be moved.

For a universal-input design there is only one window. The datasheet floor is 15 k Ω and brown-out at the minimum line is the ceiling at 31.8 k Ω , so the 47 k Ω morphing limit never comes into play. And hang nothing on LOUT2. The pin-strap drives 300 μ A into it and needs 2.5 V, so any pull-down under about 8 k Ω reads as a request for a fixed half bridge.

7.19 Output sensing and over-voltage: the ZCD divider

The auxiliary winding is divided down to the ZCD pin, and the divider ratio alone sets both over-voltage thresholds — the absolute resistor values only set the bias current:

$$R_{ZCD,L} = \frac{2.3 \text{ V}}{I_{bias}}, \quad R_{ZCD,H} = R_{ZCD,L} \left(\frac{n_{aux}}{n_{sec}} \frac{V_{OVP1,out}}{2.3 \text{ V}} - 1 \right) \quad (47)$$

$$V_{OVP1} = \frac{2.3 \text{ V}}{n_{aux}/n_{sec}} \left(\frac{R_{ZCD,H}}{R_{ZCD,L}} + 1 \right), \quad V_{OVP2} = \frac{2.5}{2.3} V_{OVP1} \quad (48)$$

Aiming OVP1 10 % above the output gives 27.50 V as a target. With $n_{aux}/n_{sec} = 1.5$ the calculated pair is 19.17 / 321.8 k Ω . The parts fitted are **19 k Ω and 330 k Ω** . The upper one is deliberately the *next E24 value above* the calculation, so OVP1 lands clear of the ripple crest rather than on it. The result is OVP1 28.16 V and OVP2 30.61 V, and start-up hands over to the loop at 16.53 V of output.

Two ways to get this wrong, and both stop the converter completely. Swapping the two resistors inverts the ratio and OVP1 trips at well under a volt of output, so the converter never starts. And the winding polarity must make ZCD *positive* when the low side of leg 1 is on; reversed, the bridge hard switches from the first pulse.

The turns-ratio ceiling in the datasheet does not apply here. It asks that the rectified auxiliary voltage stay under the V_{CC} rating at OVP2, which is a question only when that winding supplies V_{CC} . In this design V_{CC} comes from a housekeeping rail and the winding senses only, so its voltage is free — 37.5 V nominal, 45.9 V at OVP2 — and the only remaining constraint is that the turns come out whole. Carrying that check over from a self-supplied design reports a failure that is not one.

7.20 HVSU and V_{CC}

- **HVSU connects ahead of the input bridge**, on the ac side, through one 1000 V diode per line. Behind the bridge both the X-capacitor discharge and the brown-out detection stop working, because neither can see the mains disappear.
- The V_{CC} capacitor has to hold the IC from $V_{CCon} = 17 \text{ V}$ down to $V_{CCoff} = 8 \text{ V}$ for as long as it takes the supply to take over. The start-up unit stops 120 ms after V_{CCon} is reached, whether or not the supply has taken over by

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then.

- Where a linear regulator feeds V_{CC} , put a diode between its output and the pin so the start-up charging current cannot be pushed back into it.
- A 100 nF bypass sits at the pin itself.

7.21 Controller pin rules

These follow from the datasheet and each one causes immediate misbehaviour if broken.

Pin	Rule
ISEN	No filter and no series resistor. The maximum-power law and the over-current threshold share this pin.
CFG, BM	No capacitor. Both are strapped at power-up. An out-of-range CFG resistor latches the controller off.
DRV_EN, RT	No capacitor. Both are sampled briefly every few milliseconds and a capacitor is read as a fault.
LOUT2	No pull-down if morphing is used. A few kilohms to ground reads as a fixed half bridge.
HVSU	Connect ahead of the input bridge, on the ac side. Behind the bridge, both X-capacitor discharge and brown-out detection stop working.
ZCD	Auxiliary winding polarity must make ZCD positive when the low side of leg 1 is on. Reversed, the bridge hard switches.
HOUTx, LOUTx	Logic-level outputs, not gate drivers. External half-bridge drivers are required.

Table 29: Pin rules that must not be broken.

7.22 Protections, and where each threshold is set

Every one of these is a design constant fixed by a resistor this note has already chosen, so a change made for one reason moves a threshold set for another. The table is here because the thresholds are otherwise spread over five sections.

Protection	Set by	This design
Brown-out, on the rectified mains	R_{CFG} , read at power-up	V_{BO} 84.8 V, clear of the 90 Vac minimum by 1.061
Cycle-by-cycle over-current, OCP1	R_{CS} — the same resistor as the maximum-power law	22.92 A against a composite peak of 18.32 A, a margin of 1.251
Second-level over-current, OCP2	R_{CS} , fixed ratio to OCP1	31.25 A
Output over-voltage, OVP1	the ZCD divider on the auxiliary winding	28.16 V, above the ripple crest and below OVP2
Second-level over-voltage, OVP2	the same divider	30.61 V
Capacitive-mode protection	internal; it watches the ZCD edge against the gate	the last line of defence only — the frequency floor f_{Min} is the first
Burst mode at light load	R_{BM} , read at power-up	30 k Ω ; the $2f_1$ feedback ripple is 52 mV, so see Section 6.28
Loss of ZVS at the worst corner	not a protection — a design margin	swept, not estimated; Section 6.5

Table 30: Protections and the component that sets each one.

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Two of these share one resistor. R_{CS} sets the over-current thresholds *and* the maximum-power law, because the controller reads both from the same pin. Changing it to fix a power limit moves OCP1, and the other way round. That is also why the pin tolerates no filter.

7.23 System design rules

The same material as a checklist, in the order a design meets it. Every line is argued for somewhere above; collected here so a reviewer can run down the list without re-reading the argument.

- **Design the tank at the real ends of the equivalent range.** With morphing that is 173.2 and 332.3 Vac, not 2×90 and 264. Looking only at the mains range misses both worst corners.
- **Expect λ near 0.5.** The 5-to-10 inductance ratio of a two-stage LLC will not start at low line.
- **Keep the oscillator floor above f_o ,** and treat the anti-capacitive protection as the last line, not the first.
- **Size the output bank from both conditions** — ripple and hold-up — take the larger, and then check the rms ripple current separately. The capacitance and the current are different questions.
- **Put the voltage-loop crossover in the tens of hertz,** then deal with the feedback ripple that survives by moving R_{BM} , not by speeding the loop up.
- **Measure both morphing edges,** at full load and at light load. The half-bridge edge is the worst case for gain and ZVS, the full-bridge edge for switching frequency.
- **Check the turns ratio as a reflected voltage,** $nV_{o,eff}$, never as a bare ratio — that is what keeps n and n_T from being interchanged.
- **Judge the over-current margin on the composite tank peak,** not on the reflected load current.
- **Include the $2f_l$ component** in the output capacitor ripple current, and judge a centre tap at the output node rather than per winding.
- **Carry the conduction ratio $\sqrt{d}$ into the secondary rms.** Dropping it overstates the rms by about 12 % here, and the loss — which goes as the square — by about 26 %.
- **Decide how L_r is built before specifying the transformer,** and put open-circuit and short-circuit inductance on the drawing alongside the turns.
- **Use capacitor ESR at the switching frequency,** not the 120 Hz $\tan\delta$.
- **Never round L_m up.** It lowers λ and takes the ZVS margin with it.
- **Verify ZVS by sweeping the selected tank,** not with the closed form — and if the closed form is used anyway, feed it λ_{act} and the Q of the moment, and know that its error has no fixed sign.
- **Record calculated and selected values separately,** and make every downstream check read the selected one. A sheet that quietly reads the calculated value reports margins the hardware does not have.

7.24 What to measure first on hardware

In order. The first four decide whether the design is sound at all; the rest confirm margins that are computed but never yet seen.

- **$f_{sw}(\theta)$ over a line half cycle.** This settles T_{idle} , and with it both oscillator clamps — the thinnest margin in the design. Everything else in the controller network hangs off it.
- **Cold start into the full output bank.** A 75.2 mF bank is a far heavier start-up load than a conventional design presents, and the start-up window is finite. This risk is new to this architecture, so there is no earlier design to compare with.
- **ZVS at the half-bridge morphing edge** (245 V_{pk}, full load), on the gate and mid-point waveforms. The calculation says 460 ns against 220 ns; confirm it is not hard switching, and that the adaptive dead time actually settles where it is assumed to. **Then measure the worst point of the sweep as well** — 25 % load at the 332.3 Vac equivalent corner, 401 ns — because that, not the design corner, is the figure this design is held to.

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- **Switching frequency at the full-bridge morphing edge** (235 V_{pk} , full load). Calculated 249.0 kHz; check it does not run into the VCO ceiling and get power-limited.
- **Input current and THD at 90 Vac, full load**, to put a number on the zero-crossing dead zone.
- **Output ripple against what the load actually tolerates**. The bank is sized from a specification; this is where the specification is tested.
- **Ripple current and temperature rise in the output bank** — 30.24 A rms calculated, 0.19 A per capacitor.
- **Device temperatures, primary and secondary**. This is where the accepted loss-budget shortfall is decided: measure it in **half-bridge morphing**, above 245 V_{pk} , because in full bridge the standing device is switching and dissipates far less.
- **Feedback ripple against the burst threshold** at light load.
- **An ac dropout at the worst line phase** — cut the mains at the trough of the ripple and confirm the output is still above 19 V after 12 ms.
- **A step across the morphing band**, both directions, on a programmable source. A ramp gives the loop time to follow and hides the transition.

8 References

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9 Errata and open items

Everything in this chapter is something the note works around rather than something it solves. It is last because none of it is needed to follow the design, and first to be re-read before a production release — most of it is a question that only the released datasheet or a prototype can close.

9.1 Datasheet: the draft is not self-consistent

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Item	What the draft says	What is used here, and why
Oscillator idle time T_{idle}	Section 5.3.2 says 700 ns; Table 5 back-solves to about 250 ns from the frequency expressions	250 ns. The two clamps bracket the operating range at 250 ns and stop doing so at 700 ns, so this is a measurement, not a rounding
Brown-out expression	$V_{BO} = \min(60 \text{ V}, R_{CFG} \cdot 4 \text{ V/k}\Omega)$ — read literally this is 60 V for every resistor, which contradicts the configuration table	Read as $R_{CFG} \cdot 4 \text{ V/k}\Omega$ with 15 k Ω as the floor. That is the only reading consistent with the table
Brown-out units	the threshold is a <i>peak</i> voltage; board notes and design tools quote rms	Converted explicitly wherever it is compared ($\sqrt{2} = 1.414$)
Thermal resistance	TBD	No junction temperature can be predicted from the datasheet alone; the loss budget is checked against measured rise instead
ZCD absolute maximum	lower limit given as TBD	The divider is sized from the OVP thresholds only
Driver naming in Section 5.3.1	LOUT1 and LOUT2 are transposed in one sentence against the block diagram and the configuration table	The block diagram and the table are taken as correct: LOUT2 is the pin that is strapped and held high in half bridge

Table 31: Points where the draft datasheet contradicts itself or leaves a value open. Every one of them must be re-checked against the released document.

9.2 Traps in the obvious way of building a design sheet

These are not datasheet problems. They are places where the obvious way to set a design sheet up gives an answer that is wrong in a way nothing flags, so they are worth stating as traps rather than as corrections to any particular tool.

Where	The trap	Consequence if missed
Equivalent input range	taking the maximum from the ac maximum instead of the full-bridge morphing edge	The frequency corner is never evaluated. Here it is the difference between 264 and 332.3 Vac equivalent, and it also sets λ
λ from the minimum-gain condition	the same corner again, one step earlier	The required λ comes out far too small — a factor of over a hundred in one case — and L_m with it
Line-cycle rms currents	evaluating one phase, usually $\theta = \pi/4$, and calling it the line-cycle value	The output bank ripple current and the rectifier loss are both understated. A Simpson rule over the half cycle costs nothing and is exact enough
Secondary rms above resonance	squaring a full half sine when the current is truncated	Up to 8 % at the high corner. Below resonance the factor is exactly 1, which is why it can sit unnoticed for a long time
Primary device rating	rating on the reflected load current $I_{trafo,pk}$	About 12 % under-rated — the switch carries the <i>composite</i> tank current
Loss and thermal resistance	computing loss from the 25 °C $R_{DS(on)}$ and then asking for a heatsink that holds 125 °C	The required thermal resistance comes out only about half as demanding as it really is
ZVS check	the closed-form shortcut, at the design Q and with the design λ	Error of either sign; on this tank 36 % conservative, on another about 23 % optimistic. Optimistic is the dangerous direction
Selected versus calculated	a check that reads the calculated value while the board carries the selected one	Every downstream margin is reported for a design that was not built

Table 32: Traps that produce a plausible wrong number.

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9.3 Constants used here without a derivation

- The $\pi^2/8$ in the dead-time form of the λ condition. It behaves like a correction for the fundamental content of a square wave, but no source states it. It is indicative; the binding check is the ZVS sweep.
- The **exponent 5** in the closed-form ZVS shortcut. This is a fitted number, and it is the main reason the shortcut errs unpredictably near the boundary.
- The **0.744** in the input-voltage margin factor of the compensator design. It scales the crossover target and nothing downstream is sensitive to it at the percent level.
- The **16.8 $\Omega \cdot W$** maximum-power constant is consistent with the feedback span and the multiplier gains quoted in the datasheet — $2.8 \text{ V} / 0.167 = 16.8$ — so it is derived rather than assumed, and is listed here only because the gains themselves are draft values.

9.4 Open items in this design

Item	State	What closes it
T_{idle}	two candidate values, 250 and 700 ns	Measure $f_{\text{sw}}(\theta)$ on the first board
Primary conduction loss	0.311 against the budget — not met, and accepted	No single 600 V device meets a 3 W budget in the standing position. Whether it needs a heatsink is a thermal measurement, not a calculation
Secondary loss budget	1.063 — essentially exhausted	One more device in parallel per leg recovers it; the decision waits on the measured temperature
Transformer inductance tolerance	the tank tolerates a 4.1 % fall in open-circuit inductance and the usual specification asks for $\pm 10 \%$	Either tighten the low side with the supplier, or lower R_T to lift f_{Min} first. Before the transformer is ordered
Output bank volume and height	75.2 mF, 160 parts	A mechanical question, not an electrical one, and it can send the whole architecture back to the output voltage
Standby power	burst entry assumed at 75 W	Measure, then trade R_{BM} against the feedback ripple
Efficiency assumption	$\eta_{\text{HB}} = 98 \%$ assumed	Optimistic. Taking 95 % instead moves R_{CS} and R_{ac} by about 3 %, so nothing downstream is sensitive — but it should be replaced by a measurement

Table 33: What is not settled, and what would settle it.

Every cross-check here is a consistency check, not a correctness check. The numbers agree across three independent implementations, which means the three implement the same equations — not that the equations describe the hardware. That is what the measurement list in Section 7.24 is for.